

Pricing the Sensing Budget in ρ -POMDPs, with Expected Free Energy as the Canonical Information Weight

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Background: Information-gain utilities make sensing valuable directly, but introduce a weight that is usually tuned per task. An engineer may instead need to specify expected sensing usage. **Objectives:** We calibrate the information weight of a ρ -POMDP policy family to a sensing target and identify the role of Expected Free Energy (EFE) within that family. **Methods:** We define an operational shadow price as a crossing threshold of the usage curve, distinguish it from a finite-grid bracket, and construct an endpoint mixture for targets inside usage gaps. We prove scale equivariance of the receding-horizon planner and stationary convergence of a projected controller under stated conditions. Minimizing the specified recursive, reward-based EFE objective is equivalent to maximizing reward plus information gain at $w = 1$ for observe-then-commit problems and hidden-state-preserving actions in a factored extension. The reward-information identification is exact under log scoring and otherwise depends on the declared reward convention. **Results:** Usage curves collapse after rescaling, and the controller re-adapts empirically after a tenfold reward shift. Targets selected by a predeclared rule from calibration curves are met on held-out seeds to within 0.13 observations. Calibration can nevertheless sacrifice reward relative to spending less or choosing another mixture. Comparisons against usage-capped and target-matched references distinguish the cost of extra sensing from shortfalls within the family. On real sensor records with learned likelihoods, two independently fixed targets transfer within the represented collection period, one lies below the sampled range, and later batches defeat frozen calibration. Direct-penalty and fixed-count policies achieve better classification accuracy. **Conclusions:** Usage calibration is appropriate when expected sensing usage is the design requirement. It does not maximize reward under a cap. EFE supplies an information-unit anchor under specified assumptions, while usage, reward, and success must be evaluated separately.

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1 Introduction

An engineer designing an inspection policy may need to specify how much testing the system should perform before deciding whether to repair or release an item. A standard partially observable Markov decision process (POMDP) values a test through its effect on expected task reward. A ρ -POMDP additionally rewards properties of the agent's belief, allowing information gain to have explicit value (Araya-López et al. 2010). This introduces a practical choice: how much weight should information gain receive relative to reward?

We study a different design input, an expected sensing-usage target B . Such a target is useful when sensing is itself a deliverable, as in a coverage, monitoring, or verification requirement. It specifies the average number of tests or their expected cumulative cost. It is neither a hard limit on each episode nor an instruction to maximize

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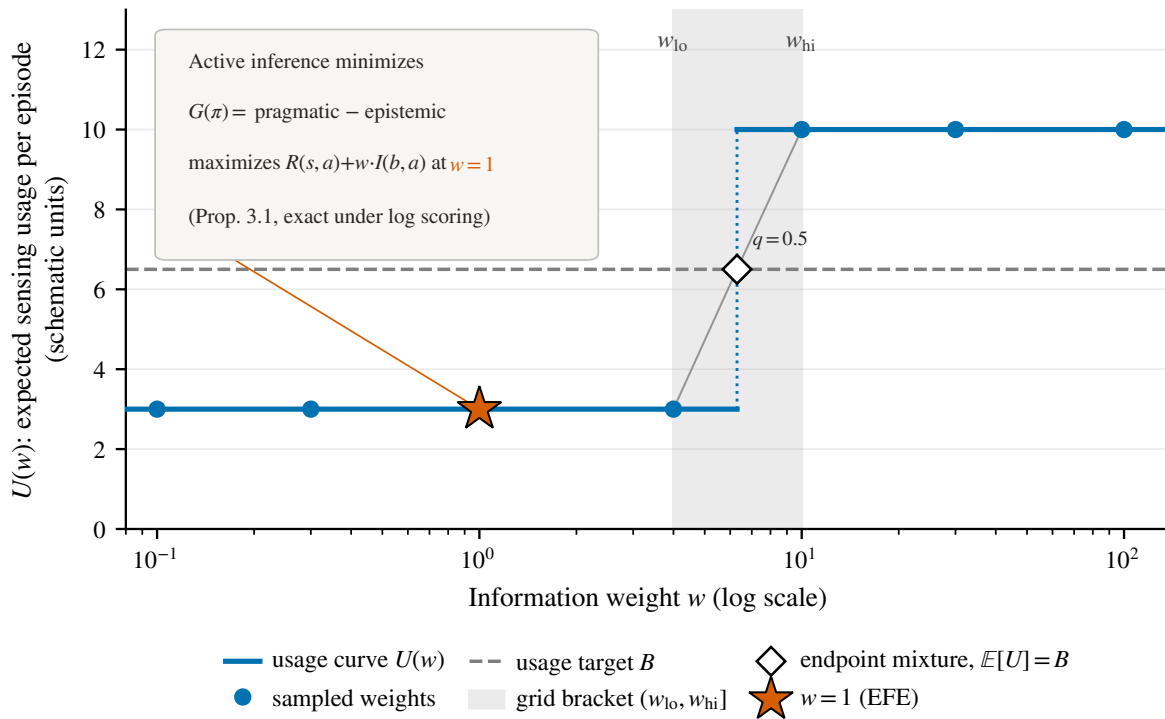


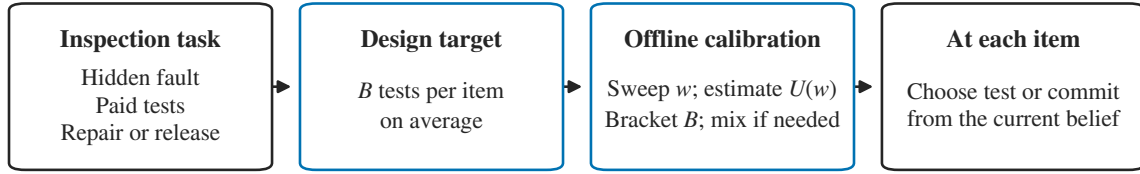
Fig. 1. An expected-usage target selects a crossing of the usage curve (schematic). Sampled grid weights bracket a jump over B . The shaded band estimates the crossing threshold, while the diamond represents a mixture of the two endpoint policies that attains B in expectation under Definition 4.1. Its horizontal position is schematic, not a third deterministic weight. The information-unit default $w = 1$ is one point on the curve and belongs to the level set of its own implicit budget. The equivalence shown in the inset uses the stated reward-based EFE objective, with exact reward identification under log scoring.

reward subject to spending at most B . Those alternatives define different design problems, and our experiments measure the reward cost of meeting a target when reward alone would justify less sensing.

For a family of policies indexed by information weight w , let $U(w)$ be the expected sensing usage. We call the weight at which this curve crosses B its *operational shadow price*, $w^*(B)$. The adjective operational matters: this price is read from a usage curve, rather than obtained as the Lagrange multiplier of a usage-cap constraint. The curve can have plateaus, jumps, and non-monotone portions. A finite grid therefore estimates a crossing by a bracket. When the selected endpoint policies straddle a usage gap, a per-episode mixture can attain the target in expectation under the conditions of Definition 4.1. The crossing threshold, its grid bracket, and this randomized policy are distinct objects.

Figures 1 and 2 connect this construction to the engineer’s workflow. The engineer supplies a model, states an expected-usage requirement, estimates the usage curve offline, and checks usage and reward on held-out episodes. During deployment the agent chooses each test or terminal decision from its current belief. It does not perform a prescribed number of tests on every item.

From an inspection requirement to a policy



Audit across items: compare mean test use with B and assess reward separately.

A usage target is an expectation, not a limit on tests for each item.

Fig. 2. From an inspection requirement to a running policy (schematic, not experimental data). For a state-preserving task with a hidden fault and paid tests, the engineer states a target B in expected tests per item. An offline sweep estimates usage $U(w)$ and brackets B . When a usage gap lies between the selected endpoint policies, mixing them across items can meet the target under Definition 4.1's conditions. The deployed agent chooses tests and terminal actions from its current belief. Usage and reward are assessed separately: meeting B does not establish reward optimality under a sensing cap.

The price has a useful response to a change of reward units. Multiplying all rewards and costs by $\alpha > 0$ and the information weight by the same factor preserves the receding-horizon planner's decisions under the assumptions of Proposition 4.2. Observation counts are unchanged and cumulative sensing costs rescale by α . An online feedback rule can also adjust the weight when observed usage differs from the target. We prove stationary convergence for its projected, decaying-step form under explicit conditions and separately evaluate re-adaptation after a reward-scale shift.

Active inference identifies a particular point in this policy family. Its Expected Free Energy (EFE) objective combines pragmatic value with expected information gain (K. Friston, Rigoli, et al. 2015; Parr and K. J. Friston 2019). For the recursive, reward-based objective specified here, minimizing EFE is equivalent to maximizing reward plus information gain at $w = 1$. The reward-information identification is exact under log scoring (Bernardo 1979). For ordinary task rewards it depends on the declared reward convention and the treatment of the preference normalizer. The coefficient is therefore an untuned default under shared units, not a scale-invariant calibration. In particular, $w = 1$ belongs to the level set of its implicit budget $U(1)$, but need not equal that budget's crossing threshold.

The paper makes three contributions. First, it formulates expected-usage calibration within a ρ -POMDP policy family, distinguishing the level set, crossing threshold, grid estimate, and endpoint mixture. It establishes scale equivariance and a conditional stationary-convergence result for online control. Second, it proves the EFE equivalence for observe-then-commit problems and extends it to the hidden-state-preserving actions of factored observation POMDPs. A two-state analysis characterizes sensing onset and an interval of weights that avoids an unnecessary second observation. Third, it evaluates calibration and the information-unit default against planning baselines and offline references, then tests learned likelihoods on real sensor records with

independently fixed targets and a chronological transfer set. Code and experiment data are available at https://github.com/PatrickAllenCooper/rho_aif.

The evidence also sets limits on these contributions. Targets selected by a predeclared rule from calibration curves are met on held-out seeds to within 0.13 observations, but the selected mixtures can sacrifice reward relative to spending less or to the best sampled reference mixture meeting the same target, and need not be the best mixtures in the weight family. The untuned $w = 1$ lies in the reward-maximizing run of sampled weights on three of five swept environments and is beaten on the other two. Its advantage over reward-only planning depends on the observation structure and available planning depth, rather than on state-space size alone. Ordinary information gain can spend the budget on uncertainty unrelated to reward, and the stated equivalence does not cover observation actions that change the hidden state. These limitations remain part of the main argument, with full controls and counterexamples in the appendices.

An abridged version was accepted at IWAI 2026 (Cooper and Velasquez 2026). It contains the EFE equivalence results and their benchmark evaluation. The budgeted formulation, scale-equivariance and controller results, budget experiments, and distractor and destructive-sensing analyses are extensions. We also revise the workshop version's claim that the advantage grows with state-space size and observation-action count, which RockSample[11,11] does not support.

Section 3 defines the model and EFE objective. Section 4 develops usage calibration, and Section 5 specifies the evaluation protocol. Sections 6 and 7 report the budget and EFE results. Section 8 explains their practical scope. Appendices retain proofs, complete tables, robustness studies, and implementation details.

2 Related Work

POMDP planning and belief-dependent rewards. POMDPs formalize decision-making under state uncertainty (Kaelbling et al. 1998; Smallwood and Sondik 1973). Point-based offline solvers approximate the value function on reachable beliefs, including PBVI, HSVI, and SARSOP (Kurniawati et al. 2008; Pineau et al. 2003; Shani et al. 2013; Smith and Simmons 2004). Online solvers plan from the current belief. POMCP uses Monte Carlo tree search with UCB1 action selection and rollout evaluation (Silver and Veness 2010), DESPOT searches regularized sparse belief trees (Ye et al. 2017), and POMCPOW uses progressive widening for continuous spaces (Sunberg and Kochenderfer 2018). Value of Information Monte Carlo planning selectively skips observation branches with low expected decision value (Laouar et al. 2026). Our planner instead adds expected information gain to the objective. These are different uses of information within planning, and our comparisons do not establish superiority over state-of-the-art POMDP solvers generally.

Araya-López et al. (2010) introduced ρ -POMDPs with belief-dependent rewards. Their ρ denotes the whole reward, whereas we reserve it for an information term added to the ordinary task reward. Convex belief rewards support convex value functions and piecewise-linear convex approximations. Expected information gain is concave in the belief, so that machinery does not apply directly. Fehr et al. (2018) treat non-convex rewards under a Lipschitz hypothesis, and Benchetrit et al. (2025) develop an anytime planner for continuous spaces. For a finite observation model, strictly positive likelihoods make expected information gain Lipschitz on the belief simplex. Zero likelihoods, including noiseless sensing, can violate this condition. Our exact tree recursion does not rely on that approximation guarantee.

POMDPs with information rewards were developed for active cooperative perception (Spaan et al. 2015), and their relation to ρ -POMDPs is discussed by Satsangi et al. (2018). Earlier preference-elicitation work selected queries by sequential value of information while retaining state-dependent rewards (Boutilier 2002). These approaches establish the motivation for valuing information explicitly. Our focus is the coefficient on that value and its relation to expected sensing usage.

Information value and experimental design. Classical value of information measures the improvement an observation can bring to a decision (Howard 1966). Bayesian experimental design instead commonly selects observations by expected information gain (Foster et al. 2021; Lindley 1956). The criteria coincide under a particular reward model: when the reward is a proper log score, the expected improvement in score is the expected information gain (Bernardo 1979). This supplies the exact reward–information identification used here. Sequential experimental design is also the closest structural analogue to our observe-then-commit setting.

Dynamic feature acquisition can greedily select features by conditional mutual information, with learned approximations to the required distributions (Covert et al. 2023). Our sensor-record study uses this established greedy rule as a fixed-count comparator. Its purpose is to test usage calibration within an information-weighted family, not introduce a new feature-selection principle.

Information-directed sampling (IDS) trades instantaneous regret against information gain through their ratio, rather than through a fixed additive coefficient (Russo and Van Roy 2014). We include an observe-then-commit IDS baseline in Appendix N. Walraven et al. (2025) connect active inference with information gathering in POMDPs. Our contribution within this setting is to define and evaluate a calibration procedure for the weight in a specified ρ -POMDP policy family.

Active inference and the EFE formulation. Active inference casts perception and action as variational inference (K. Friston 2010; Parr, Pezzulo, et al. 2022). EFE combines pragmatic and epistemic terms (Da Costa, Parr, et al. 2020; K. Friston, Rigoli, et al. 2015; Parr and K. J. Friston 2019). Its recursive sophisticated-inference form conditions future choices on future beliefs (K. Friston, Da Costa, et al. 2021). Da Costa, Sajid, et al. (2023) prove reward-Bellman optimality for finite-horizon recursive inference in the limit of precise reward-encoding preferences, with exact state estimation. That limiting result does not imply that a fixed finite information weight is reward-optimal.

Several distinct constructions occur under the EFE name. Millidge, Tschantz, and Buckley (2021) distinguish EFE from a naive future extension of variational free energy and propose Free Energy of the Expected Future (FEEF). Champion et al. (2026) examine which EFE expressions follow from common root definitions and the restrictions those roots impose on prior preferences. de Vries et al. (2025) derive EFE-based planning from a model augmented with preference and epistemic priors. The generalised free energy of Parr and K. J. Friston (2019) is another distinct functional. Our equivalence concerns only the recursive, reward-based objective defined in Section 3, with expected information gain about the hidden state as its epistemic term. It does not assert an unchanged coefficient for these alternative constructions. Broader comparisons with decision-theoretic objectives include an EFE optimality-gap analysis (Wei 2024) and a survey of formal equivalences (Sweeney et al. 2026).

Scaling work combines active inference with deep models, reinforcement-learning objectives, and Monte Carlo tree search (Fountas et al. 2020; Maisto et al. 2025; Tschantz et al. 2020). Our MCTS-EFE variant follows this direction. Its controls test exploration constants, rollout choices, and two internal components, while leaving exact beliefs, exact commit values, and EFE leaf evaluation coupled (Appendix S).

Exploration and control as inference. Control-as-inference relates maximum-entropy control to probabilistic inference (Levine 2018; Todorov 2006), with soft actor-critic and KL-minimizing control as algorithmic instances (Haarnoja, Zhou, Abbeel, et al. 2018; Rawlik et al. 2012). Its treatment of value differs from active inference (Millidge, Tschantz, Seth, et al. 2020). EFE can produce intrinsic-motivation behavior (Sajid et al. 2021), while curiosity, count-based exploration, VIME, and random network distillation use weighted exploration bonuses (Bellemare et al. 2016; Burda et al. 2019; Houthoofd et al. 2016; Pathak et al. 2017; Schmidhuber 1991). Oudeyer and Kaplan (2007) survey these intrinsic signals. Bayes-adaptive planning and Bayesian reinforcement learning primarily address uncertainty about the model (Duff 2002; Ghavamzadeh et al. 2015; Guez et al. 2013). Our planners treat the observation model as fixed, either supplied by a simulator or fitted once from training records. They do not plan to learn that model. Describing $w = 1$ as untuned still depends on a shared reward convention.

Budgets and dual control. Rational inattention constrains or prices information capacity (Matějka and McKay 2015; Sims 2003). Constrained MDPs use Lagrange multipliers on expected resource costs (Altman 1999), and constrained POMDP methods track admissible cost alongside belief (Kim et al. 2011). Automatic temperature tuning in soft actor-critic drives policy entropy toward a target (Haarnoja, Zhou, Hartikainen, et al. 2018). These provide close conceptual precedents for resource control. Our weight multiplies information gain, however, while the target concerns observation count or cost. It is not generally the multiplier of a sensing-cap constraint. Section 4.6 gives the detailed distinctions, including the difference from a dual formulation of EFE itself (Kouw 2026). The contribution is an operational calibration for this policy family, its exact scale transformation under stated assumptions, and explicit treatment of finite-grid crossings and usage gaps.

3 Methodology

We first specify the policy family whose information weight will be calibrated to a sensing target. The EFE connection identifies one member of that family under explicit reward and information-unit conventions. It does not establish reward optimality.

3.1 The ρ -POMDP Framework

A ρ -POMDP adds a belief-dependent utility $\rho : \Delta(S) \times \mathcal{A} \rightarrow \mathbb{R}$ to task reward, where $\Delta(S)$ is the probability simplex over hidden states S . Its objective is

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{\pi} \left[\sum_{t=0}^H \gamma^t (R(s_t, a_t) + \rho(b_t, a_t)) \right]. \quad (1)$$

Here S is the finite hidden-state space, $\mathcal{A}$ the action set, $b_t \in \Delta(S)$ the Bayesian belief, H the planning horizon, and γ the discount factor. The expectation runs over the initial state, state transitions, observations, and resulting belief updates under policy π . Setting $\rho = 0$ recovers a standard POMDP. Choosing information gain as ρ explicitly rewards uncertainty reduction. The equivalence below concerns $\gamma = 1$.

In the observe-then-commit class, $\mathcal{A}$ partitions into observation actions $\mathcal{A}_{\text{obs}}$, which leave the hidden state fixed and update the belief at known cost, and terminal commit actions $\mathcal{A}_{\text{com}}$, which yield state-dependent reward and emit no observation. Rewards are not observations: beliefs are conditioned on observation symbols alone. Each episode consists of observations followed by one commit. This structure isolates the information-weight question and permits analytic thresholds and near-optimal offline references. Section 3.3 extends the reduction to interleaved sensing and navigation. Appendix U.1 includes a schematic of the basic decision loop.

3.2 Expected Free Energy as ρ

At future time τ , the EFE formulation considered here combines expected negative log-preference with negative expected information gain,

$$\mathcal{G}(\pi) = -\mathbb{E}_{Q(o_{\tau}|\pi)} [\ln P(o_{\tau} | C)] - \mathbb{E}_{Q(o_{\tau}|\pi)} [D_{\text{KL}}[Q(s_{\tau} | o_{\tau}, \pi) \| Q(s_{\tau} | \pi)]] . \quad (2)$$

$Q(o_{\tau} | \pi)$ predicts observations under policy π , while $Q(s_{\tau} | \pi)$ and $Q(s_{\tau} | o_{\tau}, \pi)$ are the corresponding prior and posterior beliefs. The parameters C specify the preference distribution $P(\cdot | C)$, which is distinct from the observation model conditioned on state and action. Minimization favors preferred outcomes and informative observations. Following the recursive, closed-loop formulation associated with sophisticated inference (Da Costa, Sajid, et al. 2023; K. Friston, Da Costa, et al. 2021), we evaluate actions from the current belief and replan after every observation.

We identify the pragmatic term with negative task reward under the reward-based convention detailed below. A terminal commit has value $\mathcal{G}(\text{commit}_i | b) = -\mathbb{E}_b [R_i]$, where $R_i = R(s, \text{commit}_i)$. It has neither an epistemic

term nor a continuation term, because it emits no observation and ends the episode. An observation action k costs $c_k > 0$ and supplies expected information gain

$$I_k(b) = H(b) - \mathbb{E}_{o|obs_k}[H(b'_o)],$$

where b'_o is the Bayesian posterior and $H(b) = -\sum_s b(s) \log b(s)$ is entropy. The entropy functional $H(\cdot)$ is distinct from the planning horizon H . The recursion is

$$\mathcal{G}(\text{observe}_k | b) = c_k - I_k(b) + \mathbb{E}_o \left[\min_a \mathcal{G}(a | b'_o) \right]. \quad (3)$$

At depth H , only commit actions are available. The agent selects a minimizing action using a fixed tie-breaking rule. As policy precision $\kappa \rightarrow \infty$, the softmax $Q(\pi) \propto \exp(-\kappa \mathcal{G}(\pi))$ concentrates on policies minimizing $\mathcal{G}$. The implementation makes a deterministic selection among tied minimizers. Policy precision κ is distinct from the reward inverse temperature β below. A separately normalized preference distribution does not generally give this recursion on variable-length episodes, because its omitted normalizer can affect action choice.

3.3 Formal Equivalence with ρ -POMDPs

Writing d for search depth and setting $V = -\mathcal{G}$ gives

$$V(\text{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o \left[\max_a V(a, b'_o, d + 1) \right], \quad (4)$$

$$V(\text{commit}_i, b, d) = \mathbb{E}_b[R_i]. \quad (5)$$

At $d = H$, the value is the best commit value. Write $R(b, a) := \sum_s b(s)R(s, a)$ for expected immediate reward and $V^*(b, d) := \max_a V(a, b, d)$ for the optimal belief-state value. This is a ρ -POMDP Bellman recursion with an action-dependent belief utility, as permitted by Araya-López et al. (2010).

PROPOSITION 3.1. Define $\rho_{\text{EFE}}(b, a)$ as

$$\rho_{\text{EFE}}(b, a) = \begin{cases} I_a(b) & \text{if } a \text{ is an observation action} \\ 0 & \text{if } a \text{ is a commit action} \end{cases}$$

where $I_a(b) = H(b) - \mathbb{E}_{o|a}[H(b'_o)]$ is the expected information gain from observation action a at belief b . Assume (i) pragmatic value is identified with task reward at $\beta=1$ per reward unit, (ii) beliefs are updated by exact Bayesian conditioning, (iii) information is measured in the same units on both terms, and (iv) both recursions select actions deterministically with the same fixed tie-breaking rule. Commit actions carry $\rho_{\text{EFE}}=0$ not by stipulation but because the episode terminates on commitment, so no further observation is emitted and the epistemic term has no successor belief to score. Then for undiscounted finite horizon ($\gamma=1$), minimizing recursive EFE (Eq. 3) over horizon H produces the same policy as solving the ρ -POMDP Bellman recursion

$$V^*(b, d) = \max_a \left\{ R(b, a) + \rho_{\text{EFE}}(b, a) + \mathbb{E}_o[V^*(b'_o, d + 1)] \right\}$$

over the same horizon, with terminal commit actions carrying no continuation term and only commit actions available at $d = H$.

Negating the objective gives the Bellman maximization (proof in Appendix A). This identifies the specified recursive EFE agent with Planning+IG at $w = 1$ under the stated convention. It is a notational bridge, not a reward-optimality theorem. Unlike the $\beta \rightarrow \infty$ limit of Da Costa, Sajid, et al. (2023), it retains full information weight. It concerns the information-gain and pragmatic-value formulation among those related by Champion et al. (2026), with the scope and approximation restrictions described in Related Work and Appendix U.1.

Reward and information units. The identification in assumption (i) fixes an exchange rate rather than deriving one. With $P(o \mid C) \propto \exp(\beta R(o))$,

$$\ln P(o \mid C) = \beta R(o) - \ln Z, \quad Z = \sum_{o'} \exp(\beta R(o')).$$

Setting $\beta = 1$ assigns one nat to each reward unit. A path scoring T outcomes also accumulates $-T \ln Z$ in log-preference. Since observe-then-commit paths have different lengths, this term can be policy-dependent. It vanishes for a log-scoring reward with $Z = 1$ (Bernardo 1979), or cancels when every path scores the same number of outcomes under a common preference distribution. We do not impose that fixed-length construction. Proposition 3.1 therefore proves equivalence for the reward-based recursion as written, conditional on its pragmatic identification, rather than deriving it from general normalized preferences.

The empirical convention is different again: the implementation computes entropy in bits and runs the reported EFE agent at one reward unit per bit. Since one bit is $\ln 2$ nats, implemented $w = 1$ corresponds to $1/\ln 2 \approx 1.44$ reward units per nat. The exactly nat-canonical agent uses $w = \ln 2 \approx 0.69$ in implementation units. Appendix R compares these directly. Their reward, success, and observation count are bit-identical on Tiger, Diagnosis, Bandit, and Testbed. Tileworld changes policy, and $w = 1$ remains Pareto-dominated there. Thus the exact log-score case, the conditional recursion equivalence, and the empirical bit convention are separate claims.

The weight is also reward-scale dependent. Multiplying all rewards and costs by $k > 0$ at fixed w is behaviorally equivalent to using w/k before rescaling. The reward-maximizing weight scales with k (Appendix Q). Proposition 4.2 gives the corresponding usage identities. Whether $w = 1$ is a useful default at a fixed scale remains an empirical question.

A two-state calculation separates the weight needed to buy a first observation at $H = 1$ from the weight that buys an unnecessary second observation at $H = 2$ (Proposition U.1, Appendix U.2). The first threshold is negative when observation is already strictly instrumentally worthwhile. The second bounds a near-optimality interval under the proposition's symmetric two-state assumptions. Only Tiger among the tabulated comparisons satisfies those assumptions. Diagnosis and Tileworld use unproved two-state reductions, and Testbed lies at the boundary $|R^-| = R^+$. The interval is not a guarantee for longer receding-horizon episodes. Discounting preserves the algebraic equivalence when applied consistently to both recursions, with behavioral effects reported in Appendix K.

Extension to factored observation POMDPs. The reduction extends to interleaved sensing and navigation when both preserve the hidden state.

Definition 3.2 (Factored observation POMDP). A POMDP is a factored observation POMDP if its state decomposes as $s = (s_{\text{vis}}, s_{\text{hid}})$ where s_{vis} is fully observable and s_{hid} is hidden, and the action set partitions into (i) observation actions $\mathcal{A}_{\text{obs}}$ that produce observations about s_{hid} without changing s_{hid} (though they may change s_{vis}), (ii) navigation actions $\mathcal{A}_{\text{nav}}$ that change s_{vis} deterministically without changing s_{hid} and produce no informative observation, and (iii) exploitation actions $\mathcal{A}_{\text{exp}}$ that yield reward that may depend on s_{hid} and emit no informative observation. Throughout, rewards are not observations. The belief is conditioned on the observation symbol alone, so an exploitation action's payoff, however informative about s_{hid} it would be to an agent that conditioned on it, does not update the belief.

Observe-then-commit is the special case with no navigation and a trivial visible state. Our RockSample variant (Smith and Simmons 2004) holds rock qualities fixed, makes position and sampling status visible, and classifies checks, moves, and sample/exit actions as observation, navigation, and exploitation respectively. The original RockSample changes a good rock to bad on sampling. Our variant differs in that respect.

To state the boundary precisely, write s, s' for the hidden components, b for their belief, and O for the observation likelihood. Let

$$b'_o(s) \propto O(o \mid s, a)b(s), \quad b'_{o,T}(s') \propto \sum_s T_{\text{hid}}(s' \mid s, a)O(o \mid s, a)b(s).$$

Here the observation precedes the transition, and the known visible state is suppressed. Interpret b'_o on the post-transition state space through the identity coupling $s' = s$. The transition–observation diagnostic is

$$\Delta_T(b, a) := \mathbb{E}_o D_{\text{KL}}[b'_{o,T} \parallel b'_o].$$

It compares the correct posterior with the one obtained by assuming preservation.

PROPOSITION 3.3. *In a factored observation POMDP (Definition 3.2), let b denote the belief over s_{hid} . For any action a that preserves s_{hid} (i.e., $a \in \mathcal{A}_{\text{obs}} \cup \mathcal{A}_{\text{nav}}$), the transition–observation coupling term vanishes, $\Delta_T(b, a) = 0$, and $\rho_{\text{EFE}}(b, a) = I_a(b)$ for observation actions, $\rho_{\text{EFE}}(b, a) = 0$ for navigation actions.*

Under preservation, the two posteriors coincide, which proves the claim (Appendix U.3). Terminal exploitation has no continuation or epistemic term. Nonterminal exploitation carries no information gain here because its observation is uninformative and its payoff is not used to update the belief. A model that conditions on exploitation rewards must include the resulting information gain. Proposition 3.3 covers observation and navigation actions, not arbitrary exploitation transitions.

State preservation is an exact hypothesis. Slow drift alone does not ensure a small diagnostic: a perfect sensor gives a point-mass posterior, and an arbitrarily small probability of a state change can place mass outside its support, making Δ_T infinite. Example U.2 shows a complete action-ranking reversal from using obsolete information.

Destructive sensing can reverse the action ranking. In Example U.2, a perfect test reveals the old state and then destroys it. At $w = 1$ reward unit per bit and cost $c \in (1, 2)$, the preservation-based value is $2 - c > 0$, while the transition-aware value is $1 - c < 0$. Acting on the obsolete observation realizes expected reward $-c$, below immediate commitment’s zero. Appendix U.3 gives the complete example and Figure 14. The restriction is on the state-preserving reduction. Transition-aware EFE removes the illusory information credit. We have not established Δ_T as an additive value correction or a decision-error bound, and a transition-aware budget procedure remains open.

3.4 Agents

The six core agents in Table 1 share a known generative model and exact Bayesian updates. Planning, Planning+IG, and EFE use the same recursive search, so the comparison separates the objective from planning depth. Every $H > 1$ agent replans a full depth- H tree after each real observation and executes only its root action.

Table 1. Agent specifications. All use exact Bayesian belief updates over a known generative model.

Agent	ρ function	Horizon	Controls for
Myopic	$\rho = 0$	$H=1$	Weakest baseline
Planning	$\rho = 0$	$H > 1$	Planning depth
Info Gain	$w \cdot I(b)$	$H=1$	Epistemic bonus (myopic)
Planning+IG	$w \cdot I(b)$	$H > 1$	IG + planning depth
EFE	$I_a(b)$ via EFE	$H > 1$	Joint objective (Prop. 3.1)
Epistemic-only	$I_a(b)$ only	$H > 1$	Ablation: no pragmatic term

Planning+IG is the central baseline and represents the familiar information-reward construction in POMDP-IR and ρ -POMDPs (Araya-López et al. 2010; Satsangi et al. 2018; Spaan et al. 2015). By Proposition 3.1, its difference from EFE is the chosen weight. The pymdp check in Appendix H tests qualitative consistency of the epistemic term, not certified numerical equivalence (Heins et al. 2022).

Epistemic-only sets commit EFE to zero, removing reward from commit timing while retaining sensing costs and information gain. On commitment it chooses the most probable state. Myopic has no information bonus but can still observe when one-step lookahead reveals positive instrumental value of information. Additional baselines are Greedy without sensing, information-directed sampling (Appendix N), and a posterior-vote rule. The last draws 100 states and takes the majority of their individually optimal actions. It is not Thompson sampling, although the committed CSVs label it ThompsonSamplingAgent. Appendix U.4 gives the full specifications and ablation interpretation.

Info Gain and Planning+IG use success-maximizing weights w_{succ}^* in the main comparison tables, selected from $\{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$. Each candidate uses a common tuning stream at seed 7, reset between candidates, with 200 episodes (100 on Tileworld). Ties favor the smallest maximizing weight. This stream is disjoint from the five evaluation seeds, and the selected weights retain its sampling noise. Seed 7 also appears in the separate frontier battery, which evaluates no tuned baseline. The Pareto study instead reports reward-maximizing weights w_{ret}^* from its wider eleven-point grid (Section 7.2). Neither is the usage-calibrated price $w^*(B)$ defined next.

4 Budgeted ρ -POMDPs and the Price of Information

We now choose an information weight from a target for expected sensing usage. If sensing already has a known economic cost in reward units, that cost belongs directly in R . A usage target is useful when the reward scale is conventional or the design requirement is naturally stated as an average number or cumulative cost of tests. It controls sensing effort. It does not by itself maximize reward.

4.1 The Budgeted Problem and the Usage Curve

Let $R(\pi)$ and $U(\pi)$ denote expected episodic return and sensing usage. The calibration problem asks for a member of the receding-horizon Planning+IG family, or a per-episode mixture of two members, whose expected usage equals B . The separate constrained problem is

$$\max_{\pi} R(\pi) \quad \text{subject to} \quad U(\pi) \leq B. \quad (6)$$

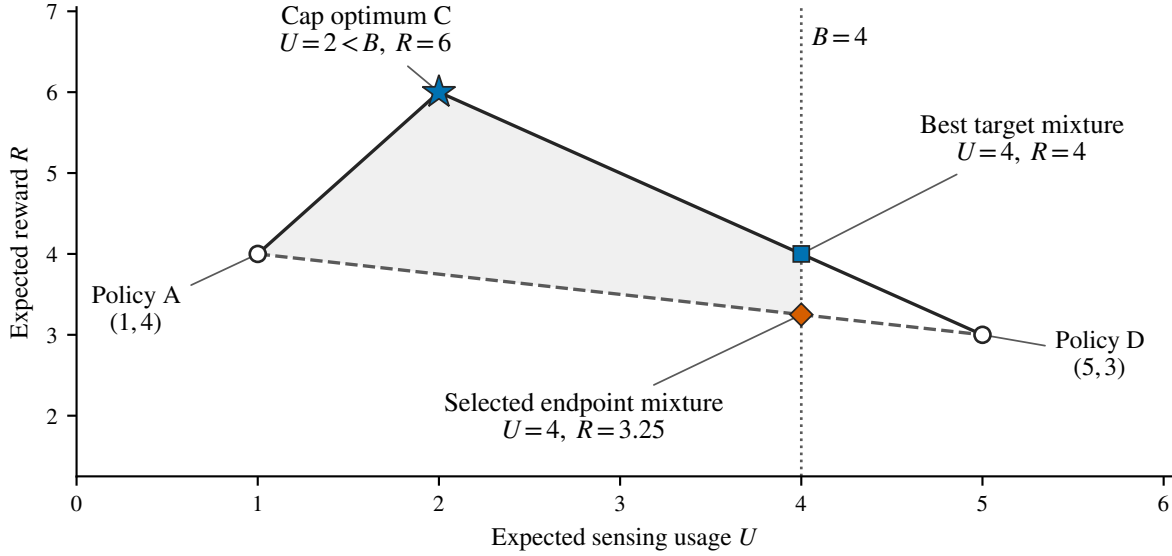
The policy class may be the fixed family Π or the broader class used by a constrained-POMDP reference. We use those references to measure the reward cost of calibration (Section 6.8). Their cap may be slack at the reward optimum.

The literal Lagrangian of (6) penalizes usage through $R(\pi) - \lambda U(\pi)$, $\lambda \geq 0$. Planning+IG instead uses $R(\pi) + wI(\pi)$, where $I(\pi)$ is expected cumulative information gain. This is an information-floor relaxation. For $w \geq 0$, an exact maximizer π_w over a fixed class solves $\max R(\pi)$ subject to $I(\pi) \geq I(\pi_w)$: optimality gives

$$R(\pi) \leq R(\pi_w) + w(I(\pi_w) - I(\pi)) \leq R(\pi_w)$$

for every floor-feasible π . This statement concerns a one-shot exact maximizer, not the implemented receding-horizon agent.

For that implemented family, let π_w^* denote the closed-loop policy using weight w and define its usage curve by $U(w) := U(\pi_w^*)$. The star here denotes the policy produced by the finite-horizon planner, not an optimizer of total episodic return. Its operational shadow price $w^*(B)$ locates a crossing of the target by this curve. It is not the usage-cap multiplier $\lambda^*(B)$, and meeting the target neither solves (6) nor guarantees the best reward among mixtures at the same usage. Figure 3 distinguishes these possibilities.



Illustrative policy points. Line segments are per-episode mixtures.
Shading marks cap-feasible mixtures.

Fig. 3. An expected-usage target and a usage cap answer different questions. This schematic uses three illustrative policies and their per-episode mixtures, not benchmark data. The target $U = B$ is the vertical line, while the shaded portion of the mixture hull satisfies the cap $U \leq B$. Suppose increasing grid weights select policies C, A, then D, so A and D form the selected crossing pair. A selected endpoint mixture can meet the target yet earn less than another mixture at the same usage. The highest-return cap-feasible policy can also leave part of the budget unused. These possibilities explain why calibration alone supplies neither an equality-constrained nor a cap-constrained reward optimum. They do not assert that every instance has either shortfall.

Discrete policy changes produce usage gaps, so a single weight need not attain B . The curve may also be nonmonotone. We estimate it on a finite grid and use the following four distinct objects to avoid treating a grid estimate as an exact inverse.

Definition 4.1 (Operational shadow price, crossing bracket, and endpoint mixture). Fix the implemented family's usage curve $U(w)$ on $w \geq 0$ and a target B .

Level set. $U^{-1}(B) = \{w \geq 0 : U(w) = B\}$ contains the weights that attain the target. It is empty at a usage gap and may contain a plateau.

Crossing threshold. Suppose the curve has finitely many upward crossings of B and at least one occurs at a finite weight. Define

$$D_B = \{w \geq 0 : U(w) < B, \exists v > w \text{ with } U(v) \geq B\}.$$

When D_B is nonempty and bounded, the operational shadow price is $w^*(B) := w^\dagger(B) := \sup D_B$, the final crossing location. Selecting the last crossing is a convention for nonmonotone curves. At a jump it is a single weight, even if no weight attains B . It is undefined if the curve never crosses from below, or has no finite final crossing. A level set can therefore exist without a shadow price.

Grid bracket. On an ordered finite grid, let w_{lo} be the largest sampled weight with $U(w_{lo}) < B$, and let w_{hi} be the first subsequent sampled weight with $U(w_{hi}) \geq B$. If both exist, the solver reports $(w_{lo}, w_{hi}]$. Its closed

enclosure $[w_{lo}, w_{hi}]$ contains the population threshold provided the grid misses no crossing, either between samples or beyond its final point. The enclosure is closed because the threshold can equal w_{lo} when usage rises immediately after it. The bracket's width measures grid resolution, not uncertainty, and its interior weights need not attain B . Without straddling endpoints, the solver flags its fallback as unbracketed, even if an unsampled population crossing exists.

Endpoint mixture. For straddling endpoints, choose w_{hi} independently at the start of each episode with probability

$$q = \frac{B - U(w_{lo})}{U(w_{hi}) - U(w_{lo})},$$

and otherwise choose w_{lo} , retaining that policy for the episode. Linearity gives expected usage exactly B for these population endpoint usages. This is one randomized policy, not an interval of scalar prices or a constrained-reward optimum. It attains gap targets that no single weight can, consistent with the need for randomization in constrained MDPs and POMDPs (Altman 1999; Kim et al. 2011).

In practice, the grid and mixture probability use estimated usages, so held-out usage need not equal B . Offline solving also labels the nearest grid point as achieving the target when $|U(w) - B| \leq \max(0.3, 0.1B)$. This numerical tolerance affects only that achievability label, never the bracket.

The closure of mixture usages is $[\inf_{w \geq 0} U(w), \sup_{w \geq 0} U(w)]$. Every interior target can be attained by mixing two straddling members. A boundary target is attained only if that endpoint usage is attained by a member. The selected crossing mixture is more restrictive: it requires the solver's upward-straddling pair. For example, a curve that descends from usage 2 to 1 admits a mixture at $B = 1.5$ but has no crossing from below. A curve whose final grid point falls below B also gives an unbracketed fallback, even if an earlier upward crossing exists. No swept curve exhibits that final-grid shape.

All reported endpoint mixtures have straddling brackets. Tiger's $B = 4$ in Table 27 and the below-range targets of Sections 6.8 and 6.9 are explicitly unbracketed. The attainable range need not begin at $U(0)$: usage is nonmonotone, and on *Tileworld-6×6* it falls from 15.64 at $w = 0$ to 14.75 at $w = 1$ (Section 7.3). Positive $U(0)$ reflects instrumental sensing under the planner's model, while belief saturation and the episode cap limit usage at large weights.

4.2 Exact Scale Equivariance

Let an α -rescaled environment multiply every pragmatic quantity, including observation costs, by $\alpha > 0$, while leaving dynamics, observations, and belief updates unchanged. Scaling w by the same factor then multiplies every candidate value by α .

PROPOSITION 4.2 (EXACT SCALE EQUIVARIANCE). *For every $\alpha > 0$ and $w \geq 0$, the receding-horizon Planning+IG agent facing the α -scaled environment with weight αw selects the same action as the agent facing the unscaled environment with weight w , at every belief, provided ties are broken by a scale-independent rule. Consequently the closed-loop policies coincide, and so do the induced distributions of every quantity that depends on the actions alone. For usage denominated in observation count this gives*

$$U_{\text{count},\alpha}(\alpha w) = U_{\text{count}}(w), \quad (7)$$

and for usage denominated in cumulative sensing cost, which sums the rescaled costs of the same actions, it gives

$$U_{\text{cost},\alpha}(\alpha w) = \alpha U_{\text{cost}}(w).$$

PROOF. For any candidate plan, the α -scaled value is $\alpha \cdot (\text{pragmatic sum}) + \alpha w \cdot (\text{IG sum}) = \alpha \cdot [\text{pragmatic sum} + w \cdot \text{IG sum}]$, i.e. exactly α times the unscaled value. Multiplying every candidate's value by the same positive constant preserves the ordering of the candidates, and the scale-independent rule compares values and a tolerance

that have both been multiplied by α , so it returns the same action. By induction over decision points, the closed-loop policies agree at every belief. Since the environment's stochastic dynamics are otherwise identical, the induced trajectory distributions agree, and with them the distribution of the observation count. Each action's cost is α times its unscaled cost, so the cumulative sensing cost of every trajectory is α times its unscaled value, which gives the cost identity. $\square$

The implementation's comparison tolerance is relative, 10^{-12} times the larger candidate magnitude, and favors the incumbent commit action. This rule scales with value. An absolute tolerance would not. Floating-point rounding need not be exactly homogeneous, so bit-identical collapse in the experiments is a measurement rather than a mathematical guarantee.

For count usage, plotting against w/α gives the same curve. Cost usage must also be divided by α . Wherever the threshold exists,

$$w_{\text{count}}^*(B; \alpha) = \alpha w_{\text{count}}^*(B; 1), \quad w_{\text{cost}}^*(B; \alpha) = \alpha w_{\text{cost}}^*(B/\alpha; 1).$$

The grid bracket rescales only when the grid itself is multiplied by α . A fixed $w = 1$ at scale α instead behaves as $w = 1/\alpha$ at scale 1, with implicit count budget $B_{\text{EFE}}(\alpha) := U(1/\alpha)$. Equivariance applies equally to a directly tuned weight on the correspondingly rescaled grid.

4.3 Monotone Comparative Statics and the Usage Staircase

Scale equivariance does not imply that usage increases with w . The general monotonicity result concerns expected information gain under exact maximization over a fixed plan set.

PROPOSITION 4.3 (MONOTONE COMPARATIVE STATICS FOR EXACT MAXIMIZERS). *Let Π be any fixed set of plans and, for $w \geq 0$, let $\pi_w \in \arg \max_{\pi \in \Pi} R(\pi) + w I(\pi)$. Then for $w_2 > w_1 \geq 0$,*

$$I(\pi_{w_2}) \geq I(\pi_{w_1}) \quad \text{and} \quad R(\pi_{w_2}) \leq R(\pi_{w_1}). \quad (8)$$

PROOF. Optimality of π_{w_1} at w_1 gives $R(\pi_{w_1}) + w_1 I(\pi_{w_1}) \geq R(\pi_{w_2}) + w_1 I(\pi_{w_2})$, and optimality of π_{w_2} at w_2 gives $R(\pi_{w_2}) + w_2 I(\pi_{w_2}) \geq R(\pi_{w_1}) + w_2 I(\pi_{w_1})$. Adding the two inequalities and simplifying yields $(w_2 - w_1)(I(\pi_{w_2}) - I(\pi_{w_1})) \geq 0$, and since $w_2 > w_1$, $I(\pi_{w_2}) \geq I(\pi_{w_1})$. Substituting this back into the first inequality gives $R(\pi_{w_1}) - R(\pi_{w_2}) \geq w_1(I(\pi_{w_2}) - I(\pi_{w_1})) \geq 0$. $\square$

These inequalities do not imply monotone observation count. A policy with fewer, more informative observations can collect more information. Moreover, a receding-horizon agent maximizes again at each reached belief, so its complete trajectory is not a single optimizer over the fixed plan set in the proposition. The empirical usage staircases can therefore dip, as they do on Bandit and Tileworld (Section 6.5).

The single observe-or-commit decision does admit an exact usage threshold. In this corollary and its onset checks, we write w_{thresh} for the w_{thresh}^* of Proposition U.1, dropping the star to distinguish it from $w^*(B)$.

COROLLARY 4.4 (PROPOSITION U.1'S ONSET THRESHOLDS ARE USAGE-STAIRCASE KNOTS). *In the two-state $H=1$ setting of Proposition U.1 with $w_{\text{thresh}} \geq 0$, equivalently $c \geq (p - \frac{1}{2})(R^+ - R^-)$, for any budget $B \in (0, 1)$ the crossing threshold of Definition 4.1 is exactly w_{thresh} , the weight at which the staircase leaves zero, and the closed enclosure $[w_{\text{lo}}, w_{\text{hi}}]$ of a finite-grid crossing bracket contains that threshold whenever the sampled usages straddle B , locating it to grid resolution rather than equalling it. When $w_{\text{thresh}} < 0$ every nonnegative weight already observes, so $U \equiv 1$ on $[0, \infty)$ and no onset knot lies in the nonnegative range. The onset boundary of the budgeted problem recovers Proposition U.1's closed form as the first knot of the usage staircase.*

Its proof is the single-step curve $U(w) = 1[w > w_{\text{thresh}}]$ under the commit-favoring tie rule, where $1[A]$ is 1 when A holds and 0 otherwise. The full argument and implementation check are in Appendix U.5. That check

runs whole receding-horizon episodes and verifies the onset, not a claim that every such episode takes exactly one observation.

4.4 Solving for the Shadow Price

We estimate $U(w)$ by multi-seed Monte Carlo on a log-spaced grid: five seeds and 100 episodes per cell on the discrete observe-then-commit domains, 40 on Tileworld- 6×6 , and lighter settings on tree-search domains (Section 6). Per-seed means give standard errors. We report the crossing bracket and the grid weight $\hat{w}(B)$ minimizing $|U - B|$, rather than a bisection root. This plotted point can lie on the lower bracket edge or, at a slack budget, below it. Count and cost usage are distinct. When all sensing actions share a cost, the two differ by a fixed conversion (Section 6.3).

4.5 Online Dual Control of the Shadow Price

A reward-scale change moves the count-budget crossing. To adjust without a fresh grid sweep, we adapt SAC's automatic temperature update (Haarnoja, Zhou, Hartikainen, et al. 2018), replacing the policy-entropy target by a sensing target. After episode t with observed usage U_t , update

$$w_{t+1} = \text{Proj}_{[0, w_{\max}]}(w_t + a_t(B - U_t)), \quad a_t = \frac{\eta_0}{1 + \delta t}, \quad (9)$$

where η_0 is the initial step, δ its decay rate, and projection clips the weight to $[0, w_{\max}]$. If $\mathbb{E}[U_t | w_t] = U(w_t)$, usage is a noisy observation of the curve. Underspending raises w and overspending lowers it. This sign corrects usage only under suitable crossing conditions.

The proposition covers the projected decaying-step controller, with finite $w_{\max}$ and $\delta > 0$. Every reported run uses $w_{\max} = 100$ and $\delta = 0.02$. No iterate exceeded 3.1, so the upper projection was present but inactive. It does not cover the released library default, which has a constant step and no upper projection, or the reset-on-shift heuristic described below.

PROPOSITION 4.5 (STATIONARY CONVERGENCE OF THE DUAL CONTROLLER). *Consider the projected decaying-step controller, the recursion (9) with a finite $w_{\max}$ and step sizes satisfying $\sum_t a_t = \infty$ and $\sum_t a_t^2 < \infty$ (the schedule in (9) with $\delta > 0$). Suppose four conditions hold. (i) The environment, agent family, and budget B are stationary, so that $\mathbb{E}[U_t | w_t] = U(w_t)$ for a fixed function U . (ii) Per-episode usage is bounded, which holds here because usage is bounded by the episode step cap, and each episode is run on an environment reset and freshly seeded independently of the past, so that the noise $U_t - U(w_t)$ is a martingale difference with bounded conditional second moment. (iii) There is a single crossing $w^* \in (0, w_{\max})$ with $U(w) < B$ for $w < w^*$ and $U(w) > B$ for $w > w^*$. (iv) The curve stays away from the budget outside every neighborhood of the crossing, that is, for every $0 < \epsilon_1 < \epsilon_2$, $\inf\{|U(w) - B| : \epsilon_1 \leq |w - w^*| \leq \epsilon_2, w \in [0, w_{\max}]\} > 0$. Then $w_t \rightarrow w^*$ with probability one, and hence in probability. Condition (iii) does not require $U(w^*) = B$, and condition (iv) does not require continuity, so the conclusion covers a gap budget in the sense of Definition 4.1, where U jumps over B at w^* without touching it. What converges in that case is the weight. The usage at the limit is $U(w^*)$, one side of the jump, so the expected usage at w_t need not converge to B . The running average of observed usage under the schedule in (9) does converge to B , $\frac{1}{n} \sum_{t < n} U_t \rightarrow B$ with probability one, gap budgets included, as Appendix U.6 proves. Attaining B in expectation with one stationary policy at a gap budget is the job of the endpoint mixture of Definition 4.1, not of this recursion.*

The update is a projected Robbins–Monro recursion (Robbins and Monro 1951), with classical almost-sure extensions due to Blum (1954) and constrained treatments in Kushner and Clark (1978). Appendix U.6 proves this bounded scalar case directly by a supermartingale argument, including discontinuous curves. Kronecker's lemma gives the running-average result for the harmonic schedule above. The theorem is conditional on one sign-consistent crossing and does not cover an arbitrary nonmonotone usage curve.

The reset-on-shift variant detects a persistent recent usage mismatch once the step size has fallen below half its initial value, then restarts the decay clock at zero. These resets break the single diminishing-step schedule, so the proposition does not justify their convergence. Section 6.2 evaluates both decay-only and reset-on-shift recovery empirically. Constant-step tracking would require separate drift and noise arguments and is not claimed here.

4.6 Positioning against Prior Art

Resource-price adaptation is classical in constrained MDPs (Altman 1999), rational inattention (Matějka and McKay 2015; Sims 2003), and SAC (Haarnoja, Zhou, Hartikainen, et al. 2018). Here one information weight controls sensing usage across episodes. It differs from an information constraint internal to EFE (Kouw 2026) and per-node constrained-POMDP dual ascent (Stocco et al. 2024). The contribution is the calibration procedure for this policy family, with count/cost equivariance, grid crossings, endpoint mixtures and conditional stationary convergence.

A constrained-POMDP solver may randomize belief-dependently and outperform a mixture that fixes one weight for each episode (Kim et al. 2011). Our references use sampled SARSOP policies and estimated feasibility, with approximation, sampling and selection uncertainty. The shallow RockSample reference does not reach EFE’s operating region. Appendix U.7 gives the detailed comparisons, including sequential experimental design.

Offline calibration requires a weight sweep. Its benefit is a requirement stated in sensing units, rather than reduced sampling. Under its hypotheses, the online controller can find a crossing without a sweep. Scale equivariance also transfers directly tuned weights on a correspondingly rescaled grid, so that benefit is not unique to usage calibration.

5 Experiments

We evaluate two claims separately. The budget experiments ask whether an information-weighted policy family can meet an expected sensing-usage target, transfer that calibration across reward scales, and re-adapt after a scale change. The EFE experiments ask where the canonical $w=1$ policy lies relative to reward-only planning, tuned information gain, and established POMDP solvers. Meeting a usage target and maximizing reward under a usage cap are different objectives, so the first battery measures both usage error and reward relative to a constrained reference.

The six main environments form a progression in sensing decisions. Tiger has one sensor and asks whether to listen again. Diagnosis and Bandit offer several tests and ask which observation to buy. Tileworld makes those tests spatial partitions of a grid. RockSample and Structural Inspection interleave sensing with movement and exploitation. The first four have a fixed hidden state and an observe-then-commit structure. The last two satisfy the factored-observation assumptions of Proposition 3.3. Complete specifications and protocols appear in Appendices E and V.

Tiger (Kaelbling et al. 1998) has two hidden states, one listen action of accuracy 0.85 and cost 1, and terminal door choices paying +10 or −100. Diagnosis has four possible conditions, two binary tests of accuracy 0.80 and cost 1, and a terminal diagnosis paying +10 or −50. Each test partitions the conditions differently. Bandit has four arms and a single hidden variable identifying the best arm. An inspection targets one arm, has accuracy 0.80, and costs 0.5. A terminal pull pays +10 for the best arm and +1 otherwise. Tileworld hides one target on a 6×6 grid. Its six noisy binary scans query bits of the target’s row or column index, each with accuracy 0.80 and cost 1. Declaring a cell pays +10 or −50.

Our RockSample variant (Smith and Simmons 2004) has known rock positions and hidden binary qualities. Checking is noisy, movement changes sensor range, and sampling and exit yield reward. It differs materially from the original benchmark. Every action costs 0.5, maps are our own, returns are undiscounted with an episode cap, exit is available from any cell, hidden rock qualities remain fixed after sampling, and sensor accuracy starts at

0.95 and reaches chance at finite range. Sample history is observable and prevents a collected rock from yielding a second reward. These choices support the fixed-hidden-state analysis and make sensing use directly interpretable, but absolute returns are not comparable to published RockSample values. Appendix V gives the six differences in full. We evaluate RS[5,3], RS[7,4], RS[7,8], and RS[11,11], where the pair denotes grid width and rock count. Hidden-state counts exclude observable position.

Structural Inspection places N components with fixed hidden nominal/faulty states on a grid. The agent moves, chooses a visual test (accuracy 0.70, cost 0.5) or detailed test (accuracy 0.90, cost 2), and makes an irreversible declaration for each component. Missing a fault costs 50, while correct declarations pay 2 or 5. We evaluate $N=8$ and $N=16$, the latter with 65,536 hidden configurations. This is a synthetic benchmark with a known model and a shared hand-coded search-leaf rule, not a deployed inspection study. Two additional boundary cases, the mild-penalty two-state Testbed and Navigation without separate observation actions, are reported in Appendices C and F. Section 6.9 separately evaluates retrospective acquisition of real sensor records with a learned model and a chronological transfer set.

Unless a battery states otherwise, the canonical seeds are {42, 123, 456, 789, 1024}. The main EFE comparison uses 1,000 episodes per seed on Tiger, Diagnosis, and Bandit, and 500 on Tileworld and Inspection. RockSample protocols appear with their tables. Means aggregate episodes across seeds, and main-table uncertainty is the SE of per-seed means. Pairwise comparisons use seed-level Welch t -tests with Holm–Bonferroni correction for the stated comparison families. Pooled episode-level statistics and bootstrap intervals are supplementary (Appendix I). A failure to reject a difference is not called equivalence. Equivalence is tested separately with fixed-margin TOST below.

The budget battery generally uses 100 episodes per evaluation cell, with each cell defined by one weight and one seed. Tileworld uses 40 episodes per usage-curve cell and 30 per scale-collapse cell, and Inspection- $N=8$ uses 30 per usage-curve cell. Different batteries have different episode counts and seeding rules, so their estimates are not interchangeable. The frontier study below explicitly addresses a consequential cross-battery difference. Appendix Z records the full reproducibility checklist, including all exceptions.

6 Experiments on Shadow Prices and Sensing Budgets

We first test scale transfer and online re-adaptation, then examine cost-denominated budgets, interleaved settings, and the measured staircases. Comparisons with unconstrained and usage-constrained references lead to the held-out calibration study. A distractor experiment then separates the amount of sensing from its relevance to reward. Additional onset and operating-point checks are summarized with appendix pointers at the end of the section.

6.1 Curve Collapse across Reward Scales

Proposition 4.2 predicts that rescaling rewards and sensing costs by α preserves a count-usage curve when the information weight is also multiplied by α . We evaluate reward scales $\alpha \in \{0.1, 1, 10\}$ on corresponding scaled weight grids. Figure 4 shows the result in raw and normalized coordinates. Every matched w/α point agrees bit-exactly on Diagnosis, Bandit, Tiger, and Tileworld. Shared per-episode random streams make exact empirical agreement possible when actions coincide. These CSVs are byte-identical under an absolute comparison tolerance in place of the relative one of Section 4.2, which does not bind at these points.

The normalized grid bracket is (0.141, 0.323] for Diagnosis at $B=8$, (3.84, 8.76] for Bandit at $B=8$, and (5.80, 20.0] for Tileworld at $B=15$, unchanged across scales. Tiger is a necessary negative control. Its usage is constant at 4.37 across the tested normalized range [0, 20], so collapse is trivial and $B=4$ is unbracketed. Equivariance preserves a curve but does not ensure that it offers a useful sensing dial. Table 27 and Appendix W.2 give the complete breadth check.

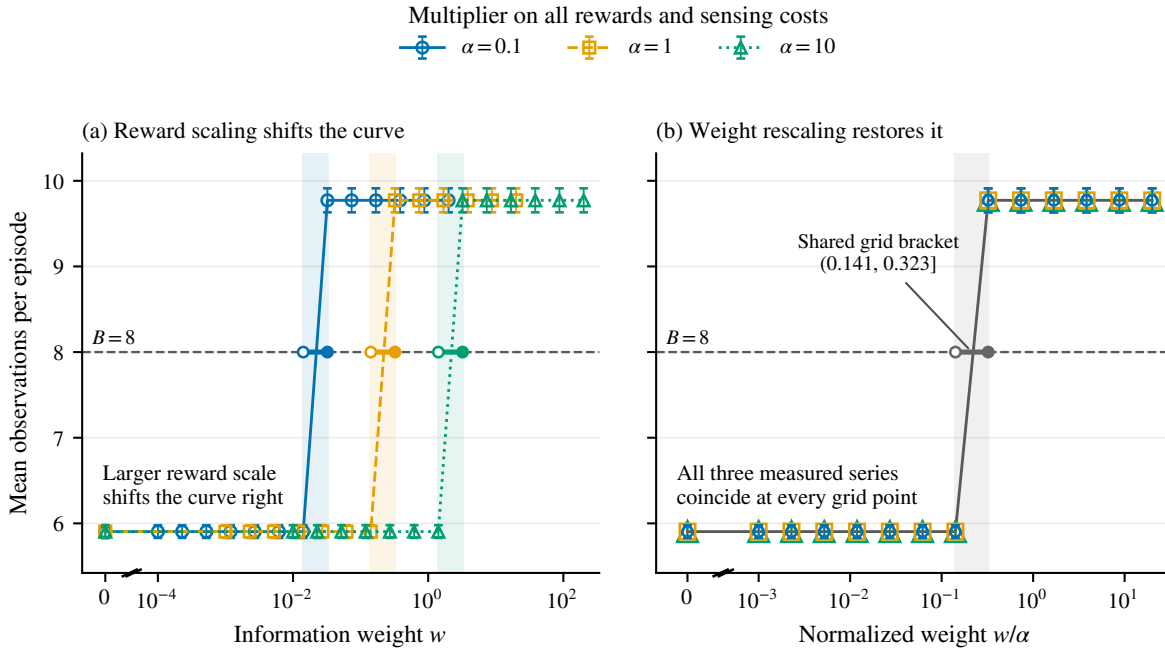


Fig. 4. Reward-scale transfer on Diagnosis. In (a), separately evaluated usage curves shift when rewards and sensing costs are multiplied by $\alpha \in \{0.1, 1, 10\}$. In (b), plotting against w/α collapses them exactly, including the $B=8$ grid bracket $(0.141, 0.323]$. Points and error bars are means ± 1 seed-level SE over five seeds and 100 episodes per seed, with shared per-episode streams. Lines guide the eye. Bands and open-closed endpoints mark grid brackets, not uncertainty intervals or policies attaining the target. The crossing between sampled points remains unresolved.

6.2 Online Re-adaptation after a Reward Rescale

We compare the projected decaying-step controller with its reset-on-shift extension on Diagnosis at target $B=8$. Both use $\eta_0=0.05$, decay $\delta=0.02$, and a tenfold reward-and-cost rescale at episode 200 of a 400-episode run. Ten shared controller seeds support a paired comparison. The extension watches a 20-episode usage window. Once the step has decayed below half its initial value, a deviation exceeding both three window standard errors and 0.5 observations restarts its decay clock. Appendix W.5 specifies the detector and steady-state summaries.

Recovery occurs when a rolling mean computed only from post-rescale episodes stays within ± 1 of B for 20 consecutive episodes. The comparison uses the restricted time $\min(T, 180)$ for every seed, where T is the zero-based post-rescale index at the start of the first qualifying hold and 180 is the last admissible start index. A seed that never completes it receives 180. Decay-only averages 157.3 episodes and reset-on-shift 51.2, a descriptive ratio of 3.07. The paired difference is 106.1 episodes with a Student- t 95% interval $[96.8, 115.4]$. All ten paired differences favor resetting. Reset-on-shift recovers on 10/10 seeds and decay-only on 9/10, so the latter's unrestricted recovery distribution is right-censored. The conditional mean among its nine recovered seeds is not the comparison estimand.

These measurements come from the per-episode-seeded runner, which also seeds the environment constructed at the rescale. Decay-only is slower than reset-on-shift on every seed, and on one of the ten it does not recover within the window. Figure 5 plots the trajectories. The mid-run rescale is nonstationary, and the reset variant

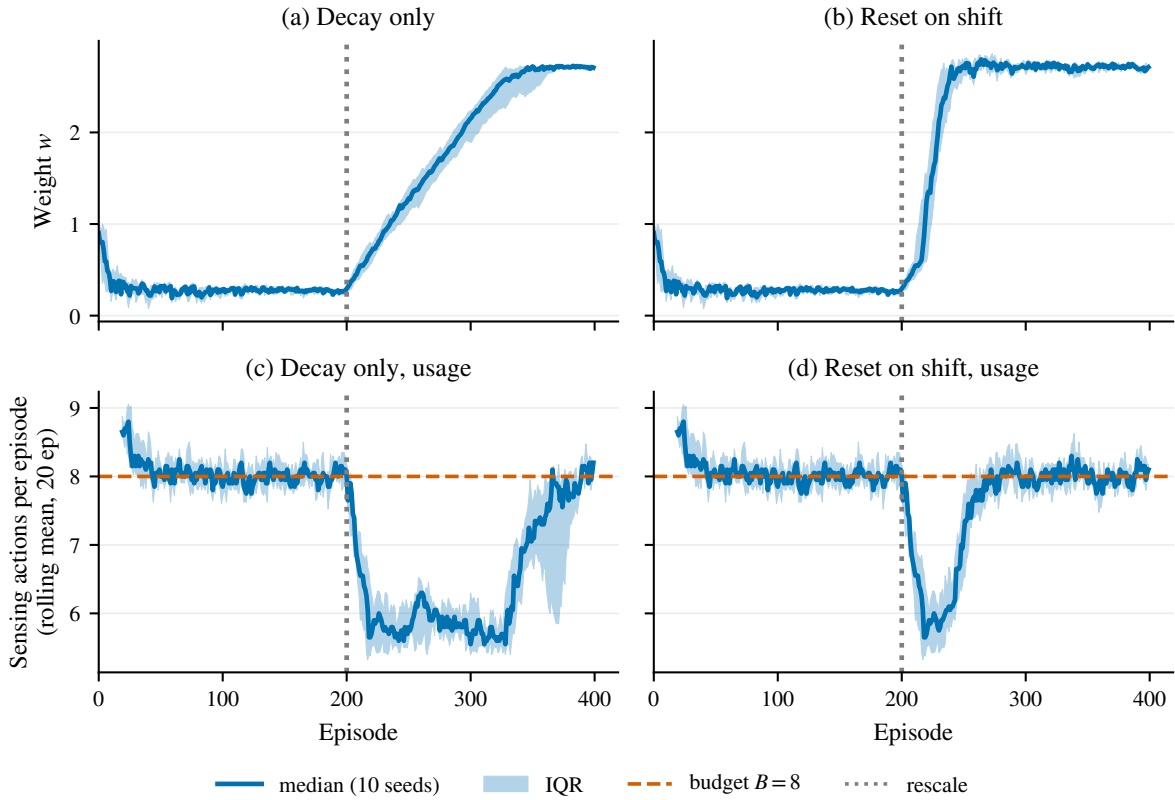


Fig. 5. Online control on Diagnosis through a tenfold reward-and-cost rescale at episode 200. Columns compare decay-only and reset-on-shift. Top panels show weight, as median and interquartile range over ten seeds. Bottom panels show rolling usage with window 20 and target $B=8$. Recovery uses post-shift-only windows and a 20-episode hold within ± 1 of the target. Restricted recovery time $\min(T, 180)$ averages 157.3 versus 51.2 episodes, with paired difference 106.1 and Student- t 95% interval $[96.8, 115.4]$. This nonstationary experiment is outside Proposition 4.5.

restarts its step schedule, so neither experiment is covered by Proposition 4.5's stationary convergence guarantee. This is empirical re-adaptation evidence on one shift scenario.

6.3 Cost-Denominated Budgets

Heterogeneous test costs make count and cost targets different design inputs. In a Diagnosis variant with equally accurate tests costing 0.5 and 2.5, the ratio $U_{\text{cost}}/U_{\text{count}}$ rises from 1.26 at weights through 10 to 1.37 and 1.41 at the two largest weights. The 12.0% spread relative to the grid mean reflects a changing test mix. Each test's information gain depends on the current belief over its partition. The cost vector alone does not predict which is more informative.

The tested budget pairs nevertheless select the same grid brackets: count/cost targets 8.86/11.50 give $(10, 31.6]$, and 13.71/19.20 give $(31.6, 100]$. Each pair is placed at the same fractional height within its curve's sampled usage range, from the smallest to the largest mean on the grid, with end margins to ensure a crossing. This is not a

fixed conversion between count and cost. The observed change in cost per test is insufficient to separate these paired crossings at this grid resolution. Figure 15 and Appendix W.1 retain the full curves and pairing rule.

6.4 RockSample and Structural Inspection as Interleaved Settings

Usage calibration also applies to the interleaved agents of Sections 7.4 and 7.5, counting check/test actions as sensing. Their sampled usage ranges are [3.00, 8.83] on RockSample[5,3], [2.00, 14.54] on RockSample[7,4], and [22.38, 58.56] on Inspection- $N = 16$. Four budgets per instance are placed inside these ranges. Figure 17 and Appendix W.4 give the brackets, uncertainty and complete grid/episode protocols.

RockSample[7,4] and Inspection- $N = 16$ have nondecreasing usage at consecutive sampled weights. RockSample[5,3] instead falls from $U(0) = 3.70 \pm 0.11$ to 3.00 ± 0.00 at $w = 0.316$ before rising. Thus even an interleaved family's attainable usage can extend below its reward-only usage. The plotted lowest RockSample[5,3] target is attainable but marked slack because its nearest grid point is $w = 0$, below its crossing bracket. These are sampled-grid observations, not bounds on untested weights.

6.5 Shadow-Price Staircases across Five Domains

The observe-then-commit domains and the smaller Inspection instance now receive the same treatment as the interleaved instances. Figure 6 reports grid estimates of $w^*(B)$ over identifiable budgets inside $[U_{\min}, U_{\max}]$ for Tiger, Diagnosis, Bandit, Tileworld-6 $\times$ 6, and Inspection- $N=8$, where $U_{\min}$ and $U_{\max}$ are the endpoints of each curve's usage range (Section 6.3). An identifiable budget is one placed strictly inside that range by the rule of that section, evenly spaced with the end margins specified in Appendix W.4, so that the grid brackets it. The monotonicity at issue is that of the underlying usage curve $U(w)$ from which each staircase is read off, taken between consecutive sampled grid weights. On Tiger, Diagnosis, and Inspection that curve is nondecreasing between every pair of consecutive grid weights. On Bandit and Tileworld it is not, showing local non-monotonicity from discrete policy switches, the first of the two obstacles noted after Proposition 4.3. We report crossing brackets and usage standard errors for every domain (Definition 4.1). The standard error attached to a bracket is that of usage at a fixed weight, and the bracket's width is set by the grid, so neither says how often a different sample of seeds would select a different bracket. We measured bracket-selection stability by resampling the five seeds with replacement 2000 times and re-solving each of the 25 budgets of Figure 6 on each resampled curve (run_bracket_stability.py, results_price_bracket_stability.csv, with the count of every bracket outcome in results_price_bracket_stability_counts.csv). The reported bracket is recovered in every resample at 19 budgets and in at least 99 percent of resamples at 21. The lowest agreement is 0.789, at Inspection's lowest budget. In all but one disagreeing resample, the alternative is a neighbouring step of the staircase or, at Tiger's lowest and highest budgets, no bracket at all. Those two Tiger budgets lie about one standard error of usage from the corresponding ends of its usage range. On Bandit and Tileworld a budget can be crossed more than once, and the reported bracket estimates the last crossing by the convention that definition adopts, so it names one threshold at grid resolution rather than displaying the reversals. Bandit's budget $B=5.87$ is the one tested case. Usage first reaches it at $w=0.139$ (6.19 observations), falls back to 5.11 at $w=0.72$, and reaches it for the last time inside the reported bracket (1.64, 3.73] (results_price_shadow_curves.csv), which does not show the earlier crossing. Tiger's price is weakly identified, consistent with Table 27, in the sense that the budget pins the price down only loosely. Instrumental listening already saturates most of the usage range at $w=0$, so the curve is flat at its floor across the low-weight range and rises in a single step at high weight. Every tested budget lands in that one step's bracket.

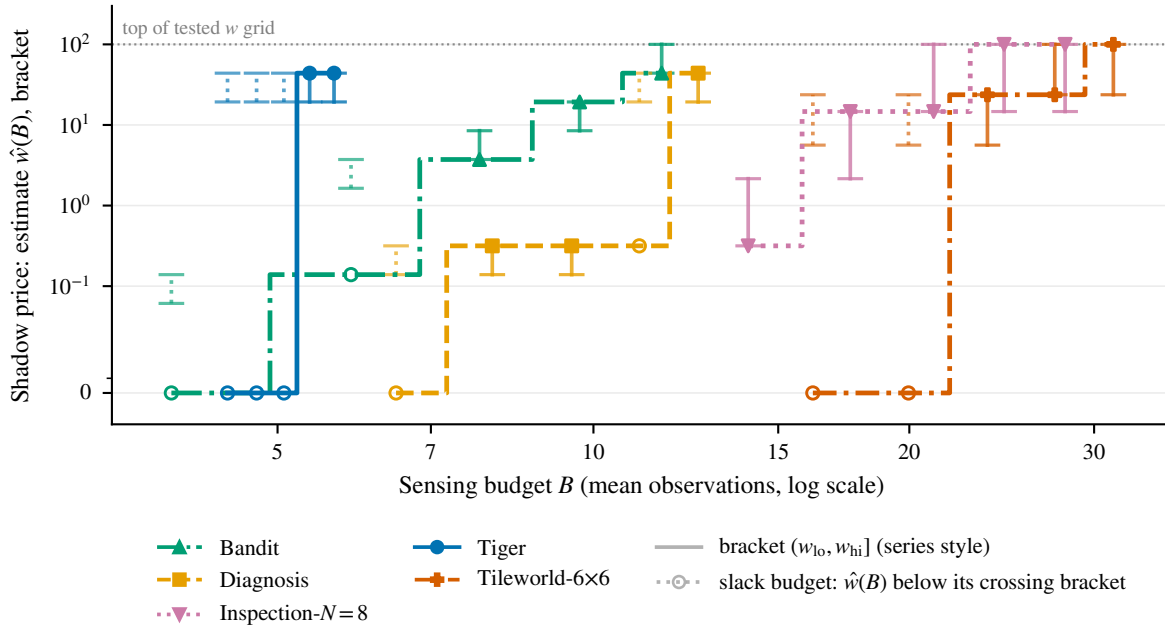


Fig. 6. Operational shadow-price staircases $w^*(B)$ across five observe-then-commit and inspection domains (Tiger, Bandit, Diagnosis, Inspection- $N=8$, Tileworld- 6×6), each in its own color, with a logarithmic budget axis and a symmetric-log weight axis that is linear near zero. Each staircase steps up as the sensing budget B rises, and the vertical bar at each step marks that budget's crossing bracket $(w_{lo}, w_{hi}]$ (Definition 4.1), capped where the bracket's upper edge coincides with the top of the tested w grid (dotted horizontal rule at 100). An open circle with a dotted bar marks a slack budget, one for which the grid weight whose usage comes nearest to the budget lies strictly below the lower edge of the crossing bracket ($w=0$ at seven of the nine marked budgets, $w=0.316$ at Diagnosis $B=11.06$, and $w=0.139$ at Bandit $B=5.87$, the twice-crossed budget discussed in the text), so the nearest-point solution returns a weight below the bracket while the dotted bar still marks the crossing. Every marker plots that nearest-point solution $\hat{w}(B)$, which can also sit on a bracket's lower edge. The usage curves behind Tiger, Diagnosis, and Inspection never step downward between consecutive sampled weights, while Bandit and Tileworld show local non-monotonicity from discrete policy switches, so their brackets mark the last crossing of each budget by the convention of Definition 4.1.

6.6 Unconstrained Reward Reference

To determine whether $w=1$ is reward-competitive at its natural operating point, we compare EFE with SARSOP (Kurniawati et al. 2008) on Tiger, Diagnosis, and Bandit. The original APPL solver is built from source, each environment is exported through a common model path, and the solve uses precision 10^{-3} at $\gamma=0.999$. The returned alpha-vector policy is evaluated in the same simulator as EFE at five seeds and 500 episodes per seed (Table 28). SARSOP is a near-optimal reference, not a certified exact solution to the reported Monte Carlo comparison. Build and export details appear in Appendix T.1.

Fixed-margin two one-sided tests (TOST) (Schuirmann 1987) assess reward equivalence within one observation's cost, ± 1.0 on Tiger and Diagnosis and ± 0.5 on Bandit. The five-seed unpaired Welch TOST rejects both nulls on all three environments after Holm correction. Its margins were chosen after a canonical-seed comparison was

inspected, so it is retrospective. The fifteen additional seeds in the $n = 20$ check were fixed after the margin and before that study ran, and all three equivalence conclusions persist.

A predeclared twenty-seed check also reseeds every episode, matching hidden states and observation draws while actions agree. It retains equivalence, with paired reward differences $+0.11 \pm 0.09$ on Diagnosis and $+0.04 \pm 0.03$ on Bandit (seed-level SE). Tiger’s policies act identically. Table 28 and Appendix W.6 give the numerical tests, intervals and archives. These results establish equivalence within the stated margins on these three environments, not universal near-optimality of $w = 1$.

6.7 A Usage-Constrained Reference

A literal usage penalty supplies a different reference from an information bonus. We replace observation reward $-\text{cost}_j$ by $-(\text{cost}_j + \lambda)$ and solve with SARSOP over a 12-point penalty grid in $[0, 100]$. This estimates policies maximizing $\mathbb{E}[R] - \lambda \mathbb{E}[U]$ over the POMDP policy space, with $\gamma=0.999$ and five seeds of 300 episodes per penalty. The penalty is the Lagrangian price of the usage cap in Equation 6. The information weight w is not that multiplier.

At budget B , the reference is the feasible envelope of the sampled policy points, obtained by maximizing their mixture reward subject to mixture usage at most B . A solution uses at most two sampled policies. Policies below the target remain eligible, and the reference is flat at the largest sampled reward once the cap is slack. This is a linear program over measured points, not equal-usage interpolation. The solver is approximate, the penalty grid is finite, and reward and usage are Monte Carlo estimates. Consequently the envelope is neither a certified population bound nor the full constrained frontier. We do not assume that finite-state strong-duality results (Altman 1999) automatically extend to the continuous belief MDP. Appendix W.7 discusses these limitations.

At EFE’s endogenous usage, the cap does not bind on any of the three environments, and estimated reward gaps are small relative to their standard errors (Table 29, Appendix W.7). The evaluated usage-matched Planning+IG weights also give the same rewards as EFE. This is a measured plateau, not an extension of Proposition 3.1 to arbitrary weights. The next study instead uses calibration-derived targets. No comparable constrained reference is available for the larger RockSample or Inspection instances.

6.8 Calibration-Derived Targets on Held-Out Seeds

The preceding comparison uses whatever sensing the weight-one policy happens to buy. We now test targets independent of that weight-one usage, chosen by a predeclared rule from each calibration curve. These targets stand in for application requirements but are not themselves application-specified. The study separates whether the procedure meets expected usage from whether doing so maximizes reward under a cap. Its original design was recorded in the producer’s docstring before results were inspected (`experiments/run_budget_frontier.py`).

Each core environment is calibrated on the canonical five seeds, 100 episodes per seed, using the 16-point weight grid and horizons of Section 6.5. We select budgets at 0.25, 0.5, and 0.75 of the measured range $[U_{\min}, U_{\max}]$, at the midpoint of its largest jump, and at half of $U_{\min}$. The last is deliberately unattainable and must be declared so. Definition 4.1 supplies the crossing bracket and endpoint-mixture probability q . All selection uses calibration data alone, and evaluation uses held-out seeds $\{7, 8, 9, 10, 11\}$ with 100 episodes each. Tiger’s gap budget duplicates its middle-range budget. Bandit’s curve is nonmonotone, so the stipulated last-crossing convention determines its bracket.

The original within-family comparator is the highest-reward single grid member whose calibration usage is at most B . Two additional LP comparators were introduced after inspecting the first results, and that amendment is disclosed in the producer. One maximizes calibration reward over mixtures with usage equal to B , and one does so with usage at most B . Each uses at most two calibration-grid weights and is evaluated on held-out seeds. Their scope is the sampled grid. Selecting calibration usage below a cap does not guarantee that the population or held-out usage meets it.

Table 2. Endpoint-mixture calibration on held-out seeds, 100 episodes per seed over five seeds. B is the calibration-derived expected-use target. U and R are held-out mixture usage and reward. All $\pm$ values are seed-level SE. The cap gap is mixture reward minus the same-stream cap-reference envelope, and the target gap is mixture reward minus the best sampled reference mixture whose usage equals B , each with seed-level SE. Reference supports and weights are fitted on those same seed means and held fixed for the SE, so intervals derived from these columns are nominal pointwise plug-in summaries, as qualified in the text. Tiger’s gap row duplicates its middle target. All three below-range targets are declared unattainable. Complete within-family and reference comparisons appear in Table 30.

Environment	Target	B	U	R	Cap gap	Target gap
Tiger	0.25	4.72	4.64 ± 0.10	5.36 ± 0.10	-0.33 ± 0.04	0.08 ± 0.04
Tiger	0.50	5.07	5.04 ± 0.14	4.96 ± 0.14	-0.72 ± 0.06	0.03 ± 0.06
Tiger	0.75	5.42	5.35 ± 0.14	4.65 ± 0.14	-1.03 ± 0.06	0.07 ± 0.06
Tiger	gap	5.07	5.04 ± 0.14	4.96 ± 0.14	-0.72 ± 0.06	0.03 ± 0.06
Diagnosis	0.25	7.72	7.70 ± 0.25	-2.14 ± 0.48	-0.03 ± 0.28	-0.03 ± 0.28
Diagnosis	0.50	9.53	9.47 ± 0.20	-2.11 ± 0.67	-0.76 ± 0.15	-0.76 ± 0.15
Diagnosis	0.75	11.35	11.47 ± 0.37	-2.31 ± 0.45	-1.10 ± 0.22	-0.05 ± 0.16
Diagnosis	gap	7.84	7.86 ± 0.25	-2.06 ± 0.47	0.00 ± 0.22	0.00 ± 0.22
Bandit	0.25	5.51	5.62 ± 0.21	6.06 ± 0.26	-0.21 ± 0.12	-0.16 ± 0.12
Bandit	0.50	7.79	7.67 ± 0.37	5.70 ± 0.25	-0.57 ± 0.09	-0.08 ± 0.06
Bandit	0.75	10.07	9.95 ± 0.36	4.95 ± 0.22	-1.32 ± 0.08	0.05 ± 0.04
Bandit	gap	4.71	4.62 ± 0.07	5.62 ± 0.21	-0.59 ± 0.20	-0.59 ± 0.20

Table 2 gives the principal held-out results. All three below-range targets are declared unattainable. At all twelve attainable targets, the selected endpoint mixture meets usage to within 0.13 observations. Errors are within one held-out SE except Bandit’s gap target, where an error of 0.09 is about 1.3 SE. The evaluated bracket endpoints and best feasible single member miss the targets by more. Endpoints miss by 0.3–1.1 observations on Tiger and 0.2–3.6 elsewhere. The selected feasible single member leaves 0.2–4.7 observations unspent on held-out data. Its usage is below the target in all twelve rows, an observed result in addition to the calibration selection fact. Conversely, the endpoint mixture’s held-out mean exceeds B on three rows, by at most 0.13. An expected-use target provides neither a hard per-episode cap nor finite-sample cap feasibility.

Meeting a target can buy useful information or impose an avoidable reward cost. On Diagnosis at $B=7.72, 9.53$, and its gap target 7.84, the mixture earns $-2.14, -2.11$, and -2.06 against the feasible member’s -2.86 . On Tiger, reward-only planning uses 4.32 observations and earns 5.68 on this held-out stream. Spending more to meet the three increasing targets reduces reward to 5.36, 4.96, and 4.65. Bandit likewise loses reward at its two larger interior targets. The endpoint procedure also need not select the best mixture meeting a target. At Bandit’s gap budget, the calibration LP mixes $w=0$ and $w=0.72$ and earns 5.94 ± 0.19 on held-out seeds, against the endpoint mixture’s 5.62 ± 0.21 . Its usage error is larger, 0.18 rather than 0.09. The chosen endpoint mixture therefore cannot stand for every mixture in the family, and its deficit at this budget cannot be attributed to planning depth alone.

A valid constrained-reward comparison must also control the evaluation stream. The original reference points were evaluated on canonical seeds with once-per-seed seeding, whereas the family uses held-out per-episode seeds. Their direct reward comparisons, retained in Appendix W.8, are descriptive cross-stream comparisons. The difference matters on Tiger. Its unpenalized reference earns 5.02 on the canonical stream but 5.68 on the held-out stream, removing the mixture’s apparent excess over the reference at the smallest budget.

We therefore added a same-stream comparison after inspecting the original results. Every saved reference policy was first checked against its committed canonical estimate to 10^{-9} , then re-evaluated on the held-out

seeds with the family's per-episode seeding, and the feasible envelope was recomputed. The Cap gap column of Table 2 pairs each seed's mixture reward with its reference reward on the same episode seeds. Reference support and mixture weights are fitted on those same held-out seed means and held fixed for the SE. The resulting Student- t intervals are nominal pointwise plug-in summaries. They omit support reselection, uncertainty in mixture weights and cap feasibility, and multiplicity across budgets. They do not establish coverage for the population constrained frontier.

Under that fitted comparison, negative 95% intervals exclude zero at eight of eleven distinct budgets, counting Tiger's duplicate once. The cap is slack at seven budgets: every Tiger target, Bandit's three interior targets, and Diagnosis's largest. At these budgets, the reference is the unpenalized SARSOP policy, the estimated unconstrained reward optimum. A gap there combines the cost of spending B with any inefficiency of the family. A predeclared target-matched reference separates the two. Subsidized SARSOP policies, with negative usage penalties, extend the sampled points above the unpenalized usage, and the reference becomes the best sampled mixture whose usage equals B (run_frontier_target_reference.py). Against it, intervals exclude zero at two of eleven distinct budgets on the held-out seeds. Replaying the held-out-fitted mixtures on ten fresh seeds again gives two shortfalls, all at budgets whose held-out cap reference binds. Fresh realized usage need not equal B . Only Bandit's gap-budget shortfall, -0.59 held out and -0.50 fresh, appears on both streams, and its held-out interval just reaches zero when the reference is matched to the mixture's realized usage rather than to B (Appendix W.8). Diagnosis's held-out shortfall at $B=9.53$ mostly pairs a policy on the usage plateau containing $w=1$ with the unpenalized SARSOP policy at nearly equal usage. The twenty-seed episode-matched check of Section 6.6 finds that pair equivalent within the margin, although on these five held-out seeds it differs by 1.01 ± 0.18 . Both references are maxima over noisy sampled points, so selection bias remains.

Tiger's paired gaps measure only extra listening, since neither the family nor any policy in either reference support makes a wrong commit in the 500 held-out episodes. A wrong commit costs 110 relative to a correct one, and the canonical reference error rate of 0.6% is worth about 0.66 expected reward, a rare-event uncertainty the nominal intervals do not capture. If the reference erred at that rate while the family never did, the smallest-budget cap shortfall would reverse and the middle one would nearly vanish.

This study supports expected-usage calibration within the tested family, including gap targets, while quantifying its limits as a reward optimizer. If the engineering requirement is to spend a specified expected sensing amount for coverage or verification that reward does not represent, the endpoint mixture directly addresses it. If the requirement is only an upper limit and reward is the objective, the feasible mixture or constrained reference is the relevant comparator. The complete three-panel table, selection details, reference construction, and paired archives are retained in Appendix W.8.

6.9 Sensor Records with Learned Likelihoods

We next test calibration on recorded observations with a learned model. The UCI Gas Sensor Array Drift dataset contains 13,910 exposures to six gases, measured by sixteen sensors (Vergara 2012; Vergara et al. 2012). Acquiring a sensor reveals its eight stored descriptors once. This is retrospective access to recordings, not selective operation of the original instruments. Before evaluating policies or inspecting usage curves, we fixed targets $B \in \{2, 4, 8\}$ accesses per case. They are researcher-defined requirements, not operator-supplied needs or measured energy budgets.

Batches 1–6 provide 3,550 training, 1,175 calibration and 1,208 test cases, stratified by batch and class. Training alone determines eight categories per sensor, class priors and smoothed categorical likelihoods. Policies assume conditional independence of sensors given gas identity, use one-observation lookahead, and cannot reacquire a sensor. Each action reveals the case's actual category. Correct classification earns 1, with a benchmark cost of 0.02 per access. All methods share this fitted model. We compare the crossing mixture with calibration-selected

equality mixtures from a direct usage-penalty family, a best equality mixture within the weight grid, and greedy conditional-information acquisition that takes exactly B sensors. Greedy information acquisition is an established approach (Covert et al. 2023), and this comparison does not reproduce a learned neural acquisition method. Appendix V.1 gives the frozen grids, splits and uncertainty procedure.

The sampled weight family cannot reach $B = 2$, whose target is below its smallest calibration usage, 3.424. The two straddled targets transfer closely to unseen cases, with mean usage 4.006 and 8.019. Their Bonferroni-adjusted 98.333% conditional bootstrap intervals for usage error are $[-0.210, 0.255]$ and $[-0.362, 0.411]$, inside the predeclared ± 0.5 tolerance. Each of the 2,000 replicates reselects the calibration crossing. None fails at $B = 4$ or 8, whereas all fail at $B = 2$. Thus two targets meet the usage criterion, and the all-three-target requirement fails.

Accurate usage does not make the selected policy competitive on classification. At $B = 4, 8$, the crossing mixtures achieve 80.23% and 76.28% accuracy, against 82.11% and 80.18% for direct equality mixtures and 81.71% and 79.47% for exact-count greedy information acquisition. The direct mixtures use 3.965 and 7.977 sensors on test cases, so their realized usage is close but not identical to the crossing mixtures'. Each nominal paired accuracy-gap interval lies below zero (Figure 7). The best equality mixtures within the weight grid also do better, at 82.12% and 78.41%. The direct and exact-count policies can serve $B = 2$, with 80.08% and 81.04% accuracy. These results favor the simpler controls in this configuration, not calibration as a general acquisition algorithm.

Frozen calibration does not transfer to the 7,977 later cases in batches 7–10. Across those batches, the $B = 4$ mixture uses 5.15–6.51 sensors and the $B = 8$ mixture uses 9.49–12.73. Accuracy also falls, reaching 14.56% and 11.56% in batch 8. The model is deliberately approximate. Even within the test period, the full-record classifier has log loss 2.237, worse than the prior's 1.674, despite higher accuracy. The study therefore demonstrates within-period usage calibration and its limits under learned likelihoods and collection shift. It provides neither deployment validation nor a general robustness advantage.

6.10 Reward-Irrelevant Sensing

An expected sensing target controls the amount of sensing, but does not ensure that it concerns the task. We test this limit with Distractor Diagnosis, whose hidden state combines the usual four-way condition with an independent binary nuisance bit. Commit rewards depend only on the condition. An added test observes only the nuisance bit, while the original tests observe only the condition. Under the independent generative process the belief stays factored, so the distractor carries exactly zero information about the condition. Appendix T.2 states and tests that property, whose scope excludes correlated priors introduced through misspecification.

We sweep ordinary Planning+IG over sixteen weights with five seeds and 100 episodes per weight and seed, refining the grid where distractor use begins. The distractor fraction is zero through $w=3.5$, 0.103 ± 0.001 at $w=4$, 0.233 ± 0.003 at $w=5$, and approximately one third at the largest weights (Figure 8). Thus the observed onset bracket is $(3.5, 4.0]$. The canonical $w=1$ agent avoids the distractor on this instance because it is below onset, not because its objective is structurally immune. At sufficiently high weights, genuine information about the nuisance bit earns more epistemic value than the test costs.

A reward-relevance variant scores information gain on the marginal belief over reward-equivalence classes. Two states belong to the same class when every commit action pays the same in them, so the partition is determined by the commit reward matrix already supplied to the agent. On this benchmark, the two states differing only in the nuisance bit share a class and the four classes are the four conditions. The planner still propagates the full joint posterior and changes only the belief on which information gain is scored. If no states are reward-equivalent, the variant reduces to ordinary information gain. This construction uses the known reward model and the benchmark's factorization, rather than learning task relevance from data.

The variant buys zero distractor tests at every swept weight. At $w=100$, ordinary information gain buys 6.46 distractor tests, 33.2% of its usage, and earns -10.48 ± 0.11 reward against the variant's -3.64 ± 0.36 . Seed-level

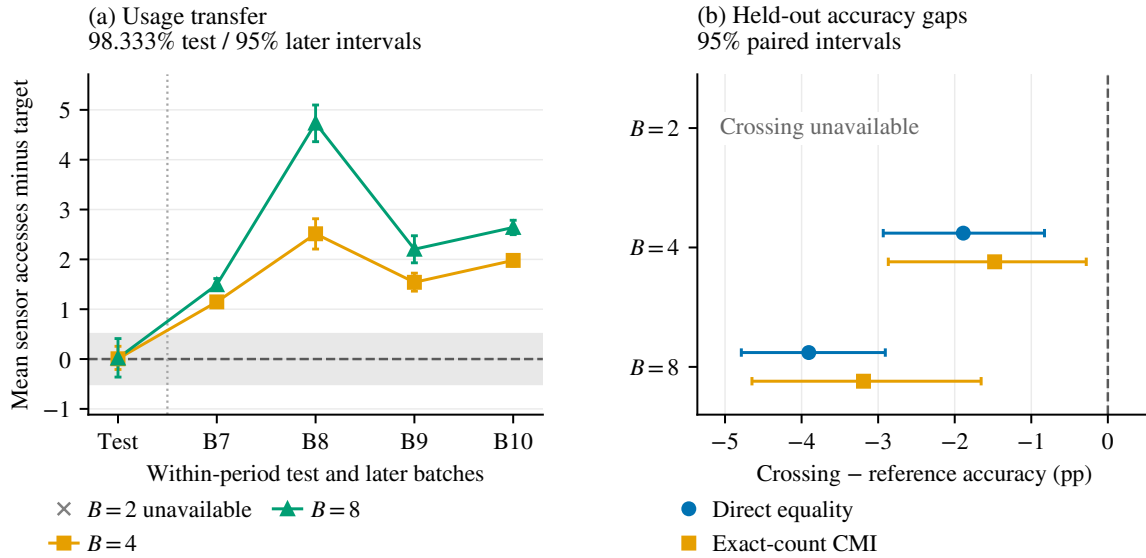


Fig. 7. Usage transfer and accuracy costs on recorded sensor cases. (a) Frozen crossing-mixture usage minus its target on within-period test cases and later batches. The gray band is the predeclared ± 0.5 access tolerance. Test intervals are 98.333% conditional bootstrap intervals with calibration reselection. Later intervals are nominal 95% intervals with selections fixed. Positions denote collection groups, not equal elapsed times. (b) Paired test accuracy gaps against direct equality mixtures and exact-count greedy conditional-information acquisition (CMI), with nominal 95% intervals including calibration reselection. Positive gaps favor the crossing mixture. Direct mixtures need not exactly meet the target on test cases. All intervals condition on one trained model and within-stratum case exchangeability. The unbracketed $B = 2$ request is retained as unavailable.

Welch tests with Holm correction over the weight-by-metric family detect usage and distractor-count differences at every tested weight past onset. Reward differences survive that correction at $w=31.6$ and 100, with $p = 1.2 \times 10^{-5}$ and 1.5×10^{-5} , but not at the smaller post-onset weights. Success rates are identical through 31.6, and the variant's nominal advantage at 100 is not significant.

Relevance weighting does not solve over-observation. At the two largest weights the variant still buys 13.16 task-relevant tests against 9.77 at $w=10$. Nor is the IDS adaptation used here immune to nuisance sensing. Its information about the optimal commit is correctly zero for the distractor, but its zero-denominator fallback uses raw state-entropy reduction and yields a distractor fraction of 0.305 ± 0.005 . This is a failure of that implementation choice, not evidence against reward-aware information measures in general. A sensing target and relevance weighting therefore address different design requirements. Appendix W.9 preserves the full curves, implementation analysis, comparison family, and remaining numerical results.

6.11 Additional Onset and Operating-Point Checks

Positive-threshold two-state instances place the observed onset within one refined grid step, 3%, above the closed-form threshold, while standard negative-threshold instances observe already at $w=0$ (Appendix W.3). The atlas collects the implicit $w=1$ budgets and two grid brackets per instance from the measured curves (Appendix T.3). These are measured operating points, not a universal predictive formula for the price.

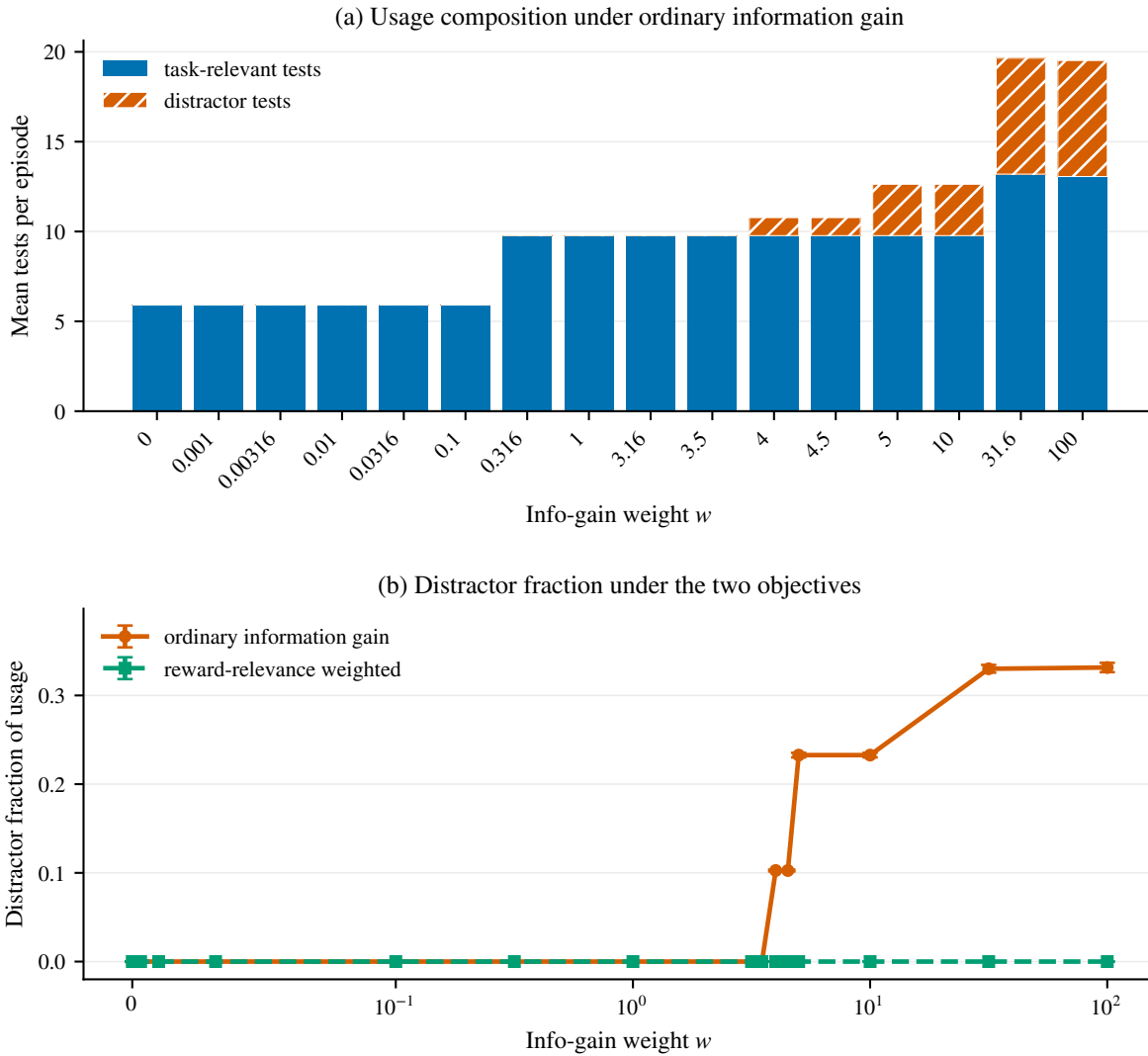


Fig. 8. Reward-irrelevant sensing on DistractorDiagnosisEnv. The upper panel decomposes ordinary Planning+IG's usage into task-relevant and distractor tests as w grows. Distractor spend is zero below the onset bracket $(3.5, 4.0]$ and saturates at roughly a third of total usage. The lower panel plots the distractor fraction for ordinary information gain against the reward-relevance-weighted variant on the same grid. The weighted variant is flat at exactly zero everywhere. Error bars are seed-level SE over 5 seeds.

7 Results

The second battery locates the canonical EFE weight within the policy family and compares it with other planners. It provides empirical context for the equivalence result, whose validity itself rests on the stated assumptions and

proof. We retain the main comparisons here and place detailed per-environment mechanisms, robustness studies, and additional tables in Appendix X.

7.1 Core Environments

Table 3 contrasts $w=1$ EFE with myopic reward maximization, same-horizon reward-only Planning, and Planning+IG tuned for success within its own class. Reward-maximizing weights are considered separately below. On Tiger, EFE and Planning produce identical trajectories. With one observation action, the information term never changes the reward-based action choice at these stakes. This is a measured property of this instance rather than a general consequence of having a single sensor.

Table 3. Core comparison, 1,000 episodes per seed over five seeds. Reward is mean $\pm$ seed-level SE. Here $w^* = w_{\text{succ}}^*$ is tuned for success, with reward tuning reported separately. Bold identifies EFE and its exact Tiger tie with Planning, not a per-column winner. Success-tuned Planning+IG buys greater accuracy at a reward cost. Full agents, pooled uncertainties, and statistical comparisons are in Appendices B and I.

Environment	Agent	Obs.	Success	Reward
Tiger $H=6, w^*=50$	Myopic	1.00	85.3%	-7.15 ± 0.41
	Planning	4.20	99.4%	$+5.19 \pm 0.21$
	Planning+IG	5.66	99.9%	$+4.21 \pm 0.08$
	EFE	4.20	99.4%	$+5.19 \pm 0.21$
Diagnosis $H=3, w^*=50$	Myopic	2.00	64.7%	-13.17 ± 0.58
	Planning	5.86	88.8%	-2.57 ± 0.20
	Planning+IG	13.16	99.2%	-3.66 ± 0.12
	EFE	9.66	97.2%	-1.37 ± 0.18
Bandit $H=2, w^*=20$	Myopic	2.06	62.2%	$+5.56 \pm 0.09$
	Planning	3.29	71.0%	$+5.75 \pm 0.08$
	Planning+IG	9.57	98.9%	$+5.11 \pm 0.04$
	EFE	5.11	86.9%	$+6.27 \pm 0.04$

On Diagnosis, EFE raises success from 88.8% to 97.2% relative to Planning and improves reward from -2.57 to -1.37 . Seed-level Welch tests give $p = 4.0 \times 10^{-8}$ for success and $p = 0.002$ for reward. On Bandit it raises success from 71.0% to 86.9% and reward from 5.75 to 6.27, with $p = 2.1 \times 10^{-5}$ and $p = 0.001$. These differences survive the stated Holm correction. Thus EFE Pareto-dominates same-horizon reward-only Planning on these two environments. Success-tuned Planning+IG reaches higher accuracy, 99.2% and 98.9%, but pays for more sensing and attains lower reward. EFE supplies an untuned trade-off, not the best value of every metric.

Two baselines limit any attribution of these outcomes solely to explicit information gain. Posterior-vote, which draws states from the belief and votes over their optimal actions without an information term, is not significantly different from EFE on any Bandit metric. Its reward is nominally higher, 6.32 against 6.27 ($p = 0.58$). It also beats EFE on Testbed, but performs worse on Diagnosis and Tiger. The Epistemic-only ablation commits immediately at chance-level success on all three core environments and Tileworld. The units force this outcome, since in bits no observation there yields as much information as it costs. Full baselines, pooled effect sizes, and seed-level tests appear in Appendices B, I, and X.1. Reward effects against Planning are small at the episode level on Diagnosis and Bandit despite statistically detectable mean differences. We do not interpret the mechanically inflated effect sizes computed from only five seed means.

POMCP provides a search-based comparison, reported in Appendix S. Its simulation count per decision is a computational budget distinct from the sensing budget B . The MCTS-EFE comparison matches simulation counts,

while a separate Tiger control matches wall-clock computation. The POMCP comparisons outside the exploration sweep use untuned exploration constants. A separate sweep selects constants on tuning seeds. The selection rule was written after inspection of an exploratory sweep on the canonical seeds and was recorded before the runs reported here. Neither the POMCP comparison nor the component ablation isolates directed observation selection as the cause of every performance difference. These qualifications remain necessary when comparing search architectures, even when the information-weighted family performs well.

7.2 Where the Information-Unit Weight Sits

We sweep $w \in \{0.01, 0.1, 0.5, 1, 2, 5, 10, 20, 50, 100, 200\}$ on Tiger, Diagnosis, Bandit, Testbed, and Tileworld. Figure 9 shows reward against success. The $w=1$ Planning+IG and EFE markers coincide under Proposition 3.1 and shared tie-breaking. The empirical question is where that common policy lies relative to other weights, not whether equivalent implementations differ.

The weight-one policy is within the reward-maximizing run of sampled weights on Tiger (0.01 through 20), Diagnosis (0.5 through 20), and Bandit (1 and 2). These are sampled plateaus, not claims about every untested weight between grid points. Testbed instead maximizes reward across sampled weights from 0.01 through 0.5. Its weight-one policy buys 6.4 percentage points of success at a reward cost of 0.11. On Tileworld, $w=20$ Pareto-dominates $w=1$, with reward -18.95 against -21.53 and success 91.8% against 72.0%. Thus the canonical weight attains the sampled reward maximum on three of five environments and is suboptimal on two.

Success reaches its maximum at the upper grid endpoint 200 on all five environments, although it dips from $w=0.5$ to $w=1$ on Bandit and Tileworld. The reward cost need not grow smoothly. On Tiger and Diagnosis, the tested weights from 1 up to the point below 50 leave reward unchanged, and the step to 50 incurs the measured cost. The grid does not exclude a cheaper untested intermediate policy. The advantage of $w=1$ here is that it selects a useful default without environment-specific search under the stated reward convention. It does not remove dependence on units, promise best accuracy, or replace calibration when a particular sensing target is required. Appendix X.2 retains the detailed trade-offs, and Appendix R compares the implementation's bit-unit default with the exactly nat-canonical weight.

7.3 Tileworld

Tileworld shows why horizon and observation structure must accompany a scaling claim. At 6×6 and $H=2$, EFE and Planning are not significantly different in reward (-21.53 against -21.39 , $p = 0.82$) or success (72.0% against 73.8%, $p = 0.17$). EFE uses fewer scans, 14.75 against 15.64 ($p = 8.8 \times 10^{-5}$). These are failures to reject reward and success differences, not TOST equivalence. The success-tuned $w=50$ agent reaches 97.3% accuracy but uses 32.25 scans and earns lower reward. Table 31 and Figure 18 retain the full comparison and an illustrative trajectory in Appendix X.3.

In a separate fixed- $H=2$ sweep (Figure 10), Planning takes no scans at 8×8 and achieves $1.4\% \pm 0.3\text{pp}$ success, while EFE achieves $69.4\% \pm 1.0\text{pp}$. The short horizon cannot value enough scans to repay their costs before committing on a large grid. The contrast is therefore with a non-scanning finite-horizon agent, not with reward-only planning at arbitrary depth. At 4×4 and 6×6 the two are within one SE, with no formal test for that sweep. The advantage appears at a horizon-and-scale threshold rather than widening uniformly with size. Its weighted comparators use fixed $w=100$, not class-specific tuning. Random and overlapping scan partitions sharply reduce all agents' absolute performance and do not separate EFE from Planning on reward or success, although EFE uses fewer scans in all three modes. Appendix X.3 preserves these robustness results and their scope.

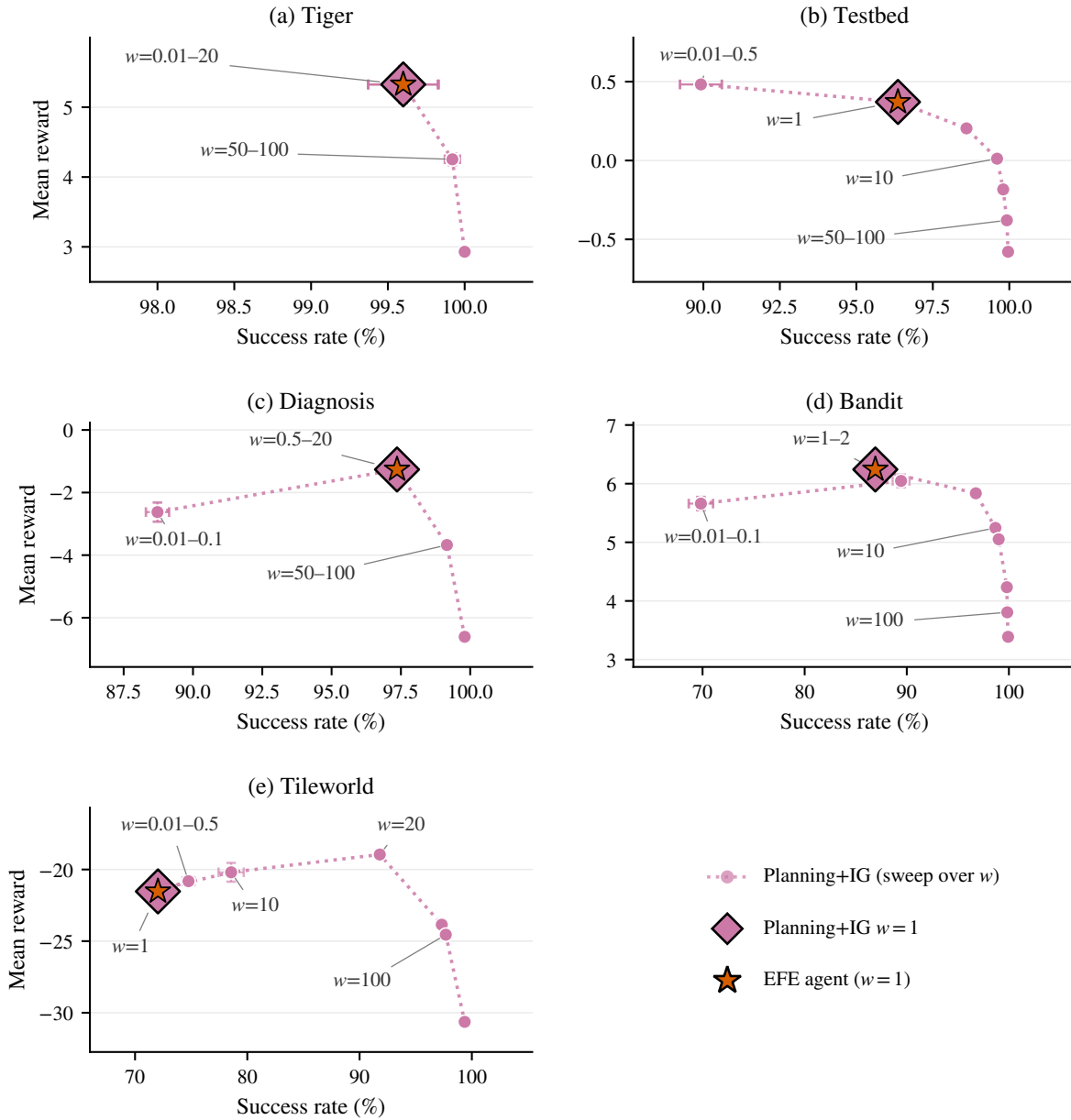


Fig. 9. Reward and success across eleven information weights from 0.01 to 200. The $w=1$ Planning+IG diamond and EFE star coincide under their equivalence assumptions and shared tie-breaking. They attain the largest sampled mean reward on Tiger, Diagnosis, and Bandit. On Testbed they trade reward for success, and on Tileworld $w=20$ dominates them on both axes. Labels identify sampled weights sharing a point. Error bars are seed-level SE, visible where larger than markers.

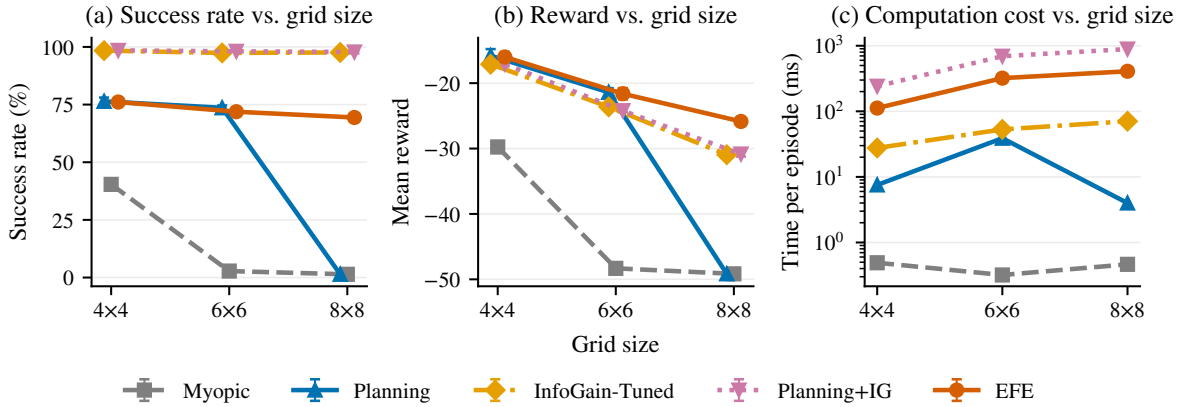


Fig. 10. Tileworld scaling ($H=2$, 200 episodes per seed $\times$ 5 seeds) across Myopic, Planning, InfoGain-Tuned, Planning+IG, and EFE. Planning collapses at 8×8 ($|S|=64$), while EFE maintains 69.4%. Planning+IG at a fixed $w=100$, not tuned in class, achieves 97.8% but at higher reward cost. Error bars in (a) and (b) are seed-level SE, tied series are offset slightly along the categorical axis for legibility, and per-episode time in (c) is a pooled mean over all episodes and seeds, shown without error bars on a log scale.

7.4 RockSample

RockSample tests the information weight within a shared factored belief-space search and a shared greedy leaf rule. On RS[5,3], EFE (16.47) and the $w=5$ comparator (16.41) are not significantly different. On RS[7,4], EFE earns 15.96 against 14.56 and leads the weighted family ($p < 10^{-5}$). On RS[7,8], EFE (21.85) and $w=5$ (23.76) are not significantly different after correction, and both outperform reward-only Planning (12.31). Raising the weight from 5 to 10 significantly hurts on two of these instances and does not significantly help on the third. Only weights 1, 5, and 10 were compared here, so this is not a dense reward-tuning claim.

The important counterexample is RS[11,11]. EFE and Planning both earn 13.23, while a standalone approach-then-check heuristic earns 28.66 ± 0.83 . That heuristic also beats both weighted agents on RS[7,8]. With this shared leaf rule, a larger state space therefore does not by itself increase the value of information weighting. At feasible search depths, distant useful checks lie beyond the tree, and the shared leaf rule values untouched rocks at zero under the uniform prior. It also charges a phantom trip to the east edge before exit, although this variant allows exit anywhere, biasing depth-capped continuations toward early exit. The cost is shared across the weighted family but limits absolute performance. Deeper POMCP does not repair this instance in the tested configurations. Behavioral traces show many distant checks, and reward declines from 12.94 at horizon 10 to -54.78 at horizon 40. Table 4 gives the full comparison. Appendix X.4 retains the leaf-rule disclosure, depth check, and heuristic comparison, alongside the extended solver analysis in Appendix M.

7.5 Structural Inspection

Inspection makes the reward-accuracy trade-off explicit. At $N=8$, EFE improves accuracy from Planning's 73.0% to 91.1% while reward falls from -17.85 to -20.95 . At $N=16$, it improves accuracy from 78.2% to 90.8%, with reward -45.80 against -45.42 , a non-significant difference ($p = 0.57$). The $w=5$ agent attains higher accuracy, 97.3%, but uses 38.3 tests against EFE's 28.1 and earns -54.74 . These results demonstrate feasible factored search on a synthetic known-model problem with a hand-coded leaf rule. They do not validate industrial deployment or

Table 4. RockSample results. RS[5,3] and RS[7,4] run 500 episodes $\times$ 10 seeds (5,000 episodes), and RS[7,8] and RS[11,11] run 100 episodes $\times$ 5 seeds (500 episodes, reduced protocol). Greedy samples without checking. POMCP is the search-tree solver at its tuned, frozen configuration (2,048 simulations, Appendix M). Rollout only runs POMCP’s approach-then-check rollout policy standalone, with no search tree. It is not significantly different from EFE on RS[5,3] and RS[7,4] and earns more on RS[7,8] and RS[11,11]. Bold marks the best mean reward per instance plus any agent not significantly different from it by a seed-level Welch t -test with Holm–Bonferroni correction, corrected within metric (complete pairwise statistics in the committed results_rocksample_*.stats.csv files). Reward reported as mean $\pm$ seed-level SE (SE of the per-seed means over 10 seeds for RS[5,3] and RS[7,4] and over 5 seeds for RS[7,8] and RS[11,11]). Steps and Checks omitted here (per-instance step and check counts are in Appendix M).

Instance	Agent	Good	Bad	Reward
RS[5,3]	Greedy	1.49	1.51	$+3.71 \pm 0.26$
	Planning ($w=0, d=3$)	0.99	0.04	$+14.28 \pm 0.13$
	Plan+IG ($w=5, d=3$)	1.44	0.01	$+16.41 \pm 0.13$
	EFE ($w=1, d=3$)	1.44	0.03	$+16.47 \pm 0.12$
	POMCP (2048 sims)	0.68	0.03	$+12.74 \pm 0.11$
	Rollout only (approach)	1.41	0.07	$+16.67 \pm 0.13$
RS[7,4]	Greedy	1.98	2.02	-1.38 ± 0.36
	Planning ($w=0, d=4$)	0.95	0.05	$+13.97 \pm 0.11$
	Plan+IG ($w=5, d=4$)	1.69	0.01	$+14.56 \pm 0.14$
	EFE ($w=1, d=4$)	1.44	0.01	$+15.96 \pm 0.15$
	POMCP (2048 sims)	0.66	0.03	$+12.01 \pm 0.06$
	Rollout only (approach)	1.88	0.10	$+15.85 \pm 0.17$
RS[7,8]	Greedy	4.02	3.98	-5.60 ± 1.03
	Planning ($w=0, d=3$)	0.48	0.03	$+12.31 \pm 0.35$
	Plan+IG ($w=5, d=3$)	3.40	0.01	$+23.76 \pm 0.58$
	EFE ($w=1, d=3$)	2.61	0.06	$+21.85 \pm 0.79$
	POMCP (2048 sims)	1.71	0.10	$+15.07 \pm 0.45$
	Rollout only (approach)	3.83	0.19	$+28.33 \pm 0.82$
RS[11,11]	Greedy	5.48	5.52	-18.90 ± 1.04
	Planning ($w=0, d=2$)	0.48	0.03	$+13.23 \pm 0.18$
	Plan+IG ($w=5, d=2$)	0.50	0.03	$+13.18 \pm 0.15$
	EFE ($w=1, d=2$)	0.48	0.03	$+13.23 \pm 0.18$
	POMCP (2048 sims)	0.49	0.03	$+12.94 \pm 0.24$
	Rollout only (approach)	5.22	0.23	$+28.66 \pm 0.83$

show that EFE uniquely optimizes either reward or accuracy. Table 5 reports both instances, with the qualified two-state interpretation in Appendix X.5.

Across the battery, the canonical weight is useful when the reward-only policy leaves valuable, representable sensing opportunities unused. Its benefit is bounded by the search horizon, sensor structure, reward convention, and objective of the comparison. It matches the sampled reward maximum on three of five swept environments, loses to other weights on two, and gains nothing over Planning on the largest RockSample instance. These limits matter alongside its positive results on Diagnosis, Bandit, and the largest fixed-horizon Tileworld grid.

Table 5. Structural Inspection results. $N=8$ and $N=16$ run 2,500 episodes each (500 per seed $\times$ 5 seeds). Reward and accuracy are reported as mean $\pm$ seed-level SE. Greedy diagnoses without testing. Bold marks the best accuracy and the best reward per instance, plus any value within sampling error of that best (seed-level Welch $p > 0.05$, uncorrected). The two are not always the same agent. EFE is a competitive untuned trade-off, not uniquely best on all metrics. Accuracy is the fraction of components correctly diagnosed.

Instance	Agent	Accuracy	Missed	Tests	Reward
$N=8$ $ S =256$	Greedy	$70.7 \pm 0.5\%$	2.34	0.0	-114.33 ± 1.96
	Planning ($w=0$)	$73.0 \pm 0.3\%$	0.07	12.7	-17.85 ± 0.23
	Planning+IG ($w=5$)	$96.3 \pm 0.1\%$	0.10	19.7	-27.98 ± 0.44
	EFE ($w=1$)	$91.1 \pm 0.2\%$	0.09	15.5	-20.95 ± 0.30
$N=16$ $ S =65,536$	Greedy	$70.5 \pm 0.4\%$	4.72	0.0	-233.09 ± 3.13
	Planning ($w=0$)	$78.2 \pm 0.2\%$	0.24	22.4	-45.42 ± 0.23
	Planning+IG ($w=5$)	$97.3 \pm 0.0\%$	0.14	38.3	-54.74 ± 0.40
	EFE ($w=1$)	$90.8 \pm 0.1\%$	0.24	28.1	-45.80 ± 0.60

8 Discussion

Choosing a sensing requirement. Expected usage is an interpretable design input when sensing itself is required. It controls neither reward efficiency nor relevance. The endpoint mixture can meet a target while earning less than another mixture at that usage, or than a policy spending less. Ordinary information gain can also pay for irrelevant uncertainty. Reward-equivalence scoring removes distractor spending in the tested factorized benchmark, but does not prevent over-observation (Section 6.10). The simulated targets are calibration-derived. The sensor-record targets were fixed independently of usage curves, but are researcher-defined rather than application requirements. Their later-batch failures require a fresh transfer check before using a frozen calibration.

Interpreting the information-unit default. The recursive EFE identity fixes $w = 1$ after reward and information units are declared. That weight belongs to the level set of $U(1)$, but need not equal its crossing threshold. Its usefulness is empirical: it lies in the reward-maximizing run of sampled weights on three of five swept environments and is beaten on two. In the sampled random two-state family, approximately 21% fail the study's chosen reward-gap criterion even at $H = 3$ (Appendix J). Larger weights can buy greater success at a reward cost, and transfer at the top of the grid can be costly (Table 32). Reward, success and usage therefore require separate evaluation.

Observation structure and planning depth. The advantage depends on sensing opportunities that the search can represent. State-space size and sensor count alone do not predict it. The largest Tileworld separation is against a fixed-horizon policy that takes no scans. On RockSample[11,11] the weighted family gains nothing over Planning, and the standalone heuristic beats the tree-search agents on the two larger instances. Navigation supplies another negative case, with no epistemic agent significantly beating the greedy mover at any tested size (Appendix F). These results do not imply superiority over reward-only planning at arbitrary depth.

At one-step depth, the recursion is myopic reward plus information gain at $w = 1$. Horizon, discounting and model accuracy independently affect its behavior (Appendices J, K, and L). The observed discount threshold is not a general regime boundary. Every tested Tiger sensor mismatch affects EFE and Planning identically, providing no EFE-specific robustness benefit. Overestimated sensor accuracy can cause premature commitment.

What the search controls establish. MCTS-EFE addresses the cost of exact belief-tree enumeration. Its main POMCP comparison matches simulation counts with different costs per simulation, and a separate Tiger control

matches computation. Tuning both solvers on disjoint seeds leaves MCTS-EFE ahead on Tiger reward and success. The Diagnosis and Tileworld gaps also survive the tested exploration constants and information-gain rollouts. However, neither in-tree information gain nor max-backup has a detectable success or reward effect after multiplicity correction at five seeds. EFE leaf evaluation, exact beliefs and exact commit-child values remain coupled, so the experiments do not isolate directed observation selection or establish general solver superiority (Appendices Y and S).

Structural and practical limits. The reduction requires hidden-state-preserving observation and navigation actions. State-changing sensing needs transition-aware predictive and posterior beliefs. The diagnostic Δ_T is neither a proved additive correction nor a decision-error bound, and perfect sensing can make it infinite after arbitrarily small drift (Example U.2, Appendix U). The simulators use known models, and Structural Inspection uses a synthetic task and a hand-coded leaf rule. The sensor-record study learns a model, but its shallow planner and conditional-independence approximation establish no general acquisition advantage. More efficient search, transition-aware budget control and validation against an operational sensing requirement remain open.

9 Conclusion

An expected sensing-usage requirement can calibrate the information-gain weight of a ρ -POMDP policy family. The operational shadow price is a crossing threshold, estimated by a grid bracket. Under stated conditions, mixing the bracket's endpoint policies meets a usage-gap target in expectation. The receding-horizon planner is scale equivariant, and a projected decaying-step controller has a stationary-convergence guarantee under explicit hypotheses. Re-adaptation after reward-scale shifts is an empirical result, including a useful reset heuristic that the theorem does not cover.

The simulated held-out study shows accurate usage calibration but also its reward cost. Real sensor records extend the test to learned likelihoods and targets fixed independently of usage curves. Two targets transfer within the represented collection period, one lies below the sampled range, and later batches defeat frozen calibration. Direct cost tuning and fixed-count information acquisition also give better test accuracy in that configuration. Calibration is therefore a way to adapt an already-selected information-weighted family to a feasible expected-use target, subject to a transfer check. It is not a reason to prefer that family when directly counting or penalizing accesses meets the design need.

The recursive EFE objective supplies an information-unit anchor at $w = 1$, with exact reward-information identification under log scoring and an explicit convention for ordinary rewards. This weight is competitive in several benchmarks and suboptimal in others. Its practical value depends on the observation structure, planning depth, and model assumptions. The resulting design procedure is to state the intended usage, calibrate within a specified policy family, and assess usage, reward, and success together.

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References

E. Altman. 1999. *Constrained Markov Decision Processes*. Chapman & Hall/CRC.

- M. Araya-López, O. Buffet, V. Thomas, and F. Charpillet. 2010. "A POMDP extension with belief-dependent rewards." In: *Advances in Neural Information Processing Systems*.
- M. Bellemare, S. Srinivasan, G. Ostrovski, T. Schaul, D. Saxton, and R. Munos. 2016. "Unifying count-based exploration and intrinsic motivation." In: *Advances in Neural Information Processing Systems*.
- R. Benchetrit, I. Lev-Yehudi, A. Zhitnikov, and V. Indelman. 2025. *Anytime incremental ρ POMDP planning in continuous spaces*. (2025). arXiv: 2502.02549.
- J. M. Bernardo. 1979. "Expected information as expected utility." *The Annals of Statistics*, 7, 3, 686–690.
- J. R. Blum. 1954. "Approximation methods which converge with probability one." *The Annals of Mathematical Statistics*, 25, 2, 382–386.
- C. Boutilier. 2002. "A POMDP formulation of preference elicitation problems." In: *AAAI Conference on Artificial Intelligence*, 239–246.
- Y. Burda, H. Edwards, A. Storkey, and O. Klimov. 2019. "Exploration by random network distillation." In: *International Conference on Learning Representations*.
- T. Champion, H. Bowman, D. Marković, and M. Grześ. 2026. "Reframing the expected free energy: Four formulations and a unification." *Neural Computation*, 38, 3, 439–469.
- P. Cooper and A. Velasquez. 2026. "Expected free energy as belief-dependent utility for ρ -POMDPs." In: *Proceedings of the 7th International Workshop on Active Inference (IWAI 2026)* (Communications in Computer and Information Science). To appear. Springer.
- I. C. Covert, W. Qiu, M. Lu, N. Y. Kim, N. J. White, and S.-I. Lee. 2023. "Learning to Maximize Mutual Information for Dynamic Feature Selection." In: *Proceedings of the 40th International Conference on Machine Learning* (Proceedings of Machine Learning Research). Vol. 202. PMLR, 6424–6447. <https://proceedings.mlr.press/v202/covert23a.html>.
- L. Da Costa, T. Parr, N. Sajid, S. Veselic, V. Neacsu, and K. Friston. 2020. "Active inference on discrete state-spaces: A synthesis." *Journal of Mathematical Psychology*, 99, 102447.
- L. Da Costa, N. Sajid, T. Parr, K. Friston, and R. Smith. 2023. "Reward maximization through discrete active inference." *Neural Computation*, 35, 5, 807–852.
- B. de Vries et al.. 2025. *Expected free energy-based planning as variational inference*. (2025). arXiv: 2504.14898.
- M. O. Duff. 2002. "Optimal learning: Computational procedures for Bayes-adaptive Markov decision processes." Ph.D. Dissertation. University of Massachusetts Amherst.
- M. Fehr, O. Buffet, V. Thomas, and J. Dibangoye. 2018. " ρ -POMDPs have Lipschitz-continuous ϵ -optimal value functions." In: *Advances in Neural Information Processing Systems*.
- A. Foster, D. R. Ivanova, I. Malik, and T. Rainforth. 2021. "Deep adaptive design: Amortizing sequential Bayesian experimental design." In: *International Conference on Machine Learning*. PMLR, 3384–3395.
- Z. Fountas, N. Sajid, P. A. M. Mediano, and K. Friston. 2020. "Deep active inference agents using Monte-Carlo methods." In: *Advances in Neural Information Processing Systems*.
- K. Friston. 2010. "The free-energy principle: a unified brain theory?" *Nature Reviews Neuroscience*, 11, 2, 127–138.
- K. Friston, L. Da Costa, D. Hafner, C. Hesp, and T. Parr. 2021. "Sophisticated inference." *Neural Computation*, 33, 3, 713–763.
- K. Friston, F. Rigoli, D. Ognibene, C. Mathys, T. FitzGerald, and G. Pezzulo. 2015. "Active inference and epistemic value." *Cognitive Neuroscience*, 6, 4, 187–214.
- M. Ghavamzadeh, S. Mannor, J. Pineau, and A. Tamar. 2015. "Bayesian reinforcement learning: A survey." *Foundations and Trends in Machine Learning*, 8, 5–6, 359–483.
- T. Gneiting and A. E. Raftery. 2007. "Strictly proper scoring rules, prediction, and estimation." *Journal of the American Statistical Association*, 102, 477, 359–378.
- A. Guez, D. Silver, and P. Dayan. 2013. "Scalable and efficient Bayes-adaptive reinforcement learning based on Monte-Carlo tree search." *Journal of Artificial Intelligence Research*, 48, 841–883.
- O. E. Gundersen, M. Helmert, and H. Hoos. 2024. "Improving reproducibility in AI research: Four mechanisms adopted by JAIR." *Journal of Artificial Intelligence Research*, 81, 1019–1041.
- T. Haarnoja, A. Zhou, P. Abbeel, and S. Levine. 2018. "Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor." In: *International Conference on Machine Learning*.
- T. Haarnoja, A. Zhou, K. Hartikainen, et al.. 2018. *Soft actor-critic algorithms and applications*. (2018). arXiv: 1812.05905.
- C. Heins, B. Millidge, D. Demekas, B. Klein, K. Friston, I. Couzin, and A. Tschantz. 2022. "pymdp: A Python library for active inference in discrete state spaces." *Journal of Open Source Software*, 7, 73, 4098.
- R. Houthoofd, X. Chen, Y. Duan, J. Schulman, F. De Turck, and P. Abbeel. 2016. "VIME: Variational information maximizing exploration." In: *Advances in Neural Information Processing Systems*.
- R. A. Howard. 1966. "Information value theory." *IEEE Transactions on Systems Science and Cybernetics*, 2, 1, 22–26.
- L. P. Kaelbling, M. L. Littman, and A. R. Cassandra. 1998. "Planning and acting in partially observable stochastic domains." *Artificial Intelligence*, 101, 1–2, 99–134.
- D. Kim, J. Lee, K.-E. Kim, and P. Poupart. 2011. "Point-based value iteration for constrained POMDPs." In: *International Joint Conference on Artificial Intelligence*, 1968–1974.

- W. M. Kouw. 2026. *Expected free energy as an information constraint on the Bethe Lagrangian*. (2026). arXiv: [2608.17167](#).
- H. Kurniawati, D. Hsu, and W. S. Lee. 2008. "SARSOP: Efficient point-based POMDP planning by approximating optimally reachable belief spaces." In: *Robotics: Science and Systems*.
- H. J. Kushner and D. S. Clark. 1978. *Stochastic Approximation Methods for Constrained and Unconstrained Systems*. Applied Mathematical Sciences. Vol. 26. Springer-Verlag, New York.
- Z. Laouar, Q. H. Ho, and Z. Sunberg. 2026. "Leveraging the value of information in POMDP planning." In: *International Conference on Automated Planning and Scheduling*, 152–160.
- S. Levine. 2018. *Reinforcement learning and control as probabilistic inference: Tutorial and review*. (2018). arXiv: [1805.00909](#).
- D. V. Lindley. 1956. "On a measure of the information provided by an experiment." *The Annals of Mathematical Statistics*, 27, 4, 986–1005.
- D. Maisto, F. Gregoretti, K. J. Friston, and G. Pezzulo. 2025. "Active inference tree search in large POMDPs." *Neurocomputing*, 623, 129319.
- F. Matějka and A. McKay. 2015. "Rational inattention to discrete choices: A new foundation for the multinomial logit model." *American Economic Review*, 105, 1, 272–298.
- B. Millidge, A. Tschantz, and C. L. Buckley. 2021. "Whence the expected free energy?" *Neural Computation*, 33, 2, 447–482.
- B. Millidge, A. Tschantz, A. K. Seth, and C. L. Buckley. 2020. "On the relationship between active inference and control as inference." In: *International Workshop on Active Inference*. Springer, 3–11.
- P.-Y. Oudeyer and F. Kaplan. 2007. "What is intrinsic motivation? A typology of computational approaches." *Frontiers in Neurorobotics*, 1, 6.
- T. Parr and K. J. Friston. 2019. "Generalised free energy and active inference." *Biological Cybernetics*, 113, 5–6, 495–513.
- T. Parr, G. Pezzulo, and K. J. Friston. 2022. *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press.
- D. Pathak, P. Agrawal, A. A. Efros, and T. Darrell. 2017. "Curiosity-driven exploration by self-supervised prediction." In: *International Conference on Machine Learning*.
- J. Pineau, G. Gordon, and S. Thrun. 2003. "Point-based value iteration: An anytime algorithm for POMDPs." In: *International Joint Conference on Artificial Intelligence*.
- K. Rawlik, M. Toussaint, and S. Vijayakumar. 2012. "On stochastic optimal control and reinforcement learning by approximate inference." In: *Robotics: Science and Systems*.
- H. Robbins and S. Monro. 1951. "A stochastic approximation method." *The Annals of Mathematical Statistics*, 22, 3, 400–407.
- D. Russo and B. Van Roy. 2014. "Learning to optimize via information-directed sampling." In: *Advances in Neural Information Processing Systems*.
- D. Russo and B. Van Roy. 2018. "Learning to optimize via information-directed sampling." *Operations Research*, 66, 1, 230–252. doi:[10.1287/opr e.2017.1663](#).
- N. Sajid, P. J. Ball, T. Parr, and K. J. Friston. 2021. "Active inference: Demystified and compared." *Neural Computation*, 33, 3, 674–712.
- Y. Satsangi, S. Whiteson, F. A. Oliehoek, and M. T. J. Spaan. 2018. "Exploiting submodular value functions for scaling up active perception." *Autonomous Robots*, 42, 2, 209–233.
- J. Schmidhuber. 1991. "A possibility for implementing curiosity and boredom in model-building neural controllers." In: *Proc. International Conference on Simulation of Adaptive Behavior*, 222–228.
- D. J. Schuurmann. 1987. "A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability." *Journal of Pharmacokinetics and Biopharmaceutics*, 15, 6, 657–680.
- G. Shani, J. Pineau, and R. Kaplow. 2013. "A survey of point-based POMDP solvers." *Autonomous Agents and Multi-Agent Systems*, 27, 1, 1–51.
- D. Silver and J. Veness. 2010. "Monte-Carlo planning in large POMDPs." In: *Advances in Neural Information Processing Systems*.
- C. A. Sims. 2003. "Implications of rational inattention." *Journal of Monetary Economics*, 50, 3, 665–690.
- R. D. Smallwood and E. J. Sondik. 1973. "The optimal control of partially observable Markov processes over a finite horizon." *Operations Research*, 21, 5, 1071–1088.
- T. Smith and R. Simmons. 2004. "Heuristic search value iteration for POMDPs." In: *Uncertainty in Artificial Intelligence*.
- M. T. J. Spaan, T. S. Veiga, and P. U. Lima. 2015. "Decision-theoretic planning under uncertainty with information rewards for active cooperative perception." *Autonomous Agents and Multi-Agent Systems*, 29, 6, 1157–1185.
- P. Stocco, S. Chundi, A. Jamgochian, and M. J. Kochenderfer. 2024. "Addressing myopic constrained POMDP planning with recursive dual ascent." In: *International Conference on Automated Planning and Scheduling*, 565–569.
- Z. N. Sunberg and M. J. Kochenderfer. 2018. "Online algorithms for POMDPs with continuous state, action, and observation spaces." In: *International Conference on Automated Planning and Scheduling*.
- P. Sweeney, J. Ruiz-Serra, and M. S. Harré. 2026. "Decision, inference, and information: Formal equivalences under active inference." *Entropy*, 28, 1, 1.
- E. Todorov. 2006. "Linearly-solvable Markov decision problems." In: *Advances in Neural Information Processing Systems*.
- M. Towers et al. 2024. *Gymnasium: A standard interface for reinforcement learning environments*. (2024). arXiv: [2407.17032](#).
- A. Tschantz, B. Millidge, A. K. Seth, and C. L. Buckley. 2020. "Reinforcement learning through active inference." In: *ICLR Workshop on Bridging AI and Cognitive Science*.
- A. Vergara. 2012. *Gas Sensor Array Drift Dataset*. Dataset. (2012). doi:[10.24432/C5RP6W](#).

- A. Vergara, S. Vembu, T. Ayhan, M. A. Ryan, M. L. Homer, and R. Huerta. 2012. “Chemical gas sensor drift compensation using classifier ensembles.” *Sensors and Actuators B: Chemical*, 166–167, 320–329. doi:10.1016/j.snb.2012.01.074.
- E. Walraven, J. Sijs, and G. J. Burghouts. 2025. “Information gathering in POMDPs using active inference.” *Autonomous Agents and Multi-Agent Systems*, 39, 1, 3.
- R. Wei. 2024. *Value of information and reward specification in active inference and POMDPs*. (2024). arXiv: 2408.06542.
- N. Ye, A. Somani, D. Hsu, and W. S. Lee. 2017. “DESPOT: Online POMDP planning with regularization.” *Journal of Artificial Intelligence Research*, 58, 231–266.

A Proof of Proposition 3.1

PROOF. Let $V_\rho(a, b, d)$ be the ρ -POMDP action value at belief b and depth d , with ρ_{EFE} defined in the proposition. Define $V(a, b, d) := -\mathcal{G}(a, b, d)$. Negation reverses ordering, so the minimizing actions for $\mathcal{G}$ are the maximizing actions for V .

For any terminal commit action i and $d \leq H$, Equation 5 gives

$$V(\text{commit}_i, b, d) = \mathbb{E}_b[R_i] = V_\rho(\text{commit}_i, b, d),$$

because $\rho_{\text{EFE}}(b, \text{commit}_i) = 0$ and there is no continuation. At depth H , only these commit actions are available, establishing the induction boundary.

For observation action k at $d < H$, negating Equation 3, with its depth arguments explicit, gives

$$V(\text{observe}_k, b, d) = -c_k + I_k(b) + \mathbb{E}_o \left[\max_{a'} V(a', b'_o, d + 1) \right].$$

The ρ -POMDP backup has the same immediate reward $-c_k$, belief utility $I_k(b)$, posterior distribution, and continuation depth. Under $\gamma = 1$, it is therefore the same expression with V_ρ replacing V . Induction on the remaining horizon $H - d$ gives $V(a, b, d) = V_\rho(a, b, d)$ for every available action and belief node. Their maximizing action sets coincide, and the shared tie-breaking rule selects the same action. Hence the policies agree throughout the finite-horizon tree and, when replanned from the same beliefs, at each real decision. $\square$

B Additional Per-Environment Results

Table 3 gives the four central agents on Tiger, Diagnosis, and Bandit. The table below adds the remaining baselines under the same 1,000-episodes-per-seed, five-seed protocol. Untuned Info Gain is identical to Myopic, and Tiger’s pymdp-AIF is identical to Posterior-vote. Complete rows remain in the environment CSVs.

Table 6. Additional core baselines. Reward uncertainty is pooled episode-level SE, unlike the seed-level SE in Table 3. Tuned Info Gain maximizes success within its own class.

Environment	Agent	Obs.	Success	Reward
Tiger	Info Gain (w=20)	4.20	99.4%	+5.19 $\pm$ 0.12
	Epistemic-only	0.00	49.7%	−45.31 $\pm$ 0.78
	Posterior-vote	2.64	96.8%	+3.79 $\pm$ 0.28
Diagnosis	Info Gain (w=100)	13.19	99.4%	−3.58 $\pm$ 0.09
	Epistemic-only	0.00	24.7%	−35.16 $\pm$ 0.37
	Posterior-vote	5.83	88.4%	−2.78 $\pm$ 0.27
Bandit	Info Gain (w=20)	9.54	98.8%	+5.12 $\pm$ 0.04
	Epistemic-only	0.00	24.7%	+3.23 $\pm$ 0.05
	Posterior-vote	5.01	87.0%	+6.32 $\pm$ 0.05

The pooled reward SEs for Myopic, Planning, Planning+IG and EFE, in that order, are (0.55, 0.12, 0.06, 0.12) on Tiger, (0.41, 0.27, 0.10, 0.15) on Diagnosis, and (0.06, 0.06, 0.04, 0.05) on Bandit. These estimate episode-level precision rather than between-seed replicability. The two Tileworld tables retain their separate 500-episodes-per-seed protocol.

Table 7. Tileworld 6×6 (2,500 episodes at 500 per seed × 5 seeds, $H=2$). This covers every agent in the Tileworld battery (experiments/run_tileworld.py). POMCP and MCTS-EFE results on this grid, run under their own protocols, are in Table 22 and Appendix S. Posterior-vote and PyMDP-AIF were not run on the spatial environments. Bold marks EFE and any agent not significantly different from it at the seed level (Welch $p > 0.05$, uncorrected), not the single best value per column. Epistemic-only commits immediately (0 scans) rather than exhausting the step cap, matching Myopic exactly (Section 7). Reward uncertainty is pooled episode-level SE, matching the other tables of this appendix.

Agent	Scans	Success	Reward
Myopic	0.00	2.9%	-48.25 ± 0.20
Planning ($H=2$)	15.64	73.8%	-21.39 ± 0.54
Info Gain ($w=1$)	5.20	31.7%	-36.16 ± 0.56
Info Gain ($w=100$)	32.30	98.0%	-23.50 ± 0.24
Planning+IG ($w=50$)	32.25	97.3%	-23.85 ± 0.26
Epistemic-only	0.00	2.9%	-48.25 ± 0.20
EFE ($H=2$)	14.75	72.0%	-21.53 ± 0.54

Table 8. Tileworld 8×8 (2,500 episodes at 500 per seed × 5 seeds, $H=2$). The agent set and bolding rule are those of the 6×6 table, so the higher success rates of the two heavily scanning tuned agents are unbolded by design. The separate spatial-scaling sweep (Figure 10), with a different per-seed episode count, gives 69.4% for EFE and 1.4% for Planning, gaps consistent with seed-to-seed variation. Planning, untuned Info Gain, and Epistemic-only take no scans and tie Myopic exactly, because a single noisy scan over 64 cells barely narrows the belief (Planning’s depth-2 case in Appendix X.3). Reward uncertainty is pooled episode-level SE, matching the other tables of this appendix.

Agent	Scans	Success	Reward
Myopic	0.00	1.8%	-48.94 ± 0.16
Planning ($H=2$)	0.00	1.8%	-48.94 ± 0.16
Info Gain ($w=1$)	0.00	1.8%	-48.94 ± 0.16
Info Gain ($w=100$)	39.45	97.2%	-31.15 ± 0.25
Planning+IG ($w=50$)	39.49	97.5%	-31.00 ± 0.25
Epistemic-only	0.00	1.8%	-48.94 ± 0.16
EFE ($H=2$)	17.60	69.8%	-25.69 ± 0.56

C Two-State Testbed

Under symmetric $+1/-1$ rewards, EFE over-observes relative to Planning ($+0.38$ vs. $+0.48$ reward, seed-level Welch $p=1.1 \times 10^{-5}$). Success-tuned agents push this trade-off to 99.5% success at $+0.01$ reward. Posterior-vote matches Planning on all three metrics. Epistemic-only matches Planning on observations and success here because information gain clears the 0.1 observation cost, but commits immediately on Tiger, Diagnosis, Bandit, and Tileworld (results_summary.csv, Appendix B).

At the post-first-observation belief, a second observation has zero instrumental value: a disconfirming outcome returns belief to the tie point. Proposition U.1’s resulting ceiling is $w_{\text{over}}^* \approx 1.01$ reward units per nat, below

Table 9. Information-seeking testbed (5,000 episodes at 1,000 per seed $\times$ 5 seeds, $H=4$). Symmetric rewards $+1/-1$. Each tuned agent is success-tuned in its own agent class, as `tune_info_gain_weight` requires, at $w_{\text{succ}}^*=20$ for myopic Info Gain and $w_{\text{succ}}^*=10$ for Planning+IG (both recorded in `results_supplementary_tuned_weights.csv`). At their respective tuned weights the two converge to the same policy and are bit-identical on all three metrics. Bold marks the best value in the success and reward columns, with exact ties bolded together. Every row here is produced by `experiments/run_supplementary.py`.

Agent	Obs.	Success	Reward
Myopic	1.00	76.3%	$+0.43 \pm 0.01$
Planning ($H=4$)	3.15	89.8%	$+0.48 \pm 0.01$
Info Gain ($w_{\text{succ}}^*=20$)	9.79	99.5%	$+0.01 \pm 0.01$
Planning+IG ($w_{\text{succ}}^*=10$)	9.79	99.5%	$+0.01 \pm 0.01$
Posterior-vote	3.15	89.8%	$+0.48 \pm 0.01$
EFE ($H=4$)	5.50	96.3%	$+0.38 \pm 0.01$

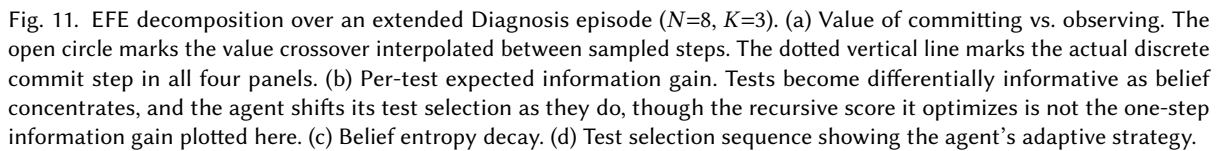
the implemented weight's 1.44 (Table 26). Testbed is a boundary case of the proposition's strict asymmetry assumption, and Appendix U.2 distinguishes this two-step calculation from measured behavior at $H=4$.

D Supplementary Behavioral Checks

Test selection. At fixed Diagnosis size $N=8$ and horizon $H=3$ (100 episodes per seed over five seeds), EFE and Planning are bit-identical with one test, within seed-level sampling error with two (43.2% vs. 43.8% success), and separated with three (96.0% vs. 82.8%). This is a threshold pattern, not steady growth with test count. The main Diagnosis battery separately shows a gap with two tests. Figure 11 shows how informative tests change within an episode. The chosen test maximizes the recursive score, not necessarily the one-step information gain in panel (b). The full scaling curves are archived in `results_showcase_obs_scaling.csv`.

Reward asymmetry. A Tiger sweep at $H=6$ uses 100 episodes per seed over the five canonical seeds, giving 500 episodes per penalty (`results_showcase_asymmetry.csv`). Across $|R^-| \in [1, 500]$, EFE increases listening from 2.7 to 5.8 without reconfiguration. Success-tuning myopic Info Gain separately at each penalty selects $w=20$ throughout, with 4.37 observations. Planning+IG reaches 5.77 at penalties 200 and 500. At penalty 1, tuned myopic Info Gain earns about 1.6 less reward than Planning. At the largest penalty, myopic Info Gain nominally beats EFE (4.61 vs. 4.23). The sweep demonstrates adaptation, not uniform dominance.

Trajectory diagnostics. Illustrative Tiger traces commit when commit value exceeds recursive observe value, without a separate stopping rule. Selected Diagnosis traces all commit correctly, after 14 observations for Planning, 29 for EFE, and 32 for tuned Info Gain. They were not selected to represent typical lengths. A separate 300-episode Diagnosis visualization ($N=8$, $K=3$) places EFE's observation-survival curve between Planning's and Planning+IG's. Entropy and cumulative-reward curves condition on episodes still running and display episode-level SE. That visualization transfers the myopic Info Gain weight selected on 100 tuning episodes to Planning+IG, unlike the main tables' own-class tuning. These diagnostic plots remain in `figures/fig_efe_trajectory.pdf`, `fig_belief_heatmap.pdf`, and `fig_efficiency_curves.pdf`. Tileworld's three-agent trace is Figure 18.



RockSample check accuracy decays with distance from the rock (0.95 at range zero, halving every 4 cells, floored at 0.5). The tabulated value is the maximum. Each RockSample instance runs under an episode step cap of $N^2 + 10K$ steps, giving 55, 89, 129, and 231 for RS[5,3], RS[7,4], RS[7,8], and RS[11,11], and the fixed rock maps are those in ROCKSAMPLE_CONFIGS (experiments/run_rocksample.py). On RS[11,11] the start cell (5, 0) coincides with a

Table 10. Full environment specifications for every environment reported in this paper’s results tables and appendices. Variant instances used outside the main results tables (Navigation 5×5 and 7×7, Tileworld 4×4, Diagnosis $N \in \{2, 8, 16\}$) differ from their tabulated siblings only in the stated size parameter. H is the planning horizon for observe-then-commit environments and the tree depth for the interleaved ones. γ is the discount used by the reporting agents. All main-text batteries are undiscounted ($\gamma=1.0$, the agent default), and γ is set to 0.999 inside the SARSOP/CPOMDP exports of Section 6.6, which require a discounted infinite-horizon model, and is otherwise varied only in the discount sweep of Appendix K and in two rows of the RockSample POMCP sensitivity battery. $|S|$ counts hidden-state configurations only, excluding fully observable components such as the RockSample agent position. Move actions are listed under observation actions only for Navigation, where they are the sole information source. Inspection additionally has four move (navigation) actions, omitted from its observation-action count.

Environment	$ S $	Obs. actions	Commit actions	Accuracy
Tiger	2	1	2	0.85
Testbed	2	1	2	0.75
Diagnosis ($N=4$)	4	2	4	0.80
Bandit ($K=4$)	4	4	4	0.80
Navigation	9	4 (move)	0 (implicit)	var.
Tileworld (6×6)	36	6 (scan)	36 (collect)	0.80
Tileworld (8×8)	64	6 (scan)	64 (collect)	0.80
Interleaved observe-act (Proposition 3.3). Undiscounted.				
RockSample				0.95
RS[5,3]	$2^3=8$	3 (check)	sample/exit	decaying
RockSample				0.95
RS[7,4]	$2^4=16$	4 (check)	sample/exit	decaying
RockSample				0.95
RS[7,8]	$2^8=256$	8 (check)	sample/exit	decaying
RockSample				0.95
RS[11,11]	$2^{11}=2,048$	11 (check)	sample/exit	decaying
Inspection ($N=8$)	$2^8=256$	2 types × 8	diagnose	0.70/0.90
Inspection ($N=16$)	$2^{16}=65,536$	2 types × 16	diagnose	0.70/0.90

rock while every other rock lies at Euclidean distance 3.16 or more, at or beyond the limit of the sensor’s useful range (accuracy roughly 0.55 at distance 3.16, and floored at chance from 3.7 cells). This geometry is behind the single-informative-check pattern of Section 7.4. Its per-instance tree depth was fixed a priori before the reported runs, not selected on measured agent performance, and the depth-3 check on RS[11,11] (Section 7.4) shows the reported pattern is insensitive to one extra ply. Inspection also gives +2 for a correct nominal call and −5 for a false alarm, with fault prior 0.3.

The Diagnosis environment uses $K = \lceil \log_2 N \rceil$ binary tests, each providing noisy information about a different partition of the state space. The Navigation environment uses proximity-based warm/cold signals with accuracy depending on Manhattan distance to the hidden goal. Tileworld projects the Diagnosis partition structure onto a 2D grid. $K = 2\lceil \log_2 n \rceil$ scans, for grid side n , partition the grid by bit-level splits of the row and column indices, enabling complete spatial disambiguation.

Table 10. Full environment specifications (continued). Sensing costs, commit rewards, and planning horizon or tree depth H with discount γ . Every RockSample action costs -0.5 . Inspection additionally pays -0.5 per move, and its other outcomes are specified below.

Environment	Obs. cost	Correct	Incorrect	H / γ
Tiger	-1.0	$+10$	-100	$6 / 1.0$
Testbed	-0.1	$+1$	-1	$4 / 1.0$
Diagnosis ($N=4$)	-1.0	$+10$	-50	$3 / 1.0$
Bandit ($K=4$)	-0.5	$+10$	$+1$	$2 / 1.0$
Navigation	-0.5	$+20$	$-$	$2 / 1.0$
Tileworld (6×6)	-1.0	$+10$	-50	$2 / 1.0$
Tileworld (8×8)	-1.0	$+10$	-50	$2 / 1.0$
Interleaved observe-act (Proposition 3.3). Undiscounted.				
RockSample				
RS[5,3]	-0.5	$+10$	-10	$3 / 1.0$
RockSample				
RS[7,4]	-0.5	$+10$	-10	$4 / 1.0$
RockSample				
RS[7,8]	-0.5	$+10$	-10	$3 / 1.0$
RockSample				
RS[11,11]	-0.5	$+10$	-10	$2 / 1.0$
Inspection				
($N=8$)	$-0.5/-2.0$	$+5$ fault	-50 missed	$3 / 1.0$
Inspection				
($N=16$)	$-0.5/-2.0$	$+5$ fault	-50 missed	$2 / 1.0$

F Navigation Results

Navigation has no separate sensing actions. Moving produces proximity-based warm/cold evidence about the hidden goal (Appendix E).

NavMyopic moves toward the highest-probability cell and has the highest mean reward and success at every size. Its advantage over NavEFE at 5×5 survives seed-level Welch tests with Holm correction within metric (reward $p=0.002$, success $p=0.010$, observations $p=0.001$). No pairwise comparison survives correction at 3×3 or 7×7 . NavEFE's nominal reward advantage over one-step NavInfoGain at 7×7 ($p=0.040$ uncorrected) also fails correction. Thus no epistemic agent significantly beats the myopic rule on these grids. This contrasts with tasks that offer several distinct sensing actions, while leaving causality between task structure and performance unresolved.

Table 11. Navigation scaling (150 episodes per seed $\times$ 5 seeds = 750 episodes per agent per grid). Step budget $3n^2$. NavEFE uses depth-limited planning $H=2$. Uncertainty is seed-level SE. See results_navigation_scaling.csv and the companion seed-level Welch tests in results_navigation_scaling_seed_stats.csv.

Grid	$ S $	Agent	Success	Mean reward
3 $\times$ 3	9	NavMyopic	90.4 $\pm$ 1.3%	+2.04 $\pm$ 1.51
		NavInfoGain	86.4 $\pm$ 1.6%	−1.37 $\pm$ 1.69
		NavEFE ($H=2$)	87.2 $\pm$ 1.7%	−1.29 $\pm$ 1.58
5 $\times$ 5	25	NavMyopic	80.7 $\pm$ 1.4%	−16.64 $\pm$ 1.54
		NavInfoGain	76.0 $\pm$ 1.5%	−23.24 $\pm$ 1.75
		NavEFE ($H=2$)	74.4 $\pm$ 1.2%	−25.76 $\pm$ 1.14
7 $\times$ 7	49	NavMyopic	65.7 $\pm$ 2.8%	−38.42 $\pm$ 2.77
		NavInfoGain	60.3 $\pm$ 1.2%	−44.28 $\pm$ 1.12
		NavEFE ($H=2$)	62.8 $\pm$ 0.7%	−40.86 $\pm$ 0.77

G Scaling Analysis

Table 12. Scaling analysis. Diagnosis $N = 2$ to 16 (2,500 episodes at 500 per seed $\times$ 5 seeds, $H=2$). Bold marks the best success and reward per N and the tests of those rows, plus any row within sampling error of it (seed-level Welch $p > 0.05$, uncorrected, results_scaling_stats.csv). At $N=8$ and $N=16$ the EFE and Planning rows are statistical ties.

N	Agent	Tests	Success	Reward
2	Myopic	1.00	80.3%	−2.83
	Planning	2.91	94.1%	+3.56
	Info Gain	1.00	80.3%	−2.83
	EFE	2.91	94.1%	+3.56
4	Myopic	2.00	64.5%	−13.29
	Planning	5.83	88.8%	−2.58
	Info Gain	2.00	64.5%	−13.29
	EFE	5.83	88.8%	−2.58
8	Myopic	3.00	51.1%	−22.33
	Planning	8.79	83.9%	−8.44
	Info Gain	3.00	51.1%	−22.33
	EFE	8.83	84.1%	−8.38
16	Myopic	4.00	40.8%	−29.50
	Planning	11.65	77.4%	−15.23
	Info Gain	4.00	40.8%	−29.50
	EFE	11.64	77.1%	−15.37

At $H=2$, EFE and Planning are bit-identical at $N=2, 4$ and show no consistent gap at $N=8, 16$. Their success rates are $84.08\% \pm 0.66\text{pp}$ vs. $83.92\% \pm 0.82\text{pp}$ at $N=8$ and $77.12\% \pm 0.78\text{pp}$ vs. $77.36\% \pm 0.98\text{pp}$ at $N=16$ (seed-level SE). Both outperform Myopic and untuned Info Gain. The contrast with EFE’s +8.3pp advantage over Planning at $H=3$ (Table 3) is consistent with a dependence on depth rather than state count alone. EFE’s success advantage

over Myopic grows from 13.8pp at $N=2$ to 36.3pp at $N=16$, a benefit of deeper planning shared by reward-only Planning.

H PyMDP Consistency Check

The standard pymdp Agent (Heins et al. 2022), wrapped for observe-then-commit tasks, matches myopic Info Gain at $w=1$ on Testbed under the canonical protocol (3.15 observations, 89.8% success, +0.480 reward). Both also match reward-only Planning, whereas recursive $H=4$ EFE takes 5.50 observations. Choices agree at the tested uniform, intermediate, and confident two-state beliefs, and information-gain evaluations agree qualitatively. This is a limited single-step consistency check, not certified numerical equivalence or validation of recursive EFE (results_summary.csv).

I Supplementary Statistics

Primary comparisons use independent two-sample Welch t -tests on per-seed means ($n=5$), with Holm-Bonferroni correction. Uncorrected comparisons are marked locally. Unless stated otherwise, the family comprises all pairwise agent comparisons and reported metrics within one environment or instance: $M\binom{A}{2}$ tests for A agents and M metrics. Nominal sizes are 108 for Tiger (9 agents, reward/success/observations), 84 for Diagnosis and Bandit (8 agents), and 18 per Inspection instance (4 agents, reward/accuracy/tests). Undefined zero-variance comparisons are excluded, leaving 107 defined tests for Tiger and 83 for Diagnosis. RockSample instead corrects within metric, 21 tests per instance over seven agents, because pooling metrics would loosen its non-rejection-based bolding rule. The discount sweep pools its 24 tests across environments, discounts, and metrics. The MCTS ablation corrects within metric across its three environments, and the exploration sweep within metric within each of its three comparison families (Appendix S).

The six main observe-then-commit and Tileworld stats files also report pooled episode-level equal-variance tests. Substituting Welch changes no non-degenerate decision at $\alpha=0.05$ among their 510 comparisons. RockSample and Inspection use Welch at both levels. Because episodes within a seed are not independent, pooled tests supplement the seed-level analysis. Bootstrap 95% intervals use 10,000 resamples and a fixed seed. Interpreted effect sizes are Cohen's d with pooled episode-level standard deviation.

Table 3 reports seed-level SE, while Appendix B supplies pooled SE and additional baselines. The former describes between-seed replicability, the latter episode-level precision. Companion *_stats.csv files contain tests and effect sizes. The separate Tileworld scaling battery has seed-level reward and success SEs but no Welch companion file.

Table 13. Pooled episode-level Cohen's d , positive when EFE scores higher. Five reward columns compare EFE with the named agent. The final column compares success with Planning, treating outcomes as Bernoulli. Magnitudes below 0.2, 0.5, and 0.8 are conventionally negligible, small, and medium, respectively.

Environment	Reward				Success	
	Myopic	Planning	Tuned IG	Plan.+IG	Vote	Planning
Tiger	0.44	0.00	0.00	0.14	0.09	0.00
Testbed	−0.07	−0.18	0.69	0.69	−0.18	0.26
Diagnosis	0.55	0.08	0.25	0.25	0.09	0.33
Bandit	0.18	0.14	0.36	0.37	−0.01	0.40

J Near-Optimality across Planning Horizons

To examine horizons beyond Proposition U.1’s $H=1$ onset and $H=2$ ceiling, we sample 100 two-state environments with asymmetry $\alpha \in [1, 50]$, accuracy $p \in [0.55, 0.95]$, and cost in $[0.1, 5]$. At each $H \in \{1, 2, 3\}$, we select the largest estimated mean reward $\hat{R}_{\text{best}}$ over $w \in \{0.01, 0.5, 1, 2, 5, 50\}$ and count $w=1$ as near-optimal if its estimated deficit is at most $\max(0.05|\hat{R}_{\text{best}}|, 0.5)$ reward units. Each cell uses 50 episodes at one seed ([experiments/run_neaopt_horizon.py](#)), a deliberate departure from the canonical five-seed protocol. The rates are descriptive, not per-environment estimates with uncertainty guarantees.

The pass rate rises from 30% to 58% to 79% across the three horizons (26%, 53%, 79% for $\alpha \geq 10$, Figure 12). Thus $w=1$ still fails on 21% at $H=3$, despite a lenient absolute tolerance that dominates when reward ranges are small. This sampled trend is consistent with a benefit from recursive planning, not a general horizon guarantee. Bandit’s 86.9% success at $H=2$ is only a qualitative comparison: the two-state threshold formula does not apply to it.

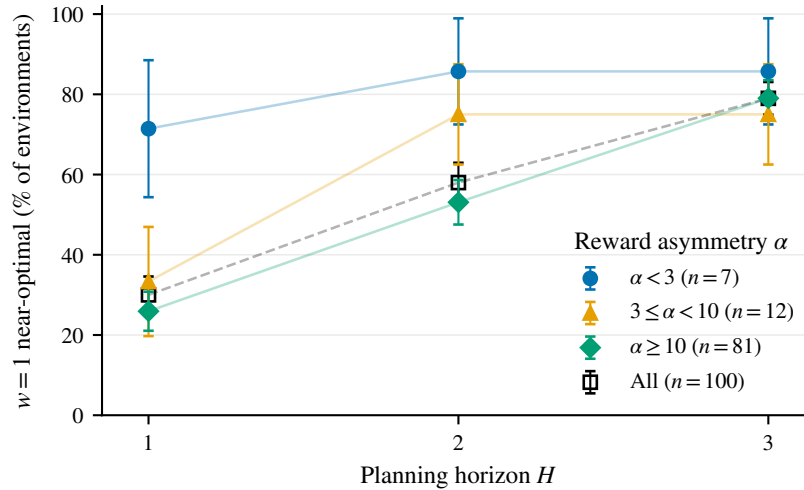


Fig. 12. Fraction of randomly generated two-state environments where $w=1$ is near-optimal, as a function of planning horizon H , stratified by reward asymmetry α . The near-optimality basin widens with horizon, consistent with multi-step planning amplifying the value of observation. 100 environments, $\alpha \in [1, 50]$, $p \in [0.55, 0.95]$, cost $\in [0.1, 5]$. Markers show the three measured horizons $H \in \{1, 2, 3\}$ with binomial standard-error bars, the legend reports the number of environments in each α stratum, and the light guide lines connect measured points only.

Agreement across planning horizons. Table 14 stratifies the same 100 environments by asymmetry, accuracy, and cost. Eight of nine strata have higher pass rates at $H=3$ than at $H=1$, most sharply at $p \geq 0.85$ (17% $\rightarrow$ 90%). Low accuracy ($p < 0.65$, 25 environments) is the exception (80% $\rightarrow$ 84% $\rightarrow$ 80%). Low asymmetry ($\alpha < 3$, 7 environments) gives 71% $\rightarrow$ 86% $\rightarrow$ 86%.

Agreement means passing at $H=1$ and at least one deeper horizon, or failing at all three. There are 44 agreements: 30 passes recurring deeper (26 at both deeper horizons, four at just one, environment IDs 2, 42, 63, 95) and 14 persistent failures. The other 56 fail at $H=1$ but pass deeper. No sampled environment passes only at $H=1$. This

Table 14. Horizon map, the fraction of randomly generated two-state environments where $w=1$ is near-optimal: its estimated reward gap from the best grid weight is at most $\max(0.05|\hat{R}_{\text{best}}|, 0.5)$ reward units, where $\hat{R}_{\text{best}}$ is the largest estimated grid reward. Results are stratified by each sampled regime axis and planning horizon H . Same 100 environments as Figure 12 (Appendix J). This is an empirical boundary map, not an adaptive-submodularity guarantee.

Regime	$H=1$	$H=2$	$H=3$
α (reward asymmetry)			
$\alpha < 3$	71%	86%	86%
$3 \leq \alpha < 10$	33%	75%	75%
$\alpha \geq 10$	26%	53%	79%
p (observation informativeness)			
$p < 0.65$	80%	84%	80%
$0.65 \leq p < 0.85$	11%	40%	71%
$p \geq 0.85$	17%	63%	90%
cost (observation cost)			
cost < 1	16%	16%	58%
$1 \leq \text{cost} < 3$	26%	55%	79%
cost ≥ 3	40%	79%	88%

$H=1$ vs. $H \geq 2$ agreement: 44% of 100 environments agree; myopic overclaims 0, underclaims 56.

boundary map neither proves monotonicity nor excludes such overclaims generally, and is not an adaptive-submodularity result. Per-environment records are in `results/results_nearopt_horizon.csv`, aggregated by `experiments/build_horizon_map.py` into `results/results_horizon_map_agreement.csv`.

K Discount Factor Sensitivity

We compare EFE with same-horizon Planning over four discounts using 500 episodes per seed and five canonical seeds (`experiments/run_discount.py`). This is lighter than Table 3’s battery. Table 15 combines numerically identical settings but retains every measured outcome.

Table 15. Discount sensitivity. Paired cells list EFE / Planning. Success gaps are percentage points, followed by seed-level Welch p . Stars denote significance after Holm correction over the battery’s 24 tests.

Environment	γ	Obs.	Success (%)	Reward	Gap (p)
Tiger	0.90	4.23/2.65	99.6/96.9	5.33/3.91	2.7 ($< 10^{-3}$)*
	0.95, 0.99, 1	4.23/4.23	99.6/99.6	5.33/5.33	0.0 (1.0)
Diagnosis	0.90, 0.95	5.83/5.86	88.8/88.7	-2.58/-2.62	0.0 (0.96)
	0.99	9.67/5.89	97.5/89.6	-1.19/-2.15	7.9 ($< 10^{-4}$)*
	1.00	9.68/5.85	97.4/88.8	-1.26/-2.59	8.6 ($< 10^{-4}$)*
Bandit	0.90, 0.95	2.04/2.04	61.9/62.0	5.55/5.56	-0.2 (0.95)
	0.99	5.16/2.38	86.9/64.5	6.24/5.61	22.4 ($< 10^{-4}$)*
	1.00	5.16/3.24	86.9/69.9	6.24/5.67	17.0 ($< 10^{-4}$)*

Tiger remains above 96% success throughout. Diagnosis and Bandit show no significant gap at $\gamma \leq 0.95$, while their EFE advantage emerges at $\gamma \geq 0.99$. On this grid, the effective horizon matters. Gaps use unrounded estimates and can differ from subtraction of the displayed percentages.

L Model Misspecification Sensitivity

We vary the agent’s observation accuracy around the true value, holding the environment fixed. Both batteries use 500 episodes per seed over five seeds (results_model_misspec.csv). Reward uncertainties below are seed-level SE.

On Tiger ($p_{\text{true}}=0.85$, $H=4$), EFE and Planning are bit-identical at every tested accuracy. For $p_{\text{agent}} \in \{0.70, 0.75, 0.80, 0.85\}$, both take 4.23 observations, succeed on 99.6%, and earn 5.33 ± 0.26 . At 0.90 and 0.95, both take 2.65 observations, succeed on 96.9%, and earn 3.91 ± 0.38 . The tested overestimates reduce sensing and performance, with no EFE-specific buffering.

Table 16. Model misspecification on Diagnosis ($p_{\text{true}}=0.80$, $H=3$, 2,500 episodes across 5 seeds). Reward reported as mean $\pm$ seed-level SE (results_model_misspec.csv).

Agent	p_{agent}	Mismatch	Obs	Success	Reward
EFE	0.65	−0.15	5.86	88.7%	-2.62 ± 0.31
EFE	0.70	−0.10	9.68	97.4%	-1.26 ± 0.34
EFE	0.75	−0.05	9.68	97.4%	-1.26 ± 0.34
EFE	0.80	± 0.00	9.68	97.4%	-1.26 ± 0.34
EFE	0.85	+0.05	5.86	88.7%	-2.62 ± 0.31
EFE	0.90	+0.10	5.86	88.7%	-2.62 ± 0.31
Planning	0.65	−0.15	5.84	89.2%	-2.35 ± 0.51
Planning	0.70	−0.10	9.71	97.6%	-1.12 ± 0.26
Planning	0.75	−0.05	9.68	97.3%	-1.29 ± 0.32
Planning	0.80	± 0.00	5.85	88.8%	-2.59 ± 0.37
Planning	0.85	+0.05	5.86	88.7%	-2.62 ± 0.31
Planning	0.90	+0.10	5.82	88.7%	-2.61 ± 0.40

Diagnosis ($p_{\text{true}}=0.80$, $H=3$) also changes in steps. Mild underestimation and calibration give EFE the same 9.68 tests and 97.4% success. Its largest underestimate and both overestimates give 5.86 tests and 88.7% success. Planning’s pattern is similar but not identical. These thresholds are therefore not specific to the epistemic term, and the experiment does not establish their mechanism.

M RockSample as an Interleaved Observe-Act Setting (Extended)

Table 17 extends Table 4 with counts, two higher weights, tuned POMCP, and its standalone rollout. Both tables use the same instance CSVs and build_rocksample_tables.py. Separate POMCP tuning, budget, horizon, and sensitivity CSVs support the diagnostics below.

Greedy samples without checking quality, taking 1.51–5.52 bad rocks per episode across instances, versus EFE’s 0.01–0.06 (seed-level Welch $p < 5 \times 10^{-8}$ at each instance). On RS[11,11], all tree-search agents instead use the low-activity policies described in Section 7.4. Appendix X.4 gives the higher-weight and heuristic comparisons. At RS[7,8], increasing w from 5 to 10 buys 0.54 more good rocks but adds 17.2 steps, each costing 0.5. More sensing need not improve return.

POMCP implementation and selection. The history tree uses UCB1, mean backup, and Monte Carlo rollout. Its depth bound shrinks near the episode cap, action masks remove wall bumps and invalid checks/samples, and subtrees are not promoted because their accumulated values use shorter horizons. These three departures depend on observable quantities. The default belief is the exact factored posterior. A literal unweighted rejection-sampling particle filter gives 12.93 ± 0.25 versus 12.81 ± 0.43 ($p=0.81$), so this comparison detects no belief-mode effect.

Table 17. RockSample extended results with the full agent roster, including POMCP at its tuned frozen configuration, POMCP’s rollout policy run standalone (Rollout only), and Planning+IG at both $w=5$ and $w=10$. Steps and checks are episode means. RS[5,3]/RS[7,4]: 500 episodes $\times$ 10 seeds. RS[7,8]/RS[11,11]: 100 episodes $\times$ 5 seeds.

Instance	Agent	Steps	Checks	Good	Bad	Reward
RS[5,3]	Greedy	12.00	0.00	1.49	1.51	$+3.71 \pm 0.26$
	POMCP (2048 sims)	7.47	2.97	0.68	0.03	$+12.74 \pm 0.11$
	Planning (d=3)	10.38	3.71	0.99	0.04	$+14.28 \pm 0.13$
	Plan+IG w=5 (d=3)	15.81	5.36	1.44	0.01	$+16.41 \pm 0.13$
	Plan+IG w=10 (d=3)	20.35	6.63	1.48	0.00	$+14.60 \pm 0.11$
	EFE w=1 (d=3)	15.26	4.79	1.44	0.03	$+16.47 \pm 0.12$
	Rollout only (approach)	13.48	3.00	1.41	0.07	$+16.67 \pm 0.13$
RS[7,4]	Greedy	22.00	0.00	1.98	2.02	-1.38 ± 0.36
	POMCP (2048 sims)	8.67	3.46	0.66	0.03	$+12.01 \pm 0.06$
	Planning (d=4)	10.00	2.00	0.95	0.05	$+13.97 \pm 0.11$
	Plan+IG w=5 (d=4)	24.55	8.15	1.69	0.01	$+14.56 \pm 0.14$
	Plan+IG w=10 (d=4)	26.82	8.80	1.78	0.01	$+14.27 \pm 0.15$
	EFE w=1 (d=4)	16.81	5.36	1.44	0.01	$+15.96 \pm 0.15$
	Rollout only (approach)	23.98	4.00	1.88	0.10	$+15.85 \pm 0.17$
RS[7,8]	Greedy	32.00	0.00	4.02	3.98	-5.60 ± 1.03
	POMCP (2048 sims)	22.18	11.77	1.71	0.10	$+15.07 \pm 0.45$
	Planning (d=3)	4.51	1.00	0.48	0.03	$+12.31 \pm 0.35$
	Plan+IG w=5 (d=3)	40.27	14.19	3.40	0.01	$+23.76 \pm 0.58$
	Plan+IG w=10 (d=3)	57.47	18.26	3.94	0.01	$+20.55 \pm 0.29$
	EFE w=1 (d=3)	27.26	9.84	2.61	0.06	$+21.85 \pm 0.79$
	Rollout only (approach)	36.02	8.00	3.83	0.19	$+28.33 \pm 0.82$
RS[11,11]	Greedy	57.00	0.00	5.48	5.52	-18.90 ± 1.04
	POMCP (2048 sims)	3.28	1.60	0.49	0.03	$+12.94 \pm 0.24$
	Planning (d=2)	2.50	1.00	0.48	0.03	$+13.23 \pm 0.18$
	Plan+IG w=5 (d=2)	3.09	1.56	0.50	0.03	$+13.18 \pm 0.15$
	Plan+IG w=10 (d=2)	3.09	1.56	0.50	0.03	$+13.18 \pm 0.15$
	EFE w=1 (d=2)	2.50	1.00	0.48	0.03	$+13.23 \pm 0.18$
	Rollout only (approach)	62.44	11.00	5.22	0.23	$+28.66 \pm 0.83$

Replacing the exit-valued depth-cap leaf by a greedy-belief leaf gives 14.52 ± 0.34 , versus the frozen 12.81 ± 0.43 and standalone rollout’s 17.16 ± 0.44 . Neither comparison survives this sensitivity battery’s per-metric Holm correction (uncorrected $p=0.015$ and 0.002).

Three disjoint tuning seeds select rollout, exploration constant, horizon, and root criterion through a two-stage coordinate grid. The configuration is frozen before evaluation. The evaluation budget of 2,048 simulations is twice the tuning budget of 1,024 and is not itself tuned. At the full protocol on RS[5,3], frozen POMCP earns 12.74 ± 0.11 against EFE’s 16.47 ± 0.12 , at 46.6 versus 1.76 ms per decision. Its standalone approach-then-check rollout earns 16.67 ± 0.13 at 0.005 ms, significantly above the search ($p=1.9 \times 10^{-14}$, surviving Holm). The position-independent 9.50 exit option makes early exit attractive under mean backup. The search exits after 7.5 steps, while its rollout follows a 13.5-step collection tour.

Budget-horizon interaction. A post-hoc RS[5,3] sweep on the canonical evaluation seeds finds frozen $H=10$ worst at both tested simulation tiers. At 16,384 simulations its reward is 12.52 ± 0.26 , versus 15.60 ± 0.37 , 15.24 ± 0.39 , and 15.62 ± 0.36 for $H=15, 25, 40$. Increasing simulations beyond 1,024 reduces reward at $H=10$ (from 13.19 to 12.52) but helps at $H=15$ (from 14.00 to 15.60). Tuning horizon at a single budget missed this interaction. The diagnostic does not re-select the frozen configuration.

On RS[11,11], the corresponding post-hoc 2,048-simulation sweep gives the opposite result. Rewards at $H=10, 15, 25, 40$ are 12.94 ± 0.24 , -7.54 ± 1.20 , -34.65 ± 1.12 , and -54.78 ± 0.86 . Every decline survives Holm (seed-level Welch $p \leq 4.3 \times 10^{-5}$). The endpoints expand from 3.3 steps and 1.6 checks to 196 steps and 144 checks. The deepest tree collects only 3.4 good rocks against the rollout's 5.2, whose checks number 11. Action costs overwhelm the collection return. The 16,384-simulation tier was not run on this instance because of its per-step cost.

Scope of the comparison. Under the RS[5,3] diagnostic's matched five-seed, 100-episode protocol, EFE earns 16.90 ± 0.45 at 1.7 ms per decision, against the best measured POMCP's roughly 15.6 at 530 ms. None of the reward differences survives Holm over nine diagnostic rows ($p=0.058$ against $H=15, 40$ and 0.024 against $H=25$ uncorrected). POMCP is also only 1.5–1.9 reward below its standalone rollout at 16,384 simulations and $H \geq 15$ (uncorrected $p=0.012$ –0.028, none surviving the seed-level 36-comparison reward family). Its $H=25$ deficit survives only the pooled-level correction. Thus the apparent search gap on this instance depends substantially on computation and tuning.

On RS[11,11], the best measured alternative, 4,096 simulations at $H=10$, earns 13.14 ± 0.15 , within sampling error of the frozen setting but 15.5 below its rollout's 28.66 ± 0.83 ($p=3.2 \times 10^{-5}$, surviving Holm at both levels). These diagnostics limit conclusions about POMCP generally. They do not alter the predeclared table or show information weighting superior to the standalone heuristic.

N IDS Baseline (Observe-then-Commit)

Our adaptation of information-directed sampling (Russo and Van Roy 2014) minimizes Δ^2/I over individual observation actions. Here Δ is the opportunity cost of observing rather than committing and I measures information about the optimal commit. Unlike IDS over randomized action distributions, this deterministic argmin lacks the general role for randomization established by Russo and Van Roy (2018). It also substitutes state-entropy gain when information about the optimal commit vanishes. The comparison therefore concerns this adaptation, not IDS in full generality.

IDS exactly ties EFE and Planning on Tiger. On Diagnosis and Bandit it attains near-ceiling success with more observations and lower reward than EFE (seed-level Welch reward $p=0.0009$ and 1.5×10^{-6}). It resembles success-tuned Planning+IG on Diagnosis and has lower mean reward on Bandit. These results favor $w=1$ for reward on the two tested multi-observation instances under the stated units, without establishing a general advantage over information-ratio methods.

Table 18. Observe-then-commit IDS and matched controls, 500 episodes per canonical seed over five seeds. Reward is mean $\pm$ seed-level SE. Planning+IG uses the own-class success-tuned weights in results_supplementary_tuned_weights.csv.

Environment	Agent	Obs.	Success	Reward
Tiger	Myopic	1.00	84.6%	-7.90 ± 0.67
	Planning	4.23	99.6%	$+5.33 \pm 0.26$
	Planning+IG (w=50)	5.66	99.9%	$+4.25 \pm 0.08$
	EFE	4.23	99.6%	$+5.33 \pm 0.26$
	IDS	4.23	99.6%	$+5.33 \pm 0.26$
Diagnosis	Myopic	2.00	64.5%	-13.29 ± 0.88
	Planning	5.85	88.8%	-2.59 ± 0.37
	Planning+IG (w=50)	13.17	99.2%	-3.68 ± 0.14
	EFE	9.68	97.4%	-1.26 ± 0.34
	IDS	13.17	99.2%	-3.63 ± 0.16
Bandit	Myopic	2.05	61.6%	$+5.52 \pm 0.13$
	Planning	3.24	69.9%	$+5.67 \pm 0.13$
	Planning+IG (w=20)	9.72	99.0%	$+5.05 \pm 0.07$
	EFE	5.16	86.9%	$+6.24 \pm 0.09$
	IDS	10.62	99.5%	$+4.65 \pm 0.07$

O Proper-Scoring Calibration

To assess terminal beliefs separately from decisions, we use the proper log score $\ln b(s^*)$ and Brier score $\sum_s (b(s) - 1[s = s^*])^2$, where s^* is the true state (Gneiting and Raftery 2007). Log-score probabilities are floored at 10^{-12} . Inspection scores average over binary component beliefs, not the joint state. These rules measure overall belief quality, combining calibration and informativeness: a uniform prior and a posterior after perfect observation can both be calibrated, despite different expected scores.

Table 19 compares Planning, EFE, and a reference over five seeds, with 500 episodes per seed except 200 for Inspection-N16. The reference on Tiger, Diagnosis, and Bandit is SARSOP’s in-simulator alpha-vector policy (Kurniawati et al. 2008). Our SARSOP build does not scale practically to Inspection, whose reference is Planning+IG at the atlas’s higher canonical budget (Appendix T.3).

Planning (reward-only) is dominated on both terminal-belief scores by every epistemic-aware agent on every instance except Tiger, where its single observation action already suffices and all three agents converge to the same policy and hence the same terminal belief. On Diagnosis, Bandit, and both Inspection instances, EFE’s terminal belief scores substantially better under both rules than reward-only Planning’s, and on the three OTC environments EFE (w=1) lies within overlapping SE bands of SARSOP’s terminal-belief scores. This is evidence that the epistemic term produces better-informed terminal beliefs, not only better terminal actions. The two need not coincide, since an agent can commit correctly from a confidently wrong belief on environments with few reachable outcomes.

Table 19. Proper-scoring calibration of the terminal belief against the true hidden state (log score in nats and Brier score, mean $\pm$ SE over 5 seeds). Reference is SARSOP for the three OTC environments and Planning+IG at the atlas’s higher canonical budget (Appendix T.3) for Structural Inspection, where the near-optimal solver used here does not scale to the instance’s state space. For Inspection, scores are averaged over binary component beliefs. Log-score probabilities are floored at 10^{-12} .

Instance	Agent	Log score	Brier score
Tiger	Planning (reward-only)	-0.035 ± 0.006	0.011 ± 0.002
Tiger	EFE ($w=1$)	-0.035 ± 0.006	0.011 ± 0.002
Tiger	SARSOP	-0.035 ± 0.006	0.011 ± 0.002
Diagnosis	Planning (reward-only)	-0.433 ± 0.018	0.199 ± 0.011
Diagnosis	EFE ($w=1$)	-0.134 ± 0.004	0.049 ± 0.002
Diagnosis	SARSOP	-0.154 ± 0.021	0.058 ± 0.010
Bandit	Planning (reward-only)	-0.901 ± 0.033	0.463 ± 0.019
Bandit	EFE ($w=1$)	-0.525 ± 0.027	0.240 ± 0.016
Bandit	SARSOP	-0.510 ± 0.034	0.231 ± 0.018
Inspection-N8	Planning (reward-only)	-0.417 ± 0.004	0.286 ± 0.003
Inspection-N8	EFE ($w=1$)	-0.245 ± 0.004	0.144 ± 0.003
Inspection-N8	Planning+IG ($w=100$)	-0.008 ± 0.001	0.002 ± 0.000
Inspection-N16	Planning (reward-only)	-0.382 ± 0.005	0.251 ± 0.003
Inspection-N16	EFE ($w=1$)	-0.257 ± 0.003	0.150 ± 0.002
Inspection-N16	Planning+IG ($w=100$)	-0.013 ± 0.003	0.004 ± 0.001

P Per-Test Value-of-Information Audit Trail

The optional audit record exposes each root action’s sensing cost, expected information gain in bits, weighted gain w IG, expected task value, total score, and selection status. Task value includes sensing cost and continuation value, including deeper information bonuses. The record is shared by PlanningInfoGainAgent, EFEAgent ($w=1$), and InspectionTreeSearchAgent. It reads already-computed quantities without changing action selection. Unit tests check that components sum to the total and that the chosen action ranks first.

At the Inspection-N8 decision in Table 20 ($w=5$), `test1` has about five times `test0`’s information gain, but its higher cost is not offset by task value, so the agent chooses `test0`. The record makes this distinction inspectable without changing the reported results.

Table 20. Per-action value-of-information audit for one representative decision (Inspection-N8, info-gain weight $w = 5$, step 2 of the episode seeded at 42). Total score is the maximized planning objective (Section 3). Task value is net of sensing cost, so Total score = Task value + $w \cdot \text{IG}$, with Cost shown for reference. The chosen action need not have the largest raw information gain once sensing cost and downstream task value are accounted for.

Action	Cost	IG (bits)	$w \cdot \text{IG}$	Task value	Total score
test0	0.50	0.021	0.106	-26.29	-26.18
test1	2.00	0.106	0.528	-26.87	-26.34
diagnose(nominal)	0.00	0.000	0.000	-26.86	-26.86
diagnose(faulty)	0.00	0.000	0.000	-31.05	-31.05

Q Reward-Rescaling Invariance

Multiplying all rewards and costs by k preserves a policy when its information weight is also multiplied by k (Proposition 4.2). A finite-grid reward sweep on Diagnosis and Bandit uses $k \in \{0.1, 1, 10\}$ and 200 episodes over each of three seeds per (k, w) pair. Table 21 records the selected weights, with ties resolved toward the smallest maximizer. The normalized ratio is constant on Bandit and varies twofold on Diagnosis’s flat plateaus. This is evidence about a finite grid, not a precise stability estimate. The full curves remain in `results_reward_scaling.csv` and `figures/fig_reward_scaling.pdf`. Section 6.1 supplies the stronger usage-curve collapse check. Fixed $w=1$ is not scale invariant.

Table 21. Sampled reward-maximizing Planning+IG weights under global reward scale k . Flat plateaus can contain several grid maximizers.

Environment	k	w_{ret}^*	w_{ret}^*/k
Diagnosis	0.1	0.1	1.0
Diagnosis	1.0	0.5	0.5
Diagnosis	10.0	5.0	0.5
Bandit	0.1	0.1	1.0
Bandit	1.0	1.0	1.0
Bandit	10.0	10.0	1.0

R Information Base and the Nat-Canonical Agent

Bit-unit $w=1$ equals $1/\ln 2 \approx 1.44$ reward units per nat. The exactly nat-canonical policy instead uses bit-unit $w=\ln 2 \approx 0.69$ (Section 3.3). A direct comparison under the Pareto sweep protocol (500 episodes per seed over five seeds, `results_nat_canonical_check.csv`) gives bit-identical reward, success, and usage on Tiger, Diagnosis, Bandit, and Testbed. Their respective reward/success pairs are 5.33/99.6%, $-1.26/97.4\%$, 6.24/86.9%, and 0.37/96.4%. This direct check matters on Bandit, where the nearest lower swept weight, 0.5, scores 6.05 rather than 6.24. Both conventions remain below Testbed’s sampled maximum 0.48.

Tileworld changes policy. Nat-canonical EFE gives -20.81 reward, 74.7% success, and 15.64 scans, versus -21.53 , 72.0%, and 14.75 at bit-unit $w=1$. Only usage differs significantly (seed-level Welch $p=9.5 \times 10^{-5}$, surviving Holm over three metrics), with reward $p=0.34$ and success $p=0.070$. Both policies remain Pareto-dominated by sampled $w=20$, whose reward is -18.95 and success 91.8%. The unit check therefore preserves the paper’s main positive and negative conclusions without implying policy invariance to information units.

S POMCP Baseline Comparison

Our observe-then-commit POMCP adaptation (Silver and Veness 2010) has two fidelity gaps. The depth-cap leaf uses the best commit reward for the sampled state, an optimistic clairvoyant value. Rollout commits instead receive their belief-expected reward, replacing a sampled payoff by its conditional expectation. Their separate policy effects are not isolated. Rollouts start from the simulated path’s updated belief, choose tests uniformly, and commit to the belief-optimal action with probability 0.3 per step. Regression tests cover observation costs and path-belief initialization. The separate RockSample implementation has neither fidelity gap (Appendix M).

The core battery uses UCB1 constant 10, fixed without environment-specific tuning, depth cap $H+3$, and simulation counts $\{500, 1000, 2000, 5000\}$. Each outer seed initializes both environment and planner randomness. Because reward scales vary, a fixed exploration constant has different effective strength across tasks. Simulation budgets do not match computation to exact EFE or Planning. Nor does this comparison isolate directed observation choice, since exact EFE also changes tree enumeration, leaf values, and commit values.

Table 22. POMCP at 1,000 simulations per decision. Five canonical seeds yield 5,000 episodes on each core environment and 1,000 on Tileworld. EFE/Planning are rerun inside this harness and reproduce Table 3 on the core tasks. Reward uncertainty is seed-level SE. Bold identifies EFE and its exact Tiger tie with Planning, not column winners.

Environment	Agent	Obs.	Success	Reward
Tiger	Planning	4.20	99.4%	+5.19 $\pm$ 0.21
	POMCP (1000)	1.64	89.2%	−3.57 $\pm$ 0.18
	EFE	4.20	99.4%	+5.19 $\pm$ 0.21
Diagnosis	Planning	5.86	88.8%	−2.57 $\pm$ 0.20
	POMCP (1000)	3.19	70.5%	−10.91 $\pm$ 0.63
	EFE	9.66	97.2%	−1.37 $\pm$ 0.18
Bandit	Planning	3.29	71.0%	+5.75 $\pm$ 0.08
	POMCP (1000)	1.34	47.2%	+4.58 $\pm$ 0.09
	EFE	5.11	86.9%	+6.27 $\pm$ 0.04
Tileworld 6×6	Planning	15.57	73.6%	−21.41 $\pm$ 0.64
	POMCP (1000)	0.05	2.6%	−48.49 $\pm$ 0.28
	EFE	14.76	71.9%	−21.62 $\pm$ 0.87

At 5,000 simulations, POMCP still reaches only $89.5 \pm 0.3\%$ success on Tiger, $71.6 \pm 0.9\%$ on Diagnosis, and $62.1 \pm 1.2\%$ on Bandit. No Tileworld budget exceeds 3.8%. Bandit’s scaling is non-monotone, with success $50.9 \pm 1.1\%$, $47.2 \pm 1.0\%$, and $51.6 \pm 1.1\%$ at 500, 1,000, and 2,000 simulations. More computation at this fixed constant therefore does not close the gaps, but this does not bound a better-tuned POMCP.

Matched computation and ablation. Appendix Y defines MCTS-EFE and reports its simulation-matched comparison. An uncontended same-machine control times MCTS-EFE(500) over five seeds and 200 episodes per seed at 13.87s, with $97.7 \pm 0.5\%$ success. Calibration at 200/500/1,000 simulations selects POMCP(661), taking 13.88s and attaining $88.0 \pm 0.6\%$ success and $−4.80 \pm 0.69$ reward. POMCP(500) takes 10.42s and gives $88.1 \pm 0.2\%$ and $−4.67 \pm 0.19$. Seed-level Welch differences between those POMCP configurations are not significant (success $p=0.89$, reward $p=0.85$), while both trail MCTS-EFE on both metrics ($p < 0.0001$). This is evidence at the tested constant and protocol. Exact EFE’s own matched-battery reference is 99.9% at $H=6$ and 64 ms per episode. MCTS-EFE’s measured Tiger success and usage remain 97.7% and 3.735 across horizons 6, 8, 10 and simulation counts 200, 500, 1000.

Table 23. MCTS-EFE component ablation under the MCTS battery’s protocol (200 episodes per seed $\times$ 5 seeds, at the same horizons and simulation budgets as the MCTS-EFE runs of `results_mcts_efe.csv`). The first row of each block, max-backup with the in-tree information gain, is the manuscript’s MCTS-EFE. Uncertainty is seed-level SE. A dagger marks a variant whose success or reward differs from that row at seed-level Welch $p < 0.05$ after Holm–Bonferroni within metric over the whole battery. No variant carries one, so no component shows a detectable effect on success or reward on any environment tested. See `results_mcts_efe_ablation.csv` and its `_stats` companion.

Environment	Backup	In-tree IG	Success	Reward	Obs.
Tiger	max	yes	$97.7 \pm 0.5\%$	$+3.73 \pm 0.53$	3.73
	mean	yes	$97.9 \pm 0.3\%$	$+4.01 \pm 0.33$	3.68
	max	no	$97.4 \pm 0.7\%$	$+3.66 \pm 0.75$	3.48
	mean	no	$97.7 \pm 0.5\%$	$+3.95 \pm 0.60$	3.52
Diagnosis ($N=4$)	max	yes	$94.2 \pm 0.5\%$	-3.26 ± 0.30	9.78
	mean	yes	$95.8 \pm 0.5\%$	-2.55 ± 0.38	10.03
	max	no	$96.3 \pm 0.6\%$	-2.49 ± 0.53	10.27
	mean	no	$95.7 \pm 0.8\%$	-2.97 ± 0.40	10.39
Tileworld-6 $\times$ 6	max	yes	$95.4 \pm 0.6\%$	-25.02 ± 0.25	32.26
	mean	yes	$96.2 \pm 0.5\%$	-24.35 ± 0.39	32.07
	max	no	$94.7 \pm 1.0\%$	-26.34 ± 0.57	33.16
	mean	no	$95.7 \pm 0.3\%$	-25.29 ± 0.22	32.71

Table 23 switches MCTS-EFE’s in-tree information gain and max-backup independently. Neither changes success or reward detectably after Holm correction within metric across the three environments. Only Tiger usage changes significantly, from 3.73 to 3.48 without in-tree information gain ($p=0.0014$). Five seeds give limited sensitivity. The fixed EFE-greedy leaf, exact beliefs, and exact commit values remain candidate explanations, so the ablation cannot attribute the advantage to information gain throughout the tree.

Exploration and rollout sweep. Table 24 sweeps constants $c \in \{1, 2, 5, 10, 20, 50, R\}$ on Tiger, Diagnosis, and Tileworld, where R is the commit reward range and MCTS-EFE’s default. This c is unrelated to sensing cost. Bandit is not swept. The informed rollout selects the largest exact one-step information gain. Outside the sweep, POMCP uses $c=10$ in Table 22 and $c=5$ in the MCTS battery. Planner randomness here follows the outer seed, whereas the earlier MCTS battery fixes the planner seed, so their POMCP rows need not coincide.

The selection rule maximizes success and then reward on seeds $\{11, 22, 33\}$ with 100 episodes each. It was recorded before selection, but both rule and grid followed inspection of the canonical-seed exploratory sweep. It selects MCTS-EFE $c=5$ on Tiger/Diagnosis and its default on Tileworld. POMCP selects $c=50$ with uniform rollouts on Tiger, $c=5$ with uniform rollouts on Diagnosis, and $c=5$ with informed rollouts on Tileworld (`select_pomcp_constants.py`). On the canonical evaluation seeds, selected MCTS-EFE retains higher Tiger success ($98.2 \pm 0.3\%$ vs. $93.2 \pm 0.5\%$, $p=0.0001$) and reward (4.33 ± 0.28 vs. 0.39 ± 0.52 , $p=0.0005$). Both survive Holm over the six selected success/reward comparisons. The selected Diagnosis and Tileworld comparisons also survive that correction. Their POMCP success rates are $65.6 \pm 1.4\%$ and $8.2 \pm 0.6\%$, against MCTS-EFE’s $94.7 \pm 0.4\%$ and $95.4 \pm 0.6\%$.

Several qualifications matter. Diagnosis’s evaluation-seed-best POMCP row ($c=2$, $69.1 \pm 1.7\%$) is not the tuning selection. Tileworld’s best uniform constant is the grid’s lower edge $c=1$, and lower constants were not tested. MCTS-EFE’s default is not reward-optimal either, with Tileworld reward improving from -25.02 ± 0.25 to -22.26 ± 0.46 at $c=5$ ($p=0.0017$). Informed POMCP rollouts leave Tiger unchanged because it has one test. At $c=5$

Table 24. POMCP exploration-constant sweep and informed rollouts under the MCTS battery’s protocol (200 episodes per seed $\times$ 5 seeds, at the same horizons and simulation budgets as the MCTS-EFE runs of results_mcts_efe.csv). R is the commit reward range (110 on Tiger, 60 elsewhere), MCTS-EFE’s default constant. The info-gain rollout takes the observation with the largest exact one-step information gain at every rollout step. POMCP’s planner stream is seeded per run from the outer seed here, which the MCTS battery left at a fixed constant, so these POMCP rows are not bit-identical to that battery’s. Uncertainty is seed-level SE. A dagger marks a row whose success or reward differs from MCTS-EFE at its default constant at seed-level Welch $p < 0.05$ after Holm–Bonferroni within metric over the whole battery. See results_pomcp_exploration_sweep.csv and its _stats companion.

Environment	Solver	c	Rollout	Success	Reward
Tiger	POMCP	1	uniform [†]	$87.3 \pm 0.4\%$	-5.51 ± 0.39
	POMCP	2	uniform [†]	$88.1 \pm 0.7\%$	-4.67 ± 0.72
	POMCP	5	uniform [†]	$89.0 \pm 0.3\%$	-3.74 ± 0.29
	POMCP	10	uniform [†]	$88.4 \pm 0.8\%$	-4.40 ± 0.87
	POMCP	20	uniform [†]	$91.5 \pm 0.5\%$	-1.24 ± 0.48
	POMCP	50	uniform [†]	$93.2 \pm 0.5\%$	$+0.39 \pm 0.52$
	POMCP	R	uniform [†]	$87.6 \pm 0.4\%$	-5.01 ± 0.39
	POMCP	5	info-gain [†]	$89.0 \pm 0.3\%$	-3.74 ± 0.29
	POMCP	R	info-gain [†]	$87.6 \pm 0.4\%$	-5.01 ± 0.39
	MCTS-EFE	R	EFE	$97.7 \pm 0.5\%$	$+3.73 \pm 0.53$
	MCTS-EFE	5	EFE	$98.2 \pm 0.3\%$	$+4.33 \pm 0.28$
Diagnosis ($N=4$)	POMCP	1	uniform [†]	$64.0 \pm 1.3\%$	-14.57 ± 0.75
	POMCP	2	uniform [†]	$69.1 \pm 1.7\%$	-11.59 ± 0.99
	POMCP	5	uniform [†]	$65.6 \pm 1.4\%$	-13.56 ± 0.85
	POMCP	10	uniform [†]	$67.4 \pm 1.0\%$	-12.59 ± 0.62
	POMCP	20	uniform [†]	$59.9 \pm 1.1\%$	-16.31 ± 0.67
	POMCP	50	uniform [†]	$50.7 \pm 1.2\%$	-21.30 ± 0.68
	POMCP	R	uniform [†]	$50.3 \pm 0.7\%$	-21.40 ± 0.44
	POMCP	5	info-gain [†]	$66.8 \pm 1.5\%$	-13.09 ± 0.87
	POMCP	R	info-gain [†]	$53.3 \pm 1.6\%$	-19.76 ± 0.98
	MCTS-EFE	R	EFE	$94.2 \pm 0.5\%$	-3.26 ± 0.30
	MCTS-EFE	5	EFE	$94.7 \pm 0.4\%$	-2.75 ± 0.31
Tileworld-6 $\times$ 6	POMCP	1	uniform [†]	$7.1 \pm 1.0\%$	-46.78 ± 0.59
	POMCP	2	uniform [†]	$3.0 \pm 0.8\%$	-48.61 ± 0.48
	POMCP	5	uniform [†]	$2.6 \pm 0.5\%$	-48.53 ± 0.31
	POMCP	10	uniform [†]	$3.1 \pm 0.3\%$	-48.16 ± 0.18
	POMCP	20	uniform [†]	$2.7 \pm 0.6\%$	-48.40 ± 0.39
	POMCP	50	uniform [†]	$3.1 \pm 0.5\%$	-48.15 ± 0.29
	POMCP	R	uniform [†]	$1.8 \pm 0.4\%$	-48.93 ± 0.24
	POMCP	5	info-gain [†]	$8.2 \pm 0.6\%$	-46.92 ± 0.38
	POMCP	R	info-gain [†]	$2.1 \pm 0.3\%$	-48.85 ± 0.18
	MCTS-EFE	R	EFE	$95.4 \pm 0.6\%$	-25.02 ± 0.25
	MCTS-EFE	5	EFE [†]	$94.9 \pm 0.7\%$	-22.26 ± 0.46

they give no significant Diagnosis success gain ($p=0.57$), and improve Tileworld success but not reward after Holm (uncorrected $p=0.0002$ and 0.012). None of these controls isolates all architectural differences.

Table 25. Atlas of operational sensing budgets across benchmark instances, giving the usage range of the Planning+IG family, the implicit EFE budget $B_{\text{EFE}} = U(w=1)$ (mean $\pm$ SE over seeds), and crossing brackets estimating $w^*(B)$ at two canonical budgets per instance. Each bracket estimates a single crossing threshold from the finite weight grid (Definition 4.1). No closed-form meta-model is implied.

Instance	$[U_{\min}, U_{\max}]$	B_{EFE}	$B_1: w^* \text{ bracket}$	$B_2: w^* \text{ bracket}$
Bandit	[3.23, 12.34]	5.11 ± 0.11	4.6: (0.0611, 0.139]	11.0: (19.3, 43.9]
Diagnosis	[5.90, 13.16]	9.77 ± 0.14	7.0: (0.139, 0.316]	12.1: (19.3, 43.9]
Inspection-N16	[22.38, 58.56]	28.02 ± 0.13	27.8: (0.316, 1.33]	53.1: (23.7, 100]
Inspection-N8	[12.70, 29.49]	15.56 ± 0.09	15.2: (0.316, 2.15]	27.0: (14.7, 100]
RS[5,3]	[3.00, 8.83]	4.79 ± 0.05	3.9: (0.316, 1.33]	8.0: (5.62, 23.7]
RS[7,4]	[2.00, 14.54]	5.37 ± 0.07	3.9: (0.316, 2.15]	12.7: (14.7, 100]
Tiger	[4.37, 5.77]	4.37 ± 0.11	4.6: (19.3, 43.9]	5.6: (19.3, 43.9]
Tileworld-6x6	[14.74, 32.73]	14.85 ± 0.13	17.4: (5.62, 23.7]	30.0: (23.7, 100]

T Budgeted ρ -POMDP Supplementary Details

T.1 SARSOP Build and Export Details

The reference uses the original APPL C++ solver `pomdpso1`, built by [tools/build_sarsop.sh](#) with four documented Apple-Silicon/clang compatibility patches. The binary is not committed. [experiments/run_sarsop_baseline.py](#) exports each discrete observe-then-commit benchmark from the agents' own state, action and observation definitions, adding an absorbing done state and discount $\gamma = 0.999$. It solves to precision 10^{-3} , parses the XML alpha vectors and evaluates their policy through the shared episode runner. [tests/test_sarsop_export.py](#) checks the exporter, parser and argmax agent without the solver binary.

T.2 Distractor-Diagnosis Environment and Its Factorization Guarantee

`DistractorDiagnosisEnv` has a four-way condition x and binary nuisance z . Each test observes one factor, terminal rewards depend only on x , and rewards are not observations. For a factored belief $b(x, z) = b_X(x)b_Z(z)$ and a nuisance likelihood $L(o | z)$, Bayes' rule gives

$$b'(x, z | o) = b_X(x) \frac{L(o | z)b_Z(z)}{\sum_{\tilde{z}} L(o | \tilde{z})b_Z(\tilde{z})}.$$

Thus the condition marginal is unchanged. Condition tests preserve the nuisance marginal by the same argument. The independent generative process preserves this factorization, but correlated beliefs, including those caused by misspecification, are outside the guarantee. [tests/test_distractor_diagnosis.py](#) checks both marginal identities numerically.

T.3 The w^* Atlas

Table 25 reports eight measured usage ranges, their implicit EFE budgets $B_{\text{EFE}} = U(1)$, and crossing brackets at two fixed relative positions within each range (Definition 4.1). [experiments/run_w_atlas.py](#) produces it from `results_price_usage_curves.csv` and `results_price_interleaved_curves.csv`. The table compares operating points, not a closed-form relationship between budget and environment parameters.

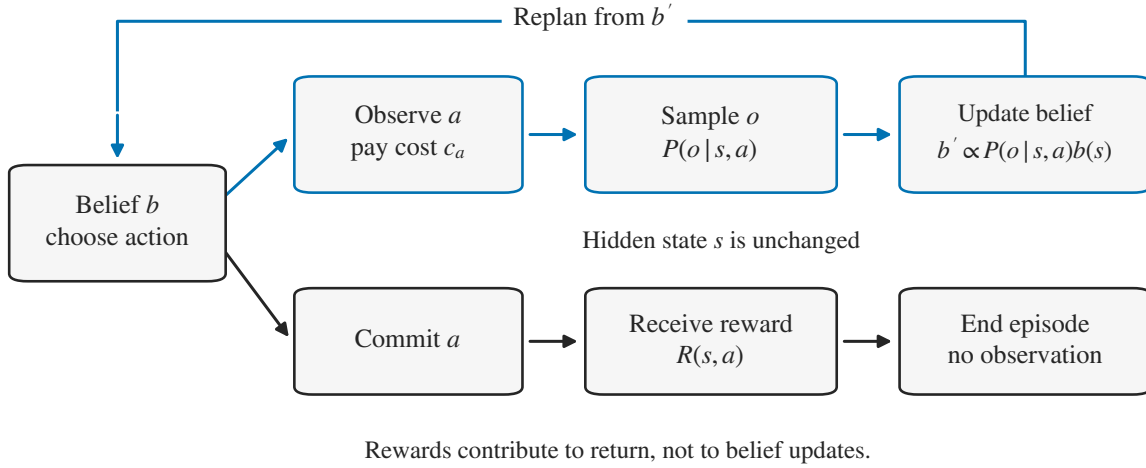


Fig. 13. The observe-then-commit decision loop, shown schematically. An observation incurs a cost and updates the belief about a hidden state that remains fixed. The agent then chooses again from the updated belief. A commit instead pays a state-dependent reward and ends the episode without emitting an observation. Rewards are not inputs to the belief update. The loop describes the episode, while the planner's finite horizon limits how far it looks ahead at each decision.

U Methodological and Budget-Theory Details

U.1 Observe-then-Commit Structure and EFE Scope

Equation 3 uses recursive closed-loop EFE associated with sophisticated inference (Da Costa, Sajid, et al. 2023; K. Friston, Da Costa, et al. 2021), specifically the information-gain and pragmatic-value formulation related by Champion et al. (2026). We do not establish its $w = 1$ coefficient for Free Energy of the Expected Future (Millidge, Tschantz, and Buckley 2021), generalised free energy (Parr and K. J. Friston 2019), epistemic-prior variational inference (de Vries et al. 2025), or the Bethe-Lagrangian formulation (Kouw 2026).

The reward-information qualification is given in Section 3.3. Log scoring gives the exact normalized case ($Z = 1$). Otherwise Proposition 3.1 concerns the reward-based recursion conditional on its pragmatic identification, since $-T \ln Z$ can depend on policy when episode lengths differ. The implementation separately adopts one reward unit per bit, and Appendix R tests the nat-canonical weight $w = \ln 2$. Proposition 4.2 handles positive reward-and-cost rescaling. None of these claims makes $w = 1$ universally reward-optimal.

U.2 Two-State Thresholds and Their Empirical Scope

PROPOSITION U.1. Consider a two-state observe-then-commit ρ -POMDP with uniform prior, one observation action of accuracy $\frac{1}{2} < p \leq 1$ and cost $c > 0$, and two commit actions paying $R^+ > 0$ when correct and $R^- < 0$ otherwise, with $|R^-| > R^+$. Write $\alpha = |R^-|/R^+$, $\eta = I_{\max}/c$ and

$$I_{\max} = \ln 2 - H_{\text{post}}(p), \quad H_{\text{post}}(p) = -p \ln p - (1 - p) \ln(1 - p).$$

Information is measured in nats. Part (a) concerns one observation followed by commitment ($H = 1$), and Part (b) allows a second observation ($H = 2$) and requires $p < 1$.

Part (a) ($H=1$ lower threshold). The minimum weight w_{thresh}^* above which the Planning+IG objective, expected reward plus w times expected information gain, prefers observing once to committing immediately is

$$w_{\text{thresh}}^* = \frac{c - (p - \frac{1}{2})(R^+ - R^-)}{I_{\text{max}}} = \frac{c - (p - \frac{1}{2})(1 + \alpha)R^+}{I_{\text{max}}}$$

For any $w > w_{\text{thresh}}^*$ the agent observes before committing at $H=1$. The threshold w_{thresh}^* is negative, making $w=1$ sufficient for at least one observation, whenever $\alpha > c/[(p - \frac{1}{2})R^+] - 1$. In particular, $w_{\text{thresh}}^* \rightarrow -\infty$ as $\alpha \rightarrow \infty$ for any fixed $p > \frac{1}{2}$ and $c > 0$.

Part (b) ($H=2$ near-optimality interval). At the post-first-observation belief $b=(p, 1-p)$, define the instrumental value of a second observation $\text{Vol}_2(b)$ and the expected information gain $I(b)$ of one further observation. The upper (over-observation) threshold is

$$w_{\text{over}}^* = \begin{cases} (c - \text{Vol}_2(b))/I(b) & \text{if } \text{Vol}_2(b) < c, \\ +\infty & \text{otherwise.} \end{cases}$$

In this symmetric model $\text{Vol}_2(b) = 0$, because a disconfirming second observation returns the posterior to the tie point. Thus the finite branch applies. If $w_{\text{thresh}}^* \leq 0$, every $w \in (w_{\text{thresh}}^*, w_{\text{over}}^*) \cap [0, \infty)$ reproduces the reward-optimal single observation. We call this the near-optimality interval. Weights above w_{over}^* buy an instrumentally unprofitable second observation. If $w_{\text{thresh}}^* > 0$, immediate commitment is reward-optimal. Weights inside the interval instead buy one observation at reward loss $c - (p - \frac{1}{2})(R^+ - R^-)$, strictly between zero and c . These statements do not extend to longer horizons. Part (a) includes perfect sensing under $0 \ln 0 = 0$, giving $I_{\text{max}} = \ln 2$. Part (b) excludes it because $I(b) = 0$ makes the upper ratio undefined.

PROOF. For Part (a), immediate commitment has value $(R^+ + R^-)/2$, while observing once and committing has value $-c + wI_{\text{max}} + pR^+ + (1-p)R^-$. Their difference is

$$(p - \frac{1}{2})(R^+ - R^-) - c + wI_{\text{max}}.$$

Solving the strict positivity condition and substituting $R^- = -\alpha R^+$ gives the stated threshold and asymmetry condition.

For Part (b), the second observation's advantage at $b = (p, 1-p)$ is $-c + wI(b) + \text{Vol}_2(b)$. A confirming observation has probability $q = p^2 + (1-p)^2$ and posterior maximum p^2/q . A disconfirming observation has probability $2p(1-p)$ and posterior maximum $1/2$. Their expected maximum is $p^2 + p(1-p) = p$, so the expected best commit reward is unchanged and $\text{Vol}_2(b) = 0$. The second observation is therefore rejected below $w_{\text{over}}^* = c/I(b)$. Combining this with Part (a) gives the stated interval and reward loss. $\square$

Only Tiger in Table 26 satisfies the proposition's two-state and strict-asymmetry hypotheses. Diagnosis and Tileworld use an unproved two-state reduction retaining observation accuracy, sensing cost and correct/incorrect rewards while discarding state and test counts. Those parameters coincide for the two domains, hence their identical rows. Testbed has $|R^-| = R^+$ and is included descriptively. The table compares these thresholds with the Pareto sweep's reward-maximizing weights.

The finite-horizon interval is descriptive for the longer-horizon sweep. Testbed's upper threshold is approximately 1.01 reward units per nat, placing nat-canonical $w = 1$ inside and implemented $w = 1$ outside. Yet at $H = 4$ the two implementations have identical reward, success and usage (results_nat_canonical_check.csv) and both over-observe relative to $w_{\text{ret}}^* \in [0.01, 0.5]$. On Tiger and Diagnosis, $w = 1$ belongs to the observed reward-maximizing range. Tileworld contradicts that extrapolation: its unique sampled optimum is $w = 20$ in bit units (28.9 per nat), above the heuristic upper threshold 7.91, and $w = 1$ is Pareto-dominated.

Table 26. Two-state thresholds in reward units per nat, with reward-maximizing weights w_{ret}^* in reward units per bit. The $w = 1$ column uses nat units, whereas implemented $w = 1$ equals $1/\ln 2 \approx 1.44$ reward units per nat. Bracketed entries are closed ranges of tied maximizing grid weights, not budget-crossing brackets. The eleven-weight grid of Section 7.2 is evaluated for 500 episodes per seed over five seeds (results_pareto_sweep.csv). Only Tiger satisfies the proposition’s hypotheses.

Environment	α	η	w_{thresh}^*	w_{over}^*	$w=1$ in $(w_{\text{thresh}}^*, w_{\text{over}}^*)$?	w_{ret}^*
Testbed	1.0	1.31	-3.06	1.01	Nat units only (see text)	[0.01, 0.5] (tied)
Tiger	10.0	0.27	-138.66	6.89	Yes	[0.01, 20] (tied)
Diagnosis	5.0	0.19	-88.20	7.91	Yes	[0.5, 20] (tied)
Tileworld	5.0	0.19	-88.20	7.91	Yes	20 (unique)

Discounting. Applying $\gamma < 1$ consistently preserves the algebraic equivalence:

$$-\mathcal{G}(\text{observe}_k, b) = -c_k + I_k(b) + \gamma \mathbb{E}_o[\max_a V(a, b'_o)].$$

Current information gain is immediate while the eventual commit reward is discounted. This increases the epistemic share along a branch but need not increase sensing, because the entire continuation is discounted relative to committing now. Diagnosis uses 5.83 observations at $\gamma = 0.95$ versus 9.68 at $\gamma = 1$ (Table 15). Appendix K gives the same-horizon EFE/Planning comparison.

U.3 State-Preservation Proof and Destructive Sensing

Example U.2 (Destructive sensing can make naive information gain harmful). Let $s \in \{0, 1\}$ have uniform prior and commit reward $r(\text{Commit}_i, s) = +1$ if $i = s$, -1 otherwise. A test a_D costs $c > 0$, reveals the old state perfectly, $O(o = s \mid s, a_D) = 1$, and then sets the state to 0, $T(s' = 0 \mid s, a_D) = 1$.

Ignoring the transition gives posterior $b'_o = \delta_o$ and one bit of information. At $w = 1$ reward unit per bit,

$$V_{\text{naive}}(a_D) = -c + 1 + 1 = 2 - c.$$

The correct posterior is instead

$$b'_{o,T}(s') \propto \sum_s T(s' \mid s, a_D) O(o \mid s, a_D) b(s) \implies b'_{o,T} = \delta_0$$

for either report. For $o = 1$ the two posteriors are disjoint point masses, so their KL divergence, and hence Δ_T , is infinite. The post-transition state is already certain before seeing the report. A transition-aware EFE calculation therefore assigns zero epistemic value and gives

$$V_{\text{true}}(a_D) = -c + \max_i r(\text{Commit}_i, 0) = 1 - c.$$

Immediate commitment has expected value zero. For $1 < c < 2$, the naive value recommends testing and the true value rejects it. Moreover, an agent that tests and commits according to the obsolete report realizes

$$-c + \frac{1}{2}r(\text{Commit}_0, 0) + \frac{1}{2}r(\text{Commit}_1, 0) = -c,$$

strictly worse than immediate commitment for every $c > 0$. The example is checked in tests/test_destructive_boundary.py.

Figure 14 isolates the timing distinction behind Example U.2.

PROOF SKETCH. When action a preserves s_{hid} , the transition on the hidden component is $T_{\text{hid}}(s'_{\text{hid}} \mid s_{\text{hid}}, a) = \delta(s'_{\text{hid}} = s_{\text{hid}})$. The belief update over s_{hid} is then $b'(s_{\text{hid}}) \propto O(o \mid s_{\text{hid}}, a) b(s_{\text{hid}})$, depending only on the observation

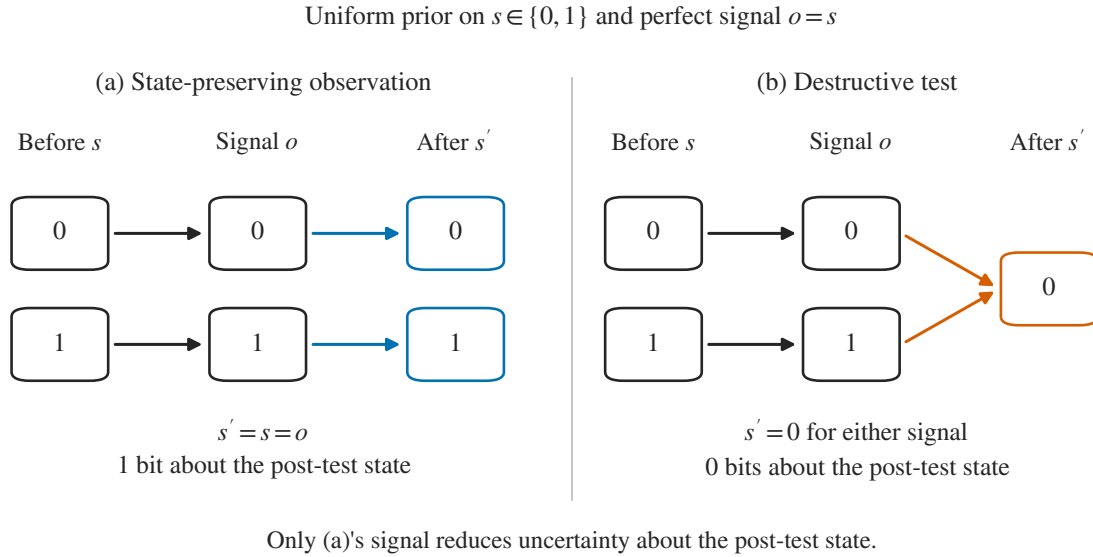


Fig. 14. An observation can describe a state that no longer exists. These schematic timelines use a perfect sensor to contrast state-preserving sensing with the destructive test of Example U.2. The sensor reports the pre-action state in both cases. Under preservation that report identifies the state used by the next decision. Under destruction both possible initial states become 0, so the correct post-action belief is $(1, 0)$ for either report. Treating a report of 1 as the current state gives an obsolete posterior. The example limits the state-preserving reduction, not transition-aware EFE.

likelihood, identical to the observe-then-commit case. The posterior that would incorporate transitions, $b'_{o,T}$, coincides with the observation-only posterior b'_o , so $D_{KL}[b'_{o,T} \| b'_o] = 0$. For observation actions, EFE reduces to cost minus information gain plus expected continuation, matching the ρ -POMDP Bellman equation with $\rho = I_a(b)$. For navigation actions, the observation is uninformative ($I_a(b) = 0$), giving $\rho = 0$. $\square$

The branch identity is preserved under arbitrary interleaving of observation and navigation actions because both preserve the hidden state. It does not cover arbitrary exploitation transitions. Terminal exploitation has no successor belief. Nonterminal exploitation has zero information gain here because its observation is uninformative and rewards are not used for belief updates. A model that conditions on payoffs must include their information gain. RockSample agents mark sampled rocks unavailable but never infer quality from the sampling payoff (rho_aif/agents/rocksample_agents.py).

The hypothesis can model non-destructive testing or spatial sensing of a quantity held fixed over the decision horizon. It must be checked for the particular action and horizon, rather than inferred from an application name. Example U.2 shows what can fail when sensing changes that quantity.

REMARK U.3 (TRANSITION-AWARE INFORMATION GAIN). Example U.2 limits the state-preserving reduction, not EFE with correct transition dynamics. Under our observe-then-transition timing, use

$$b'_{o,T}(s') \propto \sum_s T(s' | s, a) O(o | s, a) b(s)$$

and the corresponding predictive distribution when computing post-transition information gain. In the example both are concentrated at state 0, giving zero information gain and the reward-only value $1 - c$. In general genuine

information about the post-transition state can remain. A nonzero Δ_T witnesses failure of state preservation, but zero does not establish it. The diagnostic is neither an established additive value correction nor a decision-error bound. Extending the budget procedure to these dynamics, or proving a perturbation bound, remains open.

U.4 Agent Specifications and Weight Selection

Planning+IG uses EFE's tree search at the same horizon, adding $wI_a(b)$ to observation values. It represents explicit information rewards in POMDP-IR and ρ -POMDPs (Araya-López et al. 2010; Spaan et al. 2015). Satsangi et al. (2018) prove equivalence of those formulations for piecewise-linear convex belief rewards, a result we do not invoke for the concave expected-IG reward here. Proposition 3.1 identifies EFE with Planning+IG at $w=1$, so within this shared implementation tuned-weight comparisons isolate the weight. The pymdp check (Heins et al. 2022) is qualitative consistency evidence, not certified numerical equivalence (Appendix H).

Epistemic-only sets $\mathcal{G}(\text{commit}) = 0$, comparing information gain with sensing cost without reward-aware stopping. When it commits, it chooses the belief-argmax state. No observation's information gain clears its cost on Tiger, Diagnosis, Bandit, or Tileworld, so it commits from the prior at chance accuracy. Testbed's cheaper observation is the exception, where the ablation matches Planning on observations and success (Appendices B and C).

Myopic ($\rho=0, H=1$) can observe because its lookahead includes one observation followed by a commit, capturing instrumental value of information without an epistemic bonus. Greedy samples unchecked rocks in RockSample and diagnoses from the prior without tests in Inspection. The IDS adaptation minimizes squared expected regret divided by information about the optimal commit (Russo and Van Roy 2014), with its fidelity limits and results in Appendix N. Posterior-vote draws 100 states from the belief and votes over their optimal actions. It approaches a deterministic most-likely-state rule, unlike Thompson sampling, which acts optimally for one posterior draw and retains probability matching. The CSV label ThompsonSamplingAgent denotes posterior-vote.

Info Gain and Planning+IG are success-tuned within their own classes over $\{0.1, 0.5, 1, 2, 5, 10, 20, 50, 100\}$. Each candidate uses 200 episodes, or 100 on the more expensive Tileworld, on the common tuning stream 7, reset for every candidate. Ties select the smallest weight (tune_info_gain_weight, experiments/run_experiment.py). The selected w_{succ}^* therefore carries single-stream sampling noise. This stream is disjoint from the five evaluation seeds. Although seed 7 also appears in Section 6.8's held-out set, that study evaluates only the calibrated family, never a tuned baseline. Section 7.2's separate eleven-weight sweep selects reward-maximizing w_{ret}^* . Neither tuning quantity is the budget crossing $w^*(B)$ or Proposition U.1's thresholds, despite the shared star notation. Section 4 instead estimates $w^*(B)$ by adjacent grid weights whose usage crosses the target.

U.5 Proof and Numerical Check of the Usage-Onset Corollary

PROOF. At $H=1$ the only two plans are commit now and observe once then commit, and Proposition U.1 establishes the observe plan is preferred iff $w > w_{\text{thresh}}$. Consequently $U(w) = \mathbf{1}[w > w_{\text{thresh}}]$ exactly for this single observe-or-commit decision, so U is 0 below w_{thresh} and 1 above it. For any $B \in (0, 1)$ the supremum of the weights with $U(w) < B$ that precede a weight with $U(w) \geq B$ is therefore w_{thresh} , which is the stated crossing threshold, and a grid whose sampled usages straddle B returns a bracket whose closed enclosure contains w_{thresh} . The claim is about this $H=1$ decision, not about the receding-horizon usage curves of Section 6, which take many values and need not be monotone. $\square$

TestProp2OnsetExact in tests/test_budget.py checks the onset with the implemented $H = 1$ planner over complete receding-horizon episodes. It therefore asserts zero usage below the threshold and at least one observation above it, rather than the one-shot proof's exact usage of one. For a Testbed with $p = 0.6$, $c = 0.3$ and rewards ± 1 , the threshold is approximately 3.44 in bit units. The test uses weights 0.9 and 1.2 times that threshold. Appendix W.3 gives refined-grid checks on two instances.

U.6 Proof and Implementation Scope of the Online Controller

The implementation and its three variants are specified in Section 4.5. Only the projected decaying-step variant is covered below. The reported parameters are implemented by `DualWeightAgent` in `rho_aif/agents/dual_descent.py` and `dual_update` in `rho_aif/budget.py`.

The recursion originates with [Robbins and Monro \(1951\)](#), with classical almost-sure extensions due to [Blum \(1954\)](#). [Kushner and Clark \(1978\)](#) treats constrained stochastic approximation. We prove the stated bounded scalar case directly, including discontinuous usage curves. Let $\mathcal{F}_t$ be the history before episode t , $e_t = w_t - w^*$, and $M(w) = U(w) - B$. By (ii), $\mathbb{E}[U_t \mid \mathcal{F}_t] = U(w_t)$ and $(B - U_t)^2$ is bounded by a constant C . Projection onto an interval containing w^* cannot increase distance from w^* , so

$$\mathbb{E}[e_{t+1}^2 \mid \mathcal{F}_t] \leq e_t^2 - 2a_t e_t M(w_t) + C a_t^2.$$

Condition (iii) gives $e_t M(w_t) \geq 0$, also when $w_t = w^*$ without requiring $M(w^*) = 0$. Thus $Y_t = e_t^2 + C \sum_{k \geq t} a_k^2$ is a nonnegative supermartingale and converges almost surely. Summing the expected drift inequality gives $2 \mathbb{E}[\sum_t a_t e_t M(w_t)] \leq \mathbb{E}[Y_0] < \infty$, so that nonnegative series is finite almost surely. Since the tail $\sum_{k \geq t} a_k^2$ vanishes, e_t^2 has an almost-sure limit. If it were positive, the bounded interval and condition (iv) would keep $e_t M(w_t)$ bounded away from zero eventually, contradicting $\sum_t a_t = \infty$. Hence $w_t \rightarrow w^*$ almost surely.

The running-average claim follows from that convergence. Because w^* is interior and the increments $a_t(B - U_t)$ vanish, usage being bounded by (ii), the projection is inactive from some episode on, so $\sum_t a_t(B - U_t)$ converges, its partial sums being differences of the convergent iterates. Since $1/a_t = (1 + \delta t)/\eta_0$ increases to infinity, Kronecker's lemma gives $a_n \sum_{t < n} (B - U_t) \rightarrow 0$, and because $n a_n \rightarrow \eta_0/\delta > 0$ under that schedule, the one property of it the running-average claim uses, this is $\frac{1}{n} \sum_{t < n} (B - U_t) \rightarrow 0$ almost surely. The constant-step default and reset-on-shift runs require separate arguments (Section 4.5). The re-adaptation results of Section 6.2 are empirical. The proof requires a single sign-consistent crossing and does not cover arbitrary nonmonotone usage curves.

U.7 Relationship to Constrained Planning and Experimental Design

The calibrated information weight is not the multiplier of the sensing cap. Classical resource-price adaptation includes constrained MDPs ([Altman 1999](#)), Shannon-capacity and costly-information models ([Matějka and McKay 2015](#); [Sims 2003](#)), and SAC's entropy-temperature dual gradient descent ([Haarnoja, Zhou, Hartikainen, et al. 2018](#)). The last supplies the feedback pattern for Equation 9, with sensing usage replacing policy entropy.

[Kim et al. \(2011\)](#) instead plan over belief/admissible-cost pairs, enforcing costs during pruning by a minimax quadratically constrained program or an approximate linear program. Their randomized policies can outperform our episode-level mixtures, which fix one information weight throughout each episode. Our SARSOP comparison estimates a feasible envelope from sampled usage-penalty policies, not a certified optimum. Sampling error and selection bias also affect that estimate (Appendix W.7).

The same reference construction does not reach EFE's operating region on `RockSample[7,8]`. A 12-point penalty sweep with depth-3 search yields zero or one check and rewards 9.5 or 12.3, versus EFE's 9.84 checks ([experiments/run_rocksample_cpomdp_reference.py](#)). A usage penalty values a check through its short-horizon instrumental benefit, whereas the information weight credits gain locally. A deeper solver is needed for a useful reference.

Sequential Bayesian experimental design selects experiments for expected information gain ([Lindley 1956](#)), potentially through an amortized policy ([Foster et al. 2021](#)). Our objective $R + wI$ balances that gain against task reward and sensing cost, with w calibrated to expected sensing usage. It can therefore decline an informative experiment that an information-only objective selects.

Kouw (2026) derive EFE as a stationary point of constrained Bethe free energy, with a multiplier on an internal information constraint. Their constraint differs both from our policy-level cumulative-information floor (Section 4.1) and from the sensing-usage cap. Their analysis does not supply our scale identities, grid crossings, mixtures or online budget controller. Stocco et al. (2024) instead use recursive, history-dependent dual ascent within constrained-POMDP tree search. They adjust per-node multipliers within a search, whereas our controller adjusts one scalar between episodes.

Offline calibration still requires a sweep, with no sampling advantage over direct weight tuning. Its input is an expected count or cost. The controller avoids that sweep only under Proposition 4.5's conditions. Scale equivariance also transfers a directly tuned weight to αw on a rescaled grid, under the same tie assumptions. These are properties of the stated calibration procedure, not new generic duality results.

V Extended Experimental Specifications and Protocols

Sections 5 and Appendix E specify the environments, episode counts and statistical protocol. Battery-specific counts and seeding differ, so estimates are not interchangeable. For example, Tiger EFE uses 4.20 observations in Table 3 and 4.32 in Table 28. All simulators use the Gymnasium Python interface (Towers et al. 2024).

Our RockSample variant differs from Smith and Simmons (2004) in six ways. Every action costs 0.5, maps are fixed but our own, returns are undiscounted with cap $N^2 + 10K$, exit pays +10 from any cell, sampled rock qualities remain fixed, and sensor accuracy is $\max(0.5, 0.95 \cdot 2^{-d/4})$ at Euclidean distance d . The original has free moves and checks, east-edge exit, a good rock becomes bad after sampling, and its sensor starts perfect and approaches chance asymptotically. Here the sensor reaches chance at about 3.7 cells. Observable sample history prevents repeated sampling rewards. Good/bad sampling pays +10/−10, before action cost. These choices preserve hidden state but prevent comparison of absolute returns with published RockSample results. State counts include only the 2^K hidden quality vectors, not observable position.

Inspection has prior fault probability 0.3, a 0.5 move cost, and visual/detailed tests with accuracies 0.70/0.90 and costs 0.5/2. Correct nominal/fault declarations pay 2/5, missed faults cost 50 and false alarms cost 5. Each component is declared once while the agent stands on it. An episode ends after all declarations or at the step cap. Tests and moves preserve component states. This remains a synthetic known-model benchmark.

V.1 Retrospective Sensor-Acquisition Protocol

The protocol for Section 6.9 was committed before calibration or held-out evaluation. Targets, model, grids, split rules and inferential margins were fixed together. The sensor archive's ten chronological batches and consecutive eight-descriptor sensor blocks follow its primary documentation (Vergara 2012). All sixteen sensors originally operated together. This study masks access to their stored records and does not measure physical sensing costs.

Within each batch/class stratum of batches 1–6, a permutation with seed 20261001 assigns floor proportions 60% to training and 20% to calibration, with the remainder to testing. Exact duplicate feature vectors would be grouped before splitting, and any overlap with later batches excluded from the primary transfer summary. Neither duplicates nor cross-period overlaps occur in this archive. Batch/class stratification preserves represented collection periods, not independence of laboratory exposures. Batches 7–10 contain 3,613, 294, 470 and 3,600 cases, respectively. Each is evaluated separately, with an equal-batch mean also archived. No later label adjusts a prior, model, weight or mixture.

For each sensor, training-only standardization precedes eight-center k -means with ten initializations, at most 300 iterations and seed 20261001 + j for zero-based sensor index j . Add-one smoothing estimates the six-class prior and each class-conditional categorical distribution. Real sensors need not satisfy the model's conditional-independence assumption. Acquisition reveals the held-out row's encoded value, never a simulated draw from the learned likelihood. The observed-sensor mask augments the planning state. At belief b , the one-observation

acquisition value for an unobserved sensor j is

$$Q_w(j, b) = -0.02 + wI_b(Y; Z_j) + \sum_z P(z | b, j) \max_y b(y | z, j),$$

Here Y is gas identity and Z_j is the sensor category. Information is measured in bits, and the stopping value is $\max_y b(y)$. This is the $H = 1$ observe-then-commit convention of Section 3.4, replanned after each access. Numerical ties favor stopping, then the lowest sensor index, using a relative tolerance of 10^{-12} times the largest absolute eligible score or stopping value. Classification ties use the lowest class ID. No case acquires a sensor twice.

The information grid is the sorted union $\{0, \ln 2, 1\} \cup \{2^k : k = -8, \dots, 8\}$. Direct policies omit information gain and replace the acquisition cost by $0.02 + \lambda$, with $\lambda \in \{0, -0.02\} \cup \{\pm 2^k : k = -10, \dots, 1\}$. Negative effective penalties are explicit benchmark subsidies that allow equality references to spend more. Every return is nevertheless scored at the common base cost 0.02. The direct and information-grid equality references maximize estimated calibration return over per-case mixtures with expected calibration usage equal to B . Separate cap references replace equality by an upper bound. These are sampled-family linear programs, not global constrained optima. Direct equality selections at $B = 4$ and 8 mix the $\lambda = -2$ and $\lambda = 2^{-6}$ policies. The crossing rule uses weights (0.0625, 0.125) at $B = 4$ and (8, 16) at $B = 8$, with upper probabilities 0.41328 and 0.22276. Its calibration curve has one descending grid step, so monotonicity is not assumed. Any unbracketed request fails the declared crossing criterion, even if a separate policy can attain it.

The fixed-count CMI policy greedily selects the unobserved sensor with largest model conditional information gain and stops after exactly B accesses. A fixed-order control instead ranks sensors once by training-prior information. Zero-access, full-access and weights 0, 1, and $\ln 2$ are also archived. On within-period test cases, the fixed-order accuracies at $B = 2, 4, 8$ are 68.87%, 68.96% and 76.82%. The weight-one and nat-canonical policies use 6.280 and 5.985 sensors with accuracies 75.99% and 75.91%. These controls isolate aspects of this fitted model and acquisition family. They are not a comparison with state-of-the-art active feature acquisition.

Each deterministic candidate is evaluated on every relevant recorded case, with ordered acquisitions, observed categories, posterior, prediction, usage and base return archived. Mixture expectations average the candidate contributions analytically, representing a random endpoint chosen once per case. A separately seeded realization checks that implementation. Candidate evaluations do not create additional independent cases.

Within-period uncertainty uses 2,000 paired bootstrap replicates, with calibration and test rows resampled independently within their original batch/class strata. Each calibration resample reselects all crossing brackets and fitted LP supports. All policies share the resampled test cases. The three usage-error intervals use Bonferroni-adjusted 98.333% coverage, giving a nominal simultaneous 95% family under the bootstrap assumptions. The margin is ± 0.5 accesses, and all targets must pass for the study-wide criterion. Accuracy comparisons have nominal paired 95% intervals, with no family-wide superiority claim. Failed fits are retained in the denominator. No interval is reported for the 2-sensor crossing, which fails in all replicates. Both higher-target crossings and their references remain feasible throughout. Later-batch intervals resample within each batch/class while holding the original calibration selections fixed. Every interval conditions on one fitted model and assumed exchangeability within strata. It excludes retraining, unobserved dependence among exposures and variation across devices or sites.

The archive reports accuracy, balanced accuracy, class-specific usage, base return, log loss and Brier score. More access need not improve this approximate model's predictions. Full-record test accuracy is 76.32%, below reward-only planning's 82.04% at 3.385 accesses, and full-record log loss is worse than the prior's. Later shifts include changing class proportions as well as sensor responses. Their effects are not separated causally. The results support neither a deployment claim nor the premise that every usage target expresses an external need.

The source checksum, training transforms, row partitions, policies and analysis are reproducible from experiments/run_real_sensor_study.py and experiments/analyze_real_sensor_study.py, with instructions in results/real_sensor_2026-10-01/README.md. Computation is CPU only. Model fitting takes 0.44 seconds, calibration replay 1.71 seconds, held-out replay 13.68 seconds and the complete analysis 5.90 seconds on the recorded host. Peak replay process memory is approximately 186 MB. Timings describe vectorized batches, not per-case response latency. All selected policies, including failures, are retained in results_real_sensor_summary.csv.

W Extended Budget Evidence

The following studies retain the detailed protocols, auxiliary comparisons, and robustness analyses behind Section 6. The main text carries the primary calibration results and their principal limitations.

W.1 Cost-Denominated Budgets

Figure 15 gives the curves behind Section 6.3. Diagnosis tests cost 0.5 and 2.5 but retain equal accuracy. Targets are paired at the same fractional position in each curve's sampled range, with end margins ensuring a crossing. The changing test mix prevents a fixed count-to-cost conversion. Nevertheless, both tested pairs share brackets: the 12.0% spread in mean cost per test does not move either crossing across a grid boundary. Adjacent positive weights differ by $\sqrt{10}$, so this coincidence does not imply interchangeable denominations.

W.2 Curve Collapse across Reward Scales

Table 27 completes Section 6.1's check on grids scaled by $\alpha \in \{0.1, 1, 10\}$, with per-episode shared streams. All matched usage values are bit-identical and normalized brackets coincide to floating-point precision ($\leq 10^{-15}$). Tiger is a trivial collapse with no crossing at $B = 4$, showing that equivariance need not provide an adjustable usage range.

W.3 Proposition U.1 Onset on Positive-Threshold Two-State Instances

The onset check uses two variants of Appendix C's InfoSeekingEnv, with accuracy $p \in \{0.60, 0.58\}$, cost 0.3 and commit payoffs ± 1 . The threshold formula remains applicable at reward asymmetry $\alpha = 1$, although the proposition's strict-asymmetry condition does not. Thresholds are 3.44 and 7.55 reward units per bit, or 4.97 and 10.89 per nat. Usage is zero at and below each threshold and begins in $(w_{\text{thresh}}, 1.03w_{\text{thresh}}]$ on the refined grid (Figure 16). Standard Tiger and Testbed have negative thresholds and $U(0) > 0$, providing the complementary sanity checks.

W.4 Interleaved Usage Curves

Figure 17 uses five common seeds. At each grid weight, RockSample[5,3] has 50 episodes on a ten-point grid, RockSample[7,4] has 30 on eight points, and Inspection- $N = 16$ has 50 on ten points. Four budgets per instance are placed evenly inside the sampled range, with narrower end margins than the count/cost study. Usage SEs are evaluated at the plotted grid weights, not interpreted as price uncertainty.

Section 6.4 reports the ranges and RockSample[5,3]'s nonmonotone dip below $U(0)$. Only the other two sampled curves attain their minimum at $w = 0$. The lowest RockSample[5,3] target is attainable but marked slack because its nearest sampled usage occurs at $w = 0$, below the crossing bracket. None of these observations bounds untested weights. This study uses Inspection- $N = 16$, complementing the $N = 8$ staircase in Section 6.5.

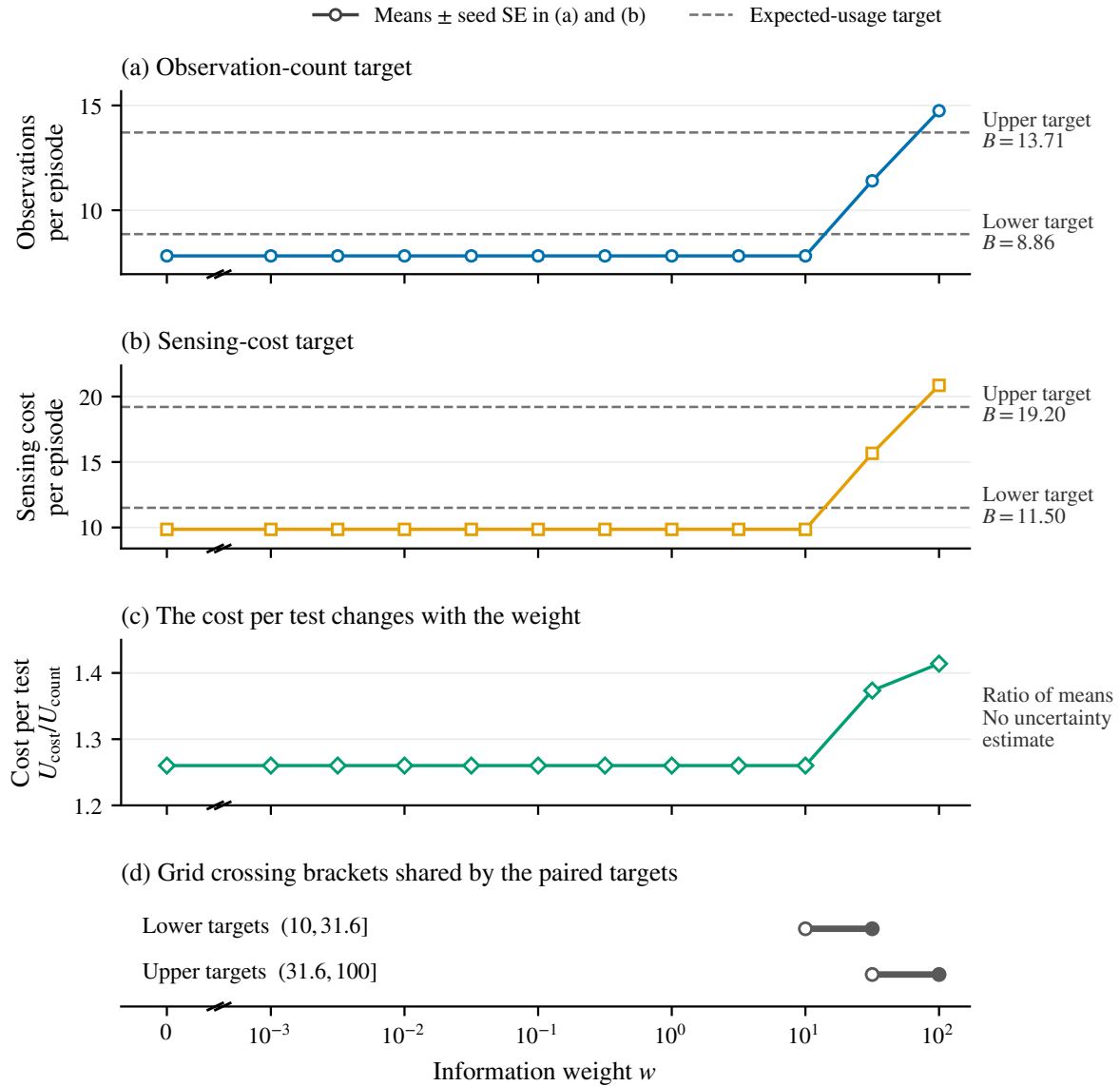
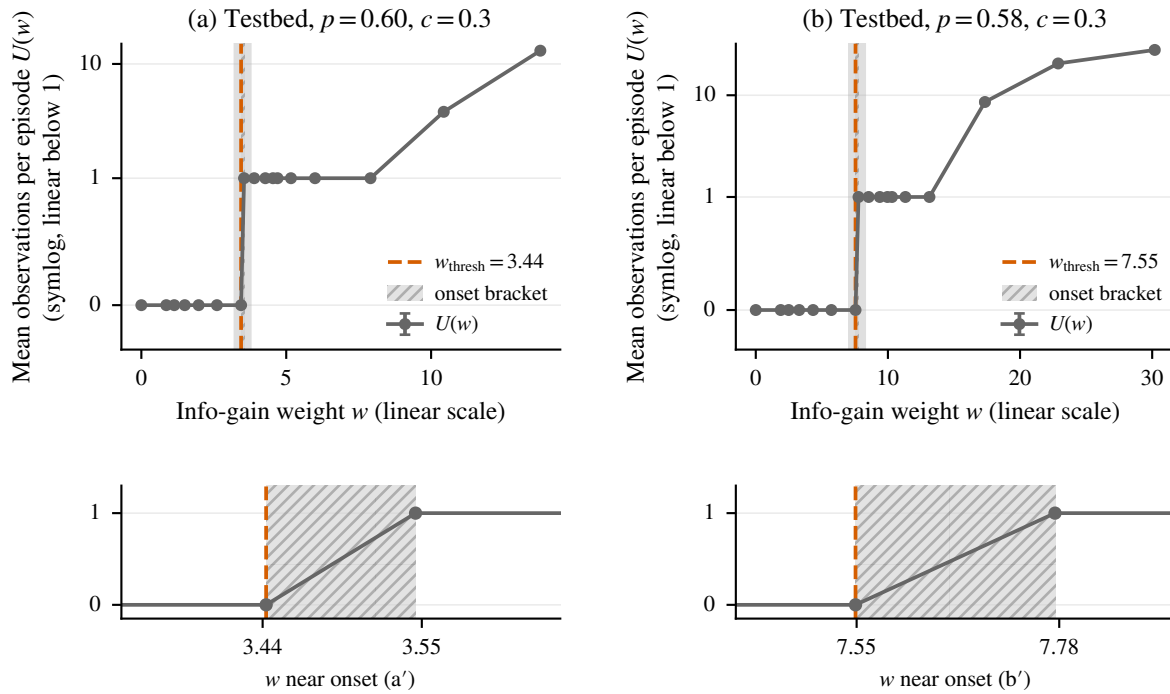


Fig. 15. Observation-count and sensing-cost targets on heterogeneous-cost Diagnosis. Panels (a) and (b) separate the two usage units, showing means ± 1 seed-level SE and the tested targets as dashed lines. The lower count/cost targets are 8.86/11.50, and the upper targets are 13.71/19.20. Panel (c) shows the ratio of mean sensing cost to mean observation count, which rises as the test mix changes. No uncertainty estimate is shown for this ratio. Panel (d) displays the paired targets' shared crossing brackets, (10, 31.6] and (31.6, 100]. These intervals measure grid resolution, not statistical uncertainty. Lines joining sampled means guide the eye.

Table 27. Curve-collapse breadth. Fraction of matched w/α points across $\alpha \in \{0.1, 1, 10\}$ within $2 \cdot \text{SE}$, and whether the B -crossing bracket coincides across scales (in w/α units).

Environment	Within $2 \cdot \text{SE}$	Max spread	Bracket at B
Diagnosis ($B=8$)	100%	0.00 obs. (bit-exact)	(0.141, 0.323] at every α
Bandit ($B=8$)	100%	0.00 obs. (bit-exact)	(3.84, 8.76] at every α
Tiger ($B=4$)	100% (trivial)	0.00 obs. (bit-exact)	unbracketed (B below observed U at every α)
Tileworld-6×6 ($B=15$)	100%	0.00 obs. (bit-exact)	(5.80, 20.0] at every α


 Fig. 16. Onset of observing on positive-threshold testbeds. The vermillion dashed vertical line marks the $H=1$ closed-form threshold (Proposition U.1) and the hatched gray band is the empirical onset bracket after grid refinement, magnified in the zoom row beneath each main panel, with the shaded strip on the main panel marking exactly the region the zoom row below it shows.

W.5 Multi-Seed Online Dual Control under Reward Rescale

Section 6.2 gives the complete detector, recovery definition and paired restricted-time comparison. All ten pairs favor reset, by at least 89 episodes. Decay-only recovers on nine seeds, two only at index 180. Its remaining seed

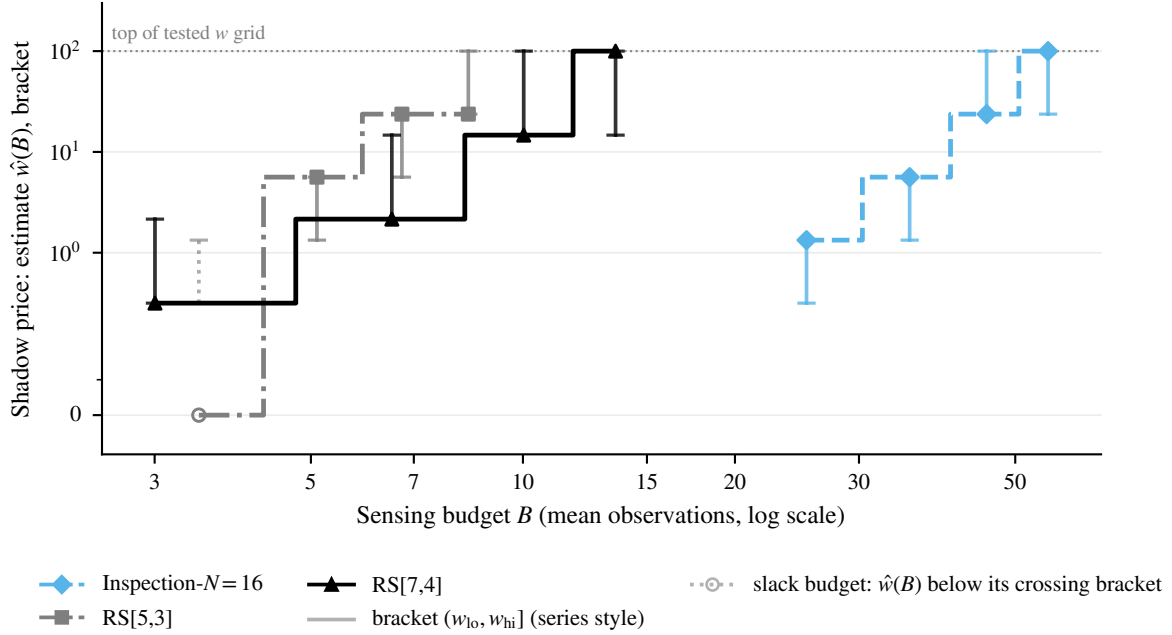


Fig. 17. Grid estimates of the shadow price on three interleaved (non-observe-then-commit) settings, RockSample[5,3], RockSample[7,4], and Inspection- $N=16$, with crossing brackets marked as vertical bars, capped where they reach the dotted rule marking the top of the tested w grid, and an open-circle marker where a budget is slack (the grid weight whose usage comes nearest to the budget sits strictly below the bracket's lower edge, here $w=0$ on the lowest RockSample[5,3] budget). Every marker plots the nearest-point solution $\hat{w}(B)$, which can sit on a bracket's lower edge. The budget axis is logarithmic and the weight axis uses a symmetric-log scale, linear near zero. The underlying usage curves $U(w)$ are monotone at every sampled grid point on RockSample[7,4] and Inspection- $N=16$, while RockSample[5,3] dips once near $w=0.316$ before rising (see text).

ends within tolerance without completing the required hold. Thus its restricted mean of 157.3 is a lower bound on these ten runs' unrestricted recovery time, not a population bound. Among the nine recovered seeds, the descriptive mean is 154.8 with normal-approximation 95% CI [142.7, 166.9]. Every reset run triggers once and recovers, giving 51.2 and [46.1, 56.3]. These separate intervals do not quantify the ratio.

Over the fixed final steady-state block, $|U - B|$ is 0.06 with CI [0.024, 0.096] under reset and 0.39 with [0.118, 0.666] under decay. Before the shift it is 0.10 with [0.051, 0.149] for both. These are normal-approximation intervals across controller seeds. Reset has lower final error on five seeds, ties on four and is worse on one. The restricted-time paired interval in the main text instead uses Student's t distribution. Metrics and the per-episode-seeded traces are archived in `results_price_dual_multiseed_metrics.csv` and its companion outputs. Neither shifted run falls under the stationary guarantee of Proposition 4.5.

W.6 SARSOP as a Near-Optimal Offline Reference

Table 28 gives the five-seed comparison. The common export, $\gamma = 0.999$ solve and alpha-vector evaluation are specified in Appendix T.1. The initial-belief value-bound gap reaches 10^{-3} in under one second per environment.

Table 28. SARSOP and EFE ($w=1$) on the three core environments, five seeds and 500 episodes per seed, $\gamma=0.999$. Values are means $\pm$ seed-level SE. Usage counts observations. Fixed-margin TOST and its retrospective/prospective scope are described in the text.

Environment	SARSOP reward	EFE reward	SARSOP usage	EFE usage
Tiger	5.061 ± 0.158	5.061 ± 0.158	4.323 ± 0.045	4.323 ± 0.045
Diagnosis	-1.452 ± 0.336	-1.217 ± 0.184	9.68 ± 0.12	9.73 ± 0.13
Bandit	6.280 ± 0.134	6.261 ± 0.112	5.16 ± 0.09	5.09 ± 0.07

The unpaired Welch TOST uses the margins of Section 6.6. The larger one-sided p -values on Tiger, Diagnosis and Bandit are 0.0010, 0.0458 and 0.0130, all surviving Holm correction. Their 90% difference intervals are $[-0.41, 0.41]$, $[-0.51, 0.98]$ and $[-0.35, 0.31]$, strictly inside the respective margins. The five-seed margins were chosen retrospectively. Fifteen additional seeds were fixed after the margin and before the $n = 20$ study, which gives $p = 4.5 \times 10^{-10}$, 6.8×10^{-7} and 6.0×10^{-11} (results_tost_sarsop.csv and results_tost_sarsop_n20_robustness.csv, each with per-seed archives).

Shared seed labels do not match episodes when policies consume random draws differently. A five-seed paired sensitivity check gives identically zero Tiger differences, and $p = 0.016$ on Diagnosis and 3.6×10^{-5} on Bandit. Its SEs of 0.24/0.03 are below the unpaired 0.38/0.18, so the unpaired analysis is conservative on these samples, not by design.

The predeclared episode-matched check resets episode e of seed s at $10^4 s + e$, matching hidden states and observation draws while actions agree. It uses the same twenty seeds, 500 episodes each and fixed margins. Tiger again acts identically. Diagnosis's paired difference is $+0.11 \pm 0.09$ with $p_{\text{TOST}} = 3.2 \times 10^{-9}$, and Bandit's is $+0.04 \pm 0.03$ with $p = 4.6 \times 10^{-12}$ (SE). Unpaired tests on those runs also support equivalence (3.8×10^{-5} and 1.8×10^{-8}). Results and per-seed inputs are in results_tost_sarsop_episode_paired.csv and its companion file.

Tiger has no usage-matched crossing on the offline grid: SARSOP's 4.32 is below the sampled floor 4.37, within one usage SE. The "usage-matched" $w = 0$ row in results_sarsop_baseline.json is therefore a low-end fallback, not an attained target.

W.7 A Near-Optimal Constrained-POMDP Reference

Section 6.7 defines the penalty sweep and feasible-envelope LP. An exact maximizer of $R - \lambda U$ is reward-optimal under a cap at its own usage. Our approximately solved, noisily evaluated points do not inherit a certified bound, and a finite penalty grid need not recover the full frontier. No zero duality gap is assumed for the continuous belief MDP. Maximizing over noisy points also introduces selection bias.

At B_{EFE} , the envelope chooses the single $\lambda = 0$ policy on Tiger and Bandit and $\lambda = 0.05$ on Diagnosis. The cap does not bind: usage equals EFE's on Tiger and is lower on the other two. In Diagnosis, reference and EFE usages are 9.785/9.824. Reward SEs of 0.12–0.39 exceed the displayed gaps. Negative estimated gaps can reflect sampling or reference approximation rather than reward beyond the population constrained optimum.

The actually evaluated usage-matched Planning+IG weights are 0.00, 0.45 and 1.07. They lie on sampled usage plateaus $[0, 19.3]$, $[0.316, 19.3]$ and $[0.72, 1.64]$ containing $w = 1$, and their archived rewards equal EFE's. This is a measurement, not an equivalence theorem for other weights. The shared feasible_envelope routine in rho_aif/budget.py produces the reference from results_cpomdp_frontier.csv, with comparisons in results_cpomdp_baseline.csv. No comparable constrained reference is available for the larger RockSample or Inspection instances.

Table 29. Estimated cap-constrained reference at EFE usage B_{EFE} . R_{ref} maximizes reward over mixtures of sampled SARSOP-Lagrangian policies with usage at most that budget. Gap is $R_{\text{ref}} - R_{\text{EFE}}$, and Gap % is $100(R_{\text{ref}} - R_{\text{EFE}})/|R_{\text{ref}}|$. Neither is a certified optimality gap. On each environment the best sampled reference is already feasible, so the cap does not bind. Values use five seeds and 300 episodes per seed. The actually evaluated usage-matched Planning+IG grid weights produce the same rewards as EFE in this run.

Environment	B_{EFE}	R_{ref}	R_{EFE}	Gap	Gap %
Tiger	4.32	5.016	5.016	0.000	0.00%
Diagnosis	9.82	-1.265	-1.224	-0.041	-3.27%
Bandit	5.10	6.314	6.320	-0.005	-0.08%

W.8 Calibration-Derived Target Budgets and What Calibration Buys

Table 30 completes the comparisons in Section 6.8. That section specifies calibration-derived targets, held-out seeds, the original endpoint and feasible-member policies, and the two LP mixtures added after inspection. All selections remain confined to the calibration grid. The best equality mixture coincides with the selected endpoint mixture at eleven of twelve targets, allowing equivalent plateau endpoints. Bandit’s gap target is the exception. The best feasible mixture equals the best feasible member at every slack target, the endpoint mixture at the three binding Diagnosis targets, and the best equality mixture at Bandit’s gap target.

Bandit’s last-crossing rule matters at $B = 5.51$. Usage first crosses between $w = 0.06$ and 0.14 , falls to 5.11 on the $[0.72, 1.64]$ plateau, then crosses again in the reported $(1.64, 3.73]$ bracket. At $B = 4.71$, the only crossing is $(0.06, 0.14]$. Tiger’s gap target duplicates its middle target, leaving eleven distinct attainable budgets.

The committed-reference column uses canonical once-per-seed streams, whereas the family uses held-out per-episode streams. It is descriptive. The same-stream reference was added after inspection, after every saved policy reproduced its committed reward to 10^{-9} . Its paired intervals hold fitted supports and probabilities fixed, omitting their selection uncertainty, cap-feasibility uncertainty and multiplicity. As detailed in the main text, the Tiger reference changes from 5.02 to 5.68 across streams. The canonical wrong-commit rate of 0.6% accounts for about 0.66 expected reward, almost the full difference. No held-out Tiger support policy errs, so its nominal intervals omit important rare-event uncertainty.

A predeclared fresh-seed replication first reproduces all twelve held-out paired gaps to 10^{-9} , then freezes the mixture/reference supports and evaluates seeds 12–21, 100 episodes each, using 9 degrees of freedom. Eight of eleven distinct cap gaps again exclude zero, but the Diagnosis rows change. Its 0.75 gap remains negative at -0.82 ± 0.28 , the 0.5 gap moves from -0.76 to $+0.02 \pm 0.29$, the smallest target becomes negative at -0.56 ± 0.24 , and the gap target narrowly includes zero at -0.46 ± 0.20 (SE). Tiger’s and Bandit’s three negative intervals recur, while Bandit’s smallest target again includes zero. One Diagnosis policy pair had already been evaluated on these fresh seeds before the replication was specified, so that pair is not blind. A wrong diagnosis changes reward by 60 , making these small samples sensitive to few errors. The replication supports the count, not the identity of deficient Diagnosis targets (results_budget_frontier_fresh_seed_replication.csv).

The predeclared target-matched reference adds eleven subsidies $\lambda = -cs$, with s from 0.1 to 0.98 and observation cost c , keeping net observation reward negative. Maximum held-out usages become 7.17 , 16.58 and 11.59 on Tiger, Diagnosis and Bandit. Equality-mixture supports are fitted on held-out usage and frozen for the fresh stream. They coincide with the cap reference at four binding budgets and include a subsidized policy at the seven slack budgets. No slack-budget target-gap interval excludes zero on either stream. The two held-out shortfalls are Diagnosis at $B = 9.53$ and Bandit’s gap target. The two fresh shortfalls are Diagnosis at $B = 7.72$ (-0.56 ± 0.24) and Bandit’s gap target (-0.50 ± 0.11). Only Bandit’s recurs.

Table 30. The calibrated family at calibration-derived target budgets, set by a predeclared rule from each calibration curve and evaluated on five held-out seeds it was not calibrated on (100 episodes per seed, seed-level SE). Budgets are set from each calibration curve’s attainable range $[U_{\min}, U_{\max}]$ at three fractions of the range, at the midpoint of the largest jump (gap), and below $U_{\min}$ (declared unattainable, not run). The mixture plays the bracket’s upper endpoint with probability q and its lower endpoint otherwise, drawn afresh each episode. Usage is the observation count. See `results_budget_frontier.csv` and `results_budget_frontier_heldout_reference.csv`.

Env.	Budget	B	q	Mixture usage	Mixture reward
Tiger	0.25 of range	4.72	0.25	4.64 ± 0.10	5.36 ± 0.10
Tiger	0.50 of range	5.07	0.50	5.04 ± 0.14	4.96 ± 0.14
Tiger	0.75 of range	5.42	0.75	5.35 ± 0.14	4.65 ± 0.14
Tiger	gap	5.07	0.50	5.04 ± 0.14	4.96 ± 0.14
Tiger	below $U_{\min}$	2.18	–	declared unattainable	
Diagnosis	0.25 of range	7.72	0.47	7.70 ± 0.25	-2.14 ± 0.48
Diagnosis	0.50 of range	9.53	0.94	9.47 ± 0.20	-2.11 ± 0.67
Diagnosis	0.75 of range	11.35	0.46	11.47 ± 0.37	-2.31 ± 0.45
Diagnosis	gap	7.84	0.50	7.86 ± 0.25	-2.06 ± 0.47
Diagnosis	below $U_{\min}$	2.95	–	declared unattainable	
Bandit	0.25 of range	5.51	0.18	5.62 ± 0.21	6.06 ± 0.26
Bandit	0.50 of range	7.79	0.29	7.67 ± 0.37	5.70 ± 0.25
Bandit	0.75 of range	10.07	0.17	9.95 ± 0.36	4.95 ± 0.22
Bandit	gap	4.71	0.50	4.62 ± 0.07	5.62 ± 0.21
Bandit	below $U_{\min}$	1.62	–	declared unattainable	

A post hoc analysis refits the equality reference at each mixture’s realized held-out usage rather than at B , using no new episodes. Every held-out Tiger mean gap then becomes zero, because rewards there equal 10 minus usage. Diagnosis still falls short at $B = 9.53$ held out and $B = 7.72$ fresh. Bandit’s gap-target interval reaches +0.001 held out but remains negative fresh. Target gaps thus include usage mismatch, and neither fitted reference certifies the population constrained frontier. These comparisons are archived in `results_budget_frontier_target_reference.csv` and its `usage_matched` companion, with subsidized points in `results_cpomdp_frontier_subsidy.csv`.

W.9 Distractor Robustness under Reward-Irrelevant Sensing

Section 6.10 gives the refined-grid protocol, curves, relevance partition and corrected comparison family. Appendix T.2 establishes the factorization guarantee. The relevance variant changes only the belief on which information gain is scored,

$$\tilde{I}_a(b) = \mathbb{E}_{o \sim Q(\cdot | b, a)} D_{\text{KL}}[Q(s_{\text{rel}} | b, o, a) \parallel Q(s_{\text{rel}} | b, a)],$$

where s_{rel} is the reward-equivalence class determined by commit payoffs. The lookahead still propagates the full joint posterior.

The remaining reward pairs, ordinary versus relevance-weighted information gain, are $-2.45/-1.45$ at $w = 4$ and $-4.31/-1.45$ at both $w = 5$ and 10. These nominal improvements do not survive Holm correction over all weights and reported metrics. At $w = 100$, success is 98.4%/99.2%, also not significantly different. Reward differences survive only at 31.6 and 100, as reported in the main text.

Table 30. The calibrated family (continued). Held-out reward comparisons within the calibration grid. Best member is the calibration-grid weight with the highest calibration reward among those whose calibration usage is at or below B . Target mixture is the reward-maximizing mixture of calibration-grid weights whose calibration usage equals B , and feasible mixture the same with usage at most B , each a linear program with at most two support weights, selected on the calibration data and run on the held-out seeds.

Env.	Budget	Best member	Target mixture	Feasible mixture
Tiger	0.25 of range	5.68 ± 0.08	5.36 ± 0.10	5.68 ± 0.08
Tiger	0.50 of range	5.68 ± 0.08	4.96 ± 0.14	5.68 ± 0.08
Tiger	0.75 of range	5.68 ± 0.08	4.65 ± 0.14	5.68 ± 0.08
Tiger	gap	5.68 ± 0.08	4.96 ± 0.14	5.68 ± 0.08
Tiger	below $U_{\min}$	declared unattainable		
Diagnosis	0.25 of range	-2.86 ± 0.43	-2.14 ± 0.48	-2.14 ± 0.48
Diagnosis	0.50 of range	-2.86 ± 0.43	-2.11 ± 0.67	-2.11 ± 0.67
Diagnosis	0.75 of range	-2.22 ± 0.68	-2.31 ± 0.45	-2.22 ± 0.68
Diagnosis	gap	-2.86 ± 0.43	-2.06 ± 0.47	-2.06 ± 0.47
Diagnosis	below $U_{\min}$	declared unattainable		
Bandit	0.25 of range	6.16 ± 0.23	6.06 ± 0.26	6.16 ± 0.23
Bandit	0.50 of range	6.16 ± 0.23	5.70 ± 0.25	6.16 ± 0.23
Bandit	0.75 of range	6.16 ± 0.23	4.95 ± 0.22	6.16 ± 0.23
Bandit	gap	5.51 ± 0.10	5.94 ± 0.19	5.94 ± 0.19
Bandit	below $U_{\min}$	declared unattainable		

The IDS adaptation’s `info_astar` correctly vanishes for the distractor, but its near-zero-denominator fallback to `info_state` rewards nuisance information. Its distractor fraction is 0.305 ± 0.005 , versus ordinary information gain’s saturated $0.330\text{--}0.332$ (SE $0.004\text{--}0.005$). The relevance variant has no fallback and buys zero distractor tests. This isolates a flaw in that IDS implementation pattern, not in reward-aware information measures generally. Appendix N reports its broader results.

Table 30. The calibrated family (continued). Constrained-reference comparisons. The reference is the estimated feasible envelope of the near-optimal constrained-POMDP reference of Section 6.7, the largest reward attainable by mixing its sampled points subject to expected usage at most B , flat at the largest sampled reward where B exceeds the largest sampled usage. That reference was evaluated on the canonical seeds with once-per-seed seeding. Same-stream reference is the same envelope with every sampled reference policy re-evaluated on the held-out seeds under the family’s per-episode seeding, and paired gap is the mixture’s reward minus that envelope, per seed, with its seed-level SE. Reference support and mixture weights are fitted on those same held-out seed means and held fixed for the SE. The nominal pointwise plug-in intervals discussed in the text omit uncertainty from that fit and cap feasibility, and make no multiplicity adjustment. Target reference is the best mixture of sampled reference policies, including subsidized ones with negative usage penalties, whose held-out usage equals B , and target gap is the paired held-out gap against it (results_budget_frontier_target_reference.csv).

Env.	Budget	Reference	Same-stream reference	Paired gap	Target reference	Target gap
Tiger	0.25 of range	5.02	5.68	-0.33 ± 0.04	5.28	0.08 ± 0.04
Tiger	0.50 of range	5.02	5.68	-0.72 ± 0.06	4.93	0.03 ± 0.06
Tiger	0.75 of range	5.02	5.68	-1.03 ± 0.06	4.58	0.07 ± 0.06
Tiger	gap	5.02	5.68	-0.72 ± 0.06	4.93	0.03 ± 0.06
Tiger	below $U_{\min}$	declared unattainable				
Diagnosis	0.25 of range	-2.00	-2.11	-0.03 ± 0.28	-2.11	-0.03 ± 0.28
Diagnosis	0.50 of range	-1.35	-1.35	-0.76 ± 0.15	-1.35	-0.76 ± 0.15
Diagnosis	0.75 of range	-1.27	-1.21	-1.10 ± 0.22	-2.26	-0.05 ± 0.16
Diagnosis	gap	-1.95	-2.06	0.00 ± 0.22	-2.06	0.00 ± 0.22
Diagnosis	below $U_{\min}$	declared unattainable				
Bandit	0.25 of range	6.31	6.27	-0.21 ± 0.12	6.22	-0.16 ± 0.12
Bandit	0.50 of range	6.31	6.27	-0.57 ± 0.09	5.78	-0.08 ± 0.06
Bandit	0.75 of range	6.31	6.27	-1.32 ± 0.08	4.90	0.05 ± 0.04
Bandit	gap	6.23	6.21	-0.59 ± 0.20	6.21	-0.59 ± 0.20
Bandit	below $U_{\min}$	declared unattainable				

X Extended EFE Comparisons

These analyses retain the full per-environment comparisons and behavioral qualifications summarized in Section 7.

X.1 Core Environments

Appendix B adds the baselines omitted from Table 3 and prints pooled episode-level SE rather than seed-level SE. It omits untuned Info Gain rows identical to Myopic and Tiger’s pymdp-AIF row identical to Posterior-vote. Both uncertainty summaries remain in the CSVs.

Posterior-vote is a relevant negative control. On Bandit it is not significantly different from EFE on reward (6.32/6.27, $p = 0.58$), success (87.0%/86.9%, $p = 0.97$) or usage (5.01/5.11, $p = 0.30$). On Testbed it ties Planning exactly and beats EFE. On Diagnosis and Tiger its rewards are instead -2.78 and 3.79, against EFE’s -1.37 and 5.19. Epistemic-only commits immediately at chance on Tiger, Diagnosis, Bandit and Tileworld, so removing reward-aware selection causes non-exploration in this implementation.

Table 13 reports pooled effect sizes. EFE/Planning reward effects are zero on Tiger and below 0.15 in magnitude on Diagnosis and Bandit, despite corrected seed-level reward and success differences on the latter two. The large Tileworld-8 $\times$ 8 reward effect ($|d| = 1.13$) compares EFE with a non-scanning planner. Equal episodes per seed make pooled and seed-level mean differences identical, so sign agreement is no independent check. Inference uses seed-level tests and intervals, not Cohen’s d computed from only five seed means.

X.2 Pareto Analysis and the Information-Unit Weight $w=1$

Section 7.2 and Figure 9 give the eleven-weight sweep on five environments, excluding RockSample and Inspection. Success peaks at the grid's upper endpoint $w = 200$ throughout. Moving to $w = 50$ on Tiger and Diagnosis gains at most 1.8 percentage points of accuracy while costing 1.07 and 2.41 reward. Bandit's $w = 5$ gains 9.8 points for 0.41 reward. Tileworld instead maximizes sampled reward at 20, while Testbed's maximizing sampled weights run from 0.01 to 0.5. These are sampled-grid trade-offs, leaving intermediate weights unresolved. The three reward-maximizing runs containing $w = 1$ do not establish success optimality or a scale-free default.

X.3 Spatial Epistemic Foraging on Tileworld

Table 31 and Figure 18 complement Section 7.3. EFE and Planning do not separate on reward ($p = 0.82$) or success ($p = 0.17$) at 6×6 , while EFE uses fewer scans (14.75/15.64, seed-level $p = 8.8 \times 10^{-5}$). These are failures to reject differences, not demonstrated equivalence. The figure's InfoGain-Tuned row is the myopic agent at its own tuned weight, distinct from the tree-search Planning+IG table row.

Table 31. Tileworld 6×6 (2,500 episodes at 500 per seed $\times$ 5 seeds, $H=2$). Tuned weight $w_{\text{succ}}^* = 50$. Reward uncertainty is seed-level SE. Bold marks EFE and any agent not significantly different from it at the seed level (Welch $p > 0.05$, uncorrected), not the single best value per column.

Agent	Scans	Success	Reward
Myopic	0.00	2.9%	-48.25 ± 0.33
Planning ($H=2$)	15.64	73.8%	-21.39 ± 0.14
Planning+IG ($w=50$)	32.25	97.3%	-23.85 ± 0.19
EFE ($H=2$)	14.75	72.0%	-21.53 ± 0.57

The separate scaling sweep fixes $H = 2$ and $w = 100$ for both weighted agents, even though one series retains the name InfoGain-Tuned. It uses the smaller episode count in Figure 10, not the full-table protocol. At 4×4 and 6×6 , EFE and Planning reward and success are within one SE without formal tests for this sweep. At 8×8 , Planning takes no scans and matches Myopic exactly (1.4% success, -49.16 reward), while EFE reaches $69.4\% \pm 1.0$ points. A two-scan lookahead cannot sufficiently concentrate the initially uniform belief over 64 cells, so this is a horizon-and-scale result rather than evidence against deeper reward planning. The fixed-100 Planning+IG series reaches 97.8% success at a substantial reward cost.

A separate 6×6 partition study uses $H = 2$, five seeds and 200 episodes per seed. Random partitions assign cells to equal halves, losing guaranteed joint identification. Overlapping partitions threshold random weighted row/column coordinates, creating correlated tilted cuts. These replace the complementary bit partitions while keeping the rest of the benchmark fixed (experiments/run_tileworld.py partition). EFE/Planning rewards, with seed-level SE, are $-21.62 \pm 0.87/-21.41 \pm 0.64$ for bitwise, $-29.31 \pm 0.43/-30.32 \pm 0.66$ for random and $-47.53 \pm 0.73/-47.66 \pm 0.68$ for overlapping partitions. Success is 71.9/73.6%, 56.1/55.1% and 8.5/8.4%, respectively. No mode separates them on reward or success. The largest reward gap, under random partitions, has Welch $p = 0.24$. EFE's scan counts are lower in all three (14.76/15.57, 12.97/13.38, 2.63/2.70). Absolute performance falls sharply with less complementary observations, so this study does not establish model-independent advantage (results_partition_sensitivity_6x6.csv).

X.4 RockSample as an Interleaved Observe-Act Setting

The weighted agents use independent per-rock beliefs and the tabled search depths. They share a leaf rule that repeatedly chooses an unsampled rock with positive belief-weighted sample value, subtracts travel cost, visits the

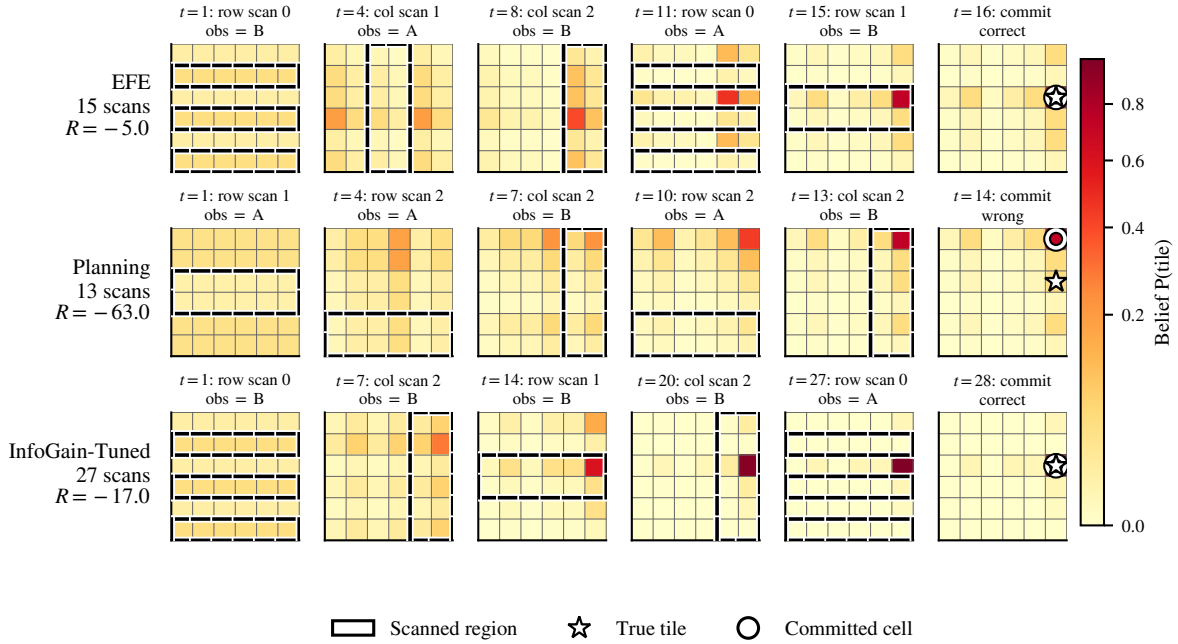


Fig. 18. Agent comparison on the same 6x6 Tileworld episode. All agents start from the identical uniform prior, stated here rather than drawn once per row. Belief probability is encoded by the shared colorbar on a square-root color scale. Black outlines mark the cell group queried by each scan, each panel's title reports that scan's noisy binary readout as obs=A or obs=B naming which half of the queried group the signal favors, the black-edged white star marks the true tile, and the black-stroked white ring marks the committed cell. Scan panels are subsampled evenly within each row, so columns are not time-aligned across rows and each panel's t gives the true step. EFE (top) commits correctly with efficient scanning. Planning (middle) commits on the wrong cell after 13 scans. InfoGain-Tuned (bottom) over-scans well past the point of diminishing returns.

best remaining rock and samples it, then exits when none remains. The leaf also charges travel to the east edge before exit, although the environment allows exit anywhere. This deflates depth-capped continuations relative to immediate exit and biases the entire family toward early exit. All weighted agents use this same leaf rule.

The standalone heuristic in Table 4 is POMCP's approach-then-check rollout without a tree. It approaches the unresolved rock with highest belief per unit distance, checks nearby at accuracy 0.95, samples above its belief threshold, abandons rocks below 0.15 and exits when none remains. It is a hand-coded policy without a search horizon. Greedy instead samples without checking. Appendix M gives the POMCP protocol and sensitivity studies.

On RS[5,3], EFE and $w = 5$ are not significantly different (16.47/16.41, Welch $p = 0.74$). On RS[7,4], EFE leads $w = 5$ (15.96/14.56, $p < 10^{-5}$) and is not significantly different from the heuristic (15.85, $p = 0.65$). On RS[7,8], EFE and $w = 5$ (21.85/23.76, uncorrected $p = 0.09$) both fall below the heuristic's 28.33 ± 0.82 , with Holm-significant $p = 4.6 \times 10^{-4}$ and 2.4×10^{-3} . These non-rejections are not equivalence tests. Raising w from 5 to 10 hurts on RS[7,8] (20.55, $p = 0.003$) and RS[5,3] (14.60, $p = 3.5 \times 10^{-9}$), surviving the per-metric Holm family, while RS[7,4] does not separate ($p = 0.18$). The three tested weights establish no dense-grid optimum.

On RS[11,11], Planning and EFE both earn 13.23 while the heuristic earns 28.66 ± 0.83 . A depth-three rerun leaves Planning/EFE at 13.23 and Planning+IG at 13.17, within one SE of the shallower results (results_rocksample_11x11_depth3_check.csv). The heuristic collects 5.22 good rocks against about half a rock for each tree-search agent. Useful distant checks require travel beyond the tested exact-search horizons. At the uniform prior, each untouched rock has zero expected sampling reward and fails the leaf's positive-value filter, compounded by the phantom exit cost. These observations support a horizon-and-leaf limitation, not absence of exploitable reward. The adverse deeper-POMCP results in Appendix M show that additional reach alone does not resolve it. Across the four RockSample instances and three Tileworld sizes, this is a mechanistic interpretation of representable sensing opportunities, not an independently tested scaling law.

X.5 Structural Inspection

The tree-search agents share a hand-coded greedy completion rule and differ only in information weight. Table 5's Missed column counts faulty components declared nominal, and Tests counts all tests per episode. EFE's accuracy gain over Planning has seed-level Welch $p < 10^{-10}$ at $N = 8$ and $p < 10^{-7}$ at $N = 16$. At $N = 16$, $w = 5$ uses 38.3 ± 0.05 tests against EFE's 28.1 ± 0.09 ($p = 1.7 \times 10^{-11}$, SE), while EFE and Planning do not separate on reward ($p = 0.57$).

Applying Proposition U.1 here is only a heuristic two-state reduction. The nominal declaration has reward asymmetry $|R^-|/R^+ = 25$, but fault declaration is symmetric, with ratio 1. The missed-fault penalty can therefore remove Part (a)'s lower-threshold restriction. Part (b)'s over-observation threshold is not evaluated, so this does not place $w = 1$ inside the full near-optimality interval or establish deployment relevance.

Y Additional Interpretation and Tree-Search Controls

This appendix retains the detailed weight-transfer, near-optimality, and tree-search analyses summarized in Section 8. The comparisons use the protocols and reference files stated locally. They describe the tested settings rather than general regime boundaries.

Y.1 The Information-Unit Weight and Tuning

Weight transfer requires matching reward/information units and the planning configuration. At $w = 100$ on Bandit, depth-two Planning+IG uses 12.36 inspections against myopic Info Gain's 11.50 (canonical five-seed Welch $p = 0.003$, experiments/run_bandit_w100_depth_comparison.py). This is evidence that a weight tuned at one depth need not retain its usage at another. Appendix K gives the separate discount sweep. Diagnosis and Bandit success gains appear at $\gamma \in \{0.99, 1\}$ but not $\{0.90, 0.95\}$, a result for those tested configurations rather than a general discount threshold.

Y.2 Zero-Shot Weight Transfer

Table 32 transfers weights tuned within the evaluated Planning+IG class. Under this shared reward convention, moderate weights lie on flat reward regions, whereas the success-tuned $w = 50$ harms reward, including on its tuning environments. This is neither scale-free calibration nor evidence that the canonical weight is optimal everywhere.

Proposition U.1's lower threshold concerns $H = 1$ and its interval concerns $H = 2$. Their two-state scope and heuristic reductions are detailed in Appendix U.2. The random-environment study of Appendix J instead uses an empirical reward-gap criterion: $w = 1$ passes on 30%, 58% and 79% of the 100 sampled environments at depths 1, 2 and 3. Thus greater depth still leaves about one fifth failing. Expected-usage calibration addresses a different requirement and remains conditional on target attainability within the selected family.

Table 32. Zero-shot weight transfer (500 episodes $\times$ 5 seeds, reward reported as mean $\pm$ seed-level SE, results_transfer.csv). Rows include $w=1$, reward-tuned weights (w_{ret}^*), and success-tuned weights (w_{succ}^*), the Planning+IG-class values recorded in results_supplementary_tuned_weights.csv, matching the evaluated agent class. Success-tuned transfer harms reward at the top of the grid, while moderate weights transfer well. Bold entries mark the best reward per column (Tiger and Diagnosis show a four-way tie among $w \in \{0.5, 1, 10, 20\}$, bit-identical, not merely within SE).

Weight	Tiger	Diagnosis	Bandit	Testbed
$w=0.5$ (w_{ret}^*)				
Diag./Testbed	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+6.05 $\pm$ 0.11	+0.48 $\pm$ 0.02
$w=1$ (EFE / w_{ret}^*)				
Tiger/Bandit	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+6.24 $\pm$ 0.09	+0.37 $\pm$ 0.01
$w=10$ (w_{succ}^*)				
Testbed	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+5.25 $\pm$ 0.06	+0.01 $\pm$ 0.01
$w=20$ (w_{succ}^*)				
Bandit	+5.33 $\pm$ 0.26	-1.26 $\pm$ 0.34	+5.05 $\pm$ 0.07	-0.18 $\pm$ 0.01
$w=50$ (w_{succ}^*)				
Tiger/Diag.	+4.25 $\pm$ 0.08	-3.68 $\pm$ 0.14	+4.23 $\pm$ 0.06	-0.38 $\pm$ 0.01

Y.3 Approximate Planning with MCTS-EFE

Exact enumeration costs $\mathcal{O}((K|O|)^H)$ for K sensing actions with $|O|$ outcomes each. MCTS-EFE uses UCB1 over observe/commit actions, enumerates each observation action's outcomes, adds exact per-node information gain, and backs up maxima. EFE-greedy leaf rollouts compare immediate commitment with the best one-step EFE observation. Commit-child values are exact belief-expected rewards. The default exploration constant is the fixed commit-reward range, 110 on Tiger, rather than the preliminary-return range used by [Silver and Veness \(2010\)](#). Appendix S reports exploration tuning, informed rollouts, the ablation and a separate compute-matched Tiger control.

The simulation-matched comparison uses 500 simulations on Tiger at $H = 10$. MCTS-EFE reaches $97.7\% \pm 0.5$ percentage points against POMCP's $88.1\% \pm 0.2$ ($p = 4.4 \times 10^{-6}$). Its cost is 14–16 ms per episode across tested horizons, versus 11–12 ms for POMCP, so matching simulations does not match computation. At $H = 5$ and 200 simulations on Diagnosis, the corresponding success rates are $94.2\% \pm 0.5$ and $65.5\% \pm 2.1$ ($p = 1.1 \times 10^{-4}$). These are seed-level SEs and Welch tests from results_mcts_eфе.csv.

On Tileworld-6 $\times$ 6, MCTS-EFE reaches $95.4\% \pm 0.6$ at both 200 and 500 simulations, compared with Exact-EFE's $71.9\% \pm 1.5$ under this battery's protocol. The gain costs reward (-25.02/-21.62) and observations (32.3/14.8). POMCP reaches only $2.6\% \pm 0.4$ and $3.9\% \pm 0.5$, with about 0.09 scans per episode. The success gaps have $p < 10^{-9}$ at both budgets. On Tiger and Diagnosis, MCTS-EFE instead falls below Exact-EFE on both success (97.7/99.9%, 94.2/97.1%) and reward (3.73/5.65, -3.26/-1.44). Thus approximate search does not uniformly improve either metric.

The controls detect no success or reward effect of either switchable component, in-tree information gain and max-backup, after Holm correction at five seeds. EFE leaf evaluation, exact beliefs and exact commit-child values remain coupled. The exploration selection uses disjoint tuning seeds but its rule was written after inspecting the canonical-seed sweep, so Tiger's tuned comparison is retrospective. These results concern the joint implementation, not isolated directed sensing or general superiority over POMDP solvers. Progressive widening methods ([Benchetrit et al. 2025](#); [Sunberg and Kochenderfer 2018](#)) are possible extensions for larger observation spaces, not evaluated improvements here.

Z Reproducibility Checklist for JAIR

This appendix answers the JAIR reproducibility checklist (Gundersen et al. 2024) for the submission version of this article, as required for review. It is not part of the accepted version.

All articles:

- (1) All claims investigated in this work are clearly stated. **Yes.**
- (2) Clear explanations are given how the work reported substantiates the claims. **Yes.** Proofs accompany the statements or appear in Appendices A and U. Empirical claims are supported by committed result files and their reported summaries.
- (3) Limitations or technical assumptions are stated clearly and explicitly. **Yes.** The statement assumptions, Example U.2, discussion and conclusion delimit the structural, reward-convention and empirical claims.
- (4) Conceptual outlines and/or pseudo-code descriptions of the AI methods introduced in this work are provided, and important implementation details are discussed. **Yes.** The recursive EFE agent (Equation 3), the usage-curve procedure (Section 4), and the online dual controller update are given in closed form in the main text, with build and export details for the SARSOP baseline in Appendix T.1.
- (5) Motivation is provided for all design choices, including algorithms, implementation choices, parameters, data sets and experimental protocols beyond metrics. **Yes.** Section 3 motivates the observe-then-commit restriction, and Section 4.6 positions every budgeted-formulation design choice against prior art.

Articles containing theoretical contributions:

Does this paper make theoretical contributions? **Yes.**

- (1) All assumptions and restrictions are stated clearly and formally. **Partially.** The definitions and propositions state their structural and regularity assumptions. The information base, bits in the implementation and nats in the canonical identification, is specified in the surrounding text rather than inside Proposition 3.1.
- (2) All novel claims are stated formally (e.g., in theorem statements). **Yes.** Propositions 3.1, U.1, 3.3, 4.2, 4.3, 4.5, Definition 4.1, and Corollary 4.4.
- (3) Proofs of all non-trivial claims are provided in sufficient detail to permit verification by readers with a reasonable degree of expertise. **Partially.** Appendix A proves Proposition 3.1. The remaining statements have proofs or case-analysis sketches at their statements or in Appendix U. The supermartingale and running-average proof covers only the projected decaying-step controller, not the released constant-step default or reset-on-shift heuristic.
- (4) Complex formalism, such as definitions or proofs, is motivated and explained clearly. **Yes.**
- (5) The use of mathematical notation and formalism serves the purpose of enhancing clarity and precision, and gratuitous use of mathematical formalism is avoided. **Yes.**
- (6) Appropriate citations are given for all non-trivial theoretical tools and techniques. **Yes.** Log scoring (Bernardo 1979), the Robbins–Monro recursion (Robbins and Monro 1951), classical almost-sure extensions (Blum 1954), background on constrained stochastic approximation (Kushner and Clark 1978), and the sophisticated-inference Bellman-optimality result (Da Costa, Sajid, et al. 2023) that the recursive agent is built on.

Articles reporting on computational experiments:

Does this paper include computational experiments? **Yes.**

- (1) All source code required for conducting experiments is included in an online appendix or will be made publicly available upon publication of the paper. **Yes.** https://github.com/PatrickAllenCooper/rho_aif. PyPI distribution as rho-aif is planned but not yet published.

- (2) The source code comes with a license that allows free usage for reproducibility purposes. **Yes.** MIT License.
- (3) The source code comes with a license that allows free usage for research purposes in general. **Yes.** MIT License.
- (4) Raw, unaggregated data from all experiments is included in an online appendix or will be made publicly available upon publication of the paper. **Partially.** Summary CSVs support every reported simulation result, but episode-level records are incomplete. Per-episode dual-control traces, per-seed SARSOP TOST results (including the $n=20$ and episode-matched checks), budget-frontier and same-stream reference rows, fresh-seed and target-matched gaps, and usage-curve/bracket-bootstrap files are committed. Appendix V.1 additionally archives every sensor-study candidate trajectory, model, split, selection and bootstrap output. UCI supplies the original measurements under a pinned checksum.
Other simulation trajectories can be regenerated using the recorded environment and README command map. Core, Tileworld, Inspection, transfer, IDS, scaling, discount and navigation summaries carry seed, episode, revision and timestamp provenance. RockSample records seeds and episode counts. Offline-reference, calibration and atlas protocols are stated in their captions or prose, while the MCTS ablation and POMCP sweep carry full provenance. Relevant summaries include seed-level SEs and Welch/Holm companions, with IDS comparisons reported inline. Reward rescaling and the single-seed, 50-episode horizon map retain point estimates only and support descriptive claims.
- (5) The unaggregated data comes with a license that allows free usage for reproducibility purposes. **Yes.** The repository's MIT License covers its code and generated simulation outputs. The external sensor measurements retain the primary source's license and attribution terms (Vergara 2012).
- (6) The unaggregated data comes with a license that allows free usage for research purposes in general. **Yes.** The sensor archive is obtained from UCI under its stated research-use and attribution terms.
- (7) If an algorithm depends on randomness, then the method used for generating random numbers and for setting seeds is described in a way sufficient to allow replication of results. **Yes.** Main simulation results use {42, 123, 456, 789, 1024}. Weight tuning uses stream 7 (Appendix U.4). The frontier uses held-out seeds 7–11 and fresh seeds 12–21, both at 100 episodes/seed. The $n=20$ SARSOP checks add seeds 2000–2014. POMCP constants use tuning seeds {11, 22, 33} at 100 episodes/seed, with evaluation on the canonical five. Counts for the three-seed reward-rescaling sweep, ten-seed dual study, ten-seed smaller RockSample instances and other batteries are stated with their results.
The benchmark, budget, RockSample, Inspection, distractor, frontier and dual runners reset each episode at $10^4 s + i$, for outer seed s and episode index i . SARSOP, CPOMDP, TOST and the observe-then-commit calibration runner instead seed once per outer stream, except for TOST's explicit –episode-seeding mode. The horizon map derives its seed from environment and horizon. POMCP's internal seed varies with the outer seed in its main comparison and exploration sweep, while the MCTS-EFE and compute-matched batteries retain a fixed internal seed (Appendix S). The dual rescale is also seeded and covered by a trajectory-reproduction regression test.
Sensor-study split/quantizer, mixture-check and bootstrap seeds are 20261001, 20261002 and 20261003, with a derived later-batch stream. Its bootstrap draws are conditional case resamples, not independent model fits. Exact reproduction uses the recorded environment or archived trajectories.
- (8) The execution environment for experiments, the computing infrastructure (hardware and software) used for running them, is described, including GPU/CPU makes and models, amount of memory (cache and RAM), make and version of operating system, and names and versions of relevant software libraries and frameworks. **Yes.** All experiments are CPU-only. The recorded host is an Apple M5 Max with 36 GB RAM, macOS 26.6.2 and Python 3.9.6. ENVIRONMENT.md records the machine and solver build, requirements-lock.txt pins package versions, and pyproject.toml gives installation requirements.

- (9) The evaluation metrics used in experiments are clearly explained and their choice is explicitly motivated. **Yes.** Success rate, mean reward, mean observation count, and usage (sensing spend) are defined at first use and motivated by the observe-then-commit setting's cost structure (Section 3).
- (10) The number of algorithm runs used to compute each result is reported. **Yes.** Five seeds per condition unless stated otherwise (e.g., ten seeds for the multi-seed dual-control study of Section 6.2). Episode counts are reported per benchmark in the seeding item above and in every results table caption.
- (11) Reported results have not been "cherry-picked" by silently ignoring unsuccessful or unsatisfactory experiments. **Yes.** Section 8 and the conclusion explicitly report where the proposed method does not help (Section 6.10's distractor robustness result, Example U.2's destructive-sensing failure mode, and cases where tuned Planning+IG exceeds EFE on particular metrics).
- (12) Analysis of results goes beyond single-dimensional summaries of performance (e.g., average, median) to include measures of variation, confidence, or other distributional information. **Yes.** Standard errors and confidence intervals are reported throughout Sections 6 and 7 (e.g., the paired Student-*t* interval for the recovery-time difference in Section 6.2, the mean $\pm$ SE usage-curve ranges of Appendix T.3).
- (13) All (hyper-)parameter settings for the algorithms/methods used in experiments have been reported, along with the rationale or method for determining them. **Yes.** Planning horizons and discount factors are reported per environment (Appendix E) and the tuned Planning+IG weight search protocol is stated in Section 3, with discount-factor sensitivity swept explicitly in Appendix K.
- (14) The number and range of (hyper-)parameter settings explored prior to conducting final experiments have been indicated, along with the effort spent on (hyper-)parameter optimisation. **Yes.** The Pareto sweep over *w* (Section 7.2) reports the full grid searched, not only the selected operating point, and the horizon sweep of Appendix J searches the horizon grid $H \in \{1, 2, 3\}$ and the weight grid $\{0.01, 0.5, 1, 2, 5, 50\}$, both stated in `experiments/run_nearopt_horizon.py`, from which its reward-optimal weight is selected.
- (15) Appropriately chosen statistical hypothesis tests are used to establish statistical significance in the presence of noise effects. **Yes.** Confidence intervals and seed-level *t*-tests (`tests/test_stats_seed.py`) are used rather than point-estimate comparisons alone. The SARSOP comparison of Section 6.6 is additionally backed by a fixed-margin TOST equivalence test (Schuirmann 1987). The near-optimal CPOMDP reference of Section 6.7 estimates the optimality gap directly via a Lagrangian-penalized SARSOP sweep, evaluated by simulation rather than a certified bound, in place of an informal consistency argument.

Articles using data sets:

Does this work rely on one or more data sets (possibly obtained from a benchmark generator or similar software artifact)? **Yes.** Section 6.9 uses the public UCI Gas Sensor Array Drift dataset (Vergara 2012; Vergara et al. 2012). The primary source, download checksum, original row identities, deterministic training/calibration/test split and later-batch holdout are archived. Appendix V.1 describes the sensor grouping, training-only transforms, fixed targets and model restrictions. There are no exact duplicate or cross-period overlapping feature vectors. Access costs are benchmark conventions, not measured physical costs. Other environments are simulators defined generatively in code (Appendix E). Our cost-modified RockSample is not the standard generator and its absolute returns are not comparable to published RockSample numbers.

Explanations on any of the answers above (optional):

The pinned environment and archived policy outputs permit reconstruction of the reported estimates. Simulation and bootstrap seeds are explicit. Floating-point results and numerical tie decisions can differ across processors and system libraries, so exact reproduction uses the recorded environment or saved trajectories. Timings describe the measured host and are not hardware-independent performance guarantees.